ANNUAL REVIEWS Further
Click here to view this article's online features:
• Download figures as PPT slides
• Navigate linked references
• Download citations
• Explore related articles
• Search keywords

6) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Structure and Kinematics of Early-Type Galaxies from Integral Field Spectroscopy

Michele Cappellari

Subdepartment of Astrophysics, Department of Physics, University of Oxford, Oxford OX1 3RH, United Kingdom; email: michele.cappellari@physics.ox.ac.uk

Annu. Rev. Astron. Astrophys. 2016. 54:597–665

The *Annual Review of Astronomy and Astrophysics* is online at astro.annualreviews.org

This article's doi:
10.1146/annurev-astro-082214-122432

Copyright © 2016 by Annual Reviews.
All rights reserved

Keywords

elliptical galaxies, S0 galaxies, galaxy evolution, galaxy formation, galaxy structure, galaxy kinematics and dynamics

Abstract

Observations of galaxy isophotes, long-slit kinematics, and high-resolution photometry suggested a possible dichotomy between two distinct classes of elliptical galaxies. But these methods are expensive for large galaxy samples. Instead, integral field spectroscopy can efficiently recognize the shape, dynamics, and stellar population of complete samples of early-type galaxies (ETGs). These studies showed that the two main classes, the fast and slow rotators, can be separated using stellar kinematics. I show that there is a dichotomy in the dynamics of the two classes. The slow rotators are weakly triaxial and dominate above $M_{\rm crit} \approx 2 \times 10^{11}\,{\rm M}_{\odot}$. Below $M_{\rm crit}$, the structure of fast rotators parallels that of spiral galaxies. There is a smooth sequence along which the age, the metal content, the enhancement in α-elements, and the weight of the stellar initial mass function all increase with the central mass density slope, or bulge mass fraction, while the molecular gas fraction correspondingly decreases. The properties of ETGs on galaxy scaling relations, in particular the $(M_*, R_{\rm e})$ diagram, and their dependence on environment, indicate two main independent channels for galaxy evolution. Fast-rotator ETGs start as star-forming disks and evolve through a channel dominated by gas accretion, bulge growth, and quenching, whereas slow rotators assemble near the centers of massive halos via intense star formation at high redshift and remain as such for the rest of their evolution via a channel dominated by gas poor mergers. This is consistent with independent studies of the galaxies redshift evolution.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Contents

1. INTRODUCTION 598
2. STRUCTURE FROM PHOTOMETRY 599
2.1. Definition of Early-Type Galaxies 599
2.2. Global Galaxy Profiles 600
2.3. Isophotal Shape 600
2.4. Nuclear Galaxy Profiles 601
3. STRUCTURE FROM KINEMATICS 602
3.1. Visual Classification of Kinematic Maps 602
3.2. Generalizing Photometry to Kinematics Maps 604
3.3. Intrinsic Shapes of Early-Type Galaxies 606
3.4. Dynamical Modeling of Stellar Kinematics 610
3.5. Understanding the $(V/\sigma, \varepsilon)$ Diagram 617
3.6. Quantitative Kinematic Classification 621
3.7. Two-Dimensional Stellar Population 626
4. LOCAL SCALING RELATIONS 627
4.1. Classic Scaling Relations 627
4.2. Understanding Scaling Relations via Integral Field Spectroscopy 630
4.3. The Mass-Size and Mass-σ Distributions 635
4.4. Parallel Sequencing of Early-Type Galaxies and Spirals 641
5. ENVIRONMENTAL TRENDS 643
5.1. Kinematic Morphology-Density Relation 643
5.2. Mass-Size Versus Environment 645
6. REDSHIFT EVOLUTION 647
6.1. Mass-Size Evolution 647
6.2. Profile Evolution 649
7. IMPLICATIONS FOR GALAXY FORMATION 649
7.1. Galaxy Evolution on the (M_*, R_e) Plane 650
7.2. Hierarchical Origins of Fast- and Slow-Rotator Early-Type Galaxies 652
8. CONCLUSIONS 654

1. INTRODUCTION

Elliptical galaxy: displays elliptical isophotes on the sky

S0 galaxy: has an outer disk like spiral galaxies but has a smooth appearance without spiral arms

Our knowledge of the structure of galaxies in general, and of early-type galaxies (ETGs; ellipticals and S0s) in particular, has evolved in parallel with the technological advances in the instrumentation used to study them. Photographic plates of galaxies formed the basis of the classic and still widely popular galaxy morphological classification scheme by Hubble (1926). Beginning in the late 1950s, rotation curves were obtained from ionized gas emission in external spiral galaxies (e.g., Burbidge et al. 1959). This revealed flat rotation curves out to large radii (Rubin & Ford 1970), an observation subsequently strengthened by radio determinations (Bosma 1978), and indicated dark matter surrounding the galaxies (see Courteau et al. 2014 for a review). Dark matter is one of the pillars of the current paradigm of how galaxies form (White & Rees 1978, Blumenthal et al. 1984).

In the 1970s, long-slit spectrographs were used to measure stellar rotation in elliptical galaxies. This revealed that massive elliptical galaxies tend to rotate slowly, whereas less-massive ellipticals and galaxy bulges appear to rotate faster (Section 3.5.1). In the 1980s, photometry of elliptical

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

galaxies using CCDs showed that faint ellipticals can have disky isophotes, suggesting the presence of stellar disks embedded in dominant stellar spheroids (Section 2.3). The *Hubble Space Telescope* (HST) showed in the 1990s that the steepness of the inner galaxy profiles is also related to other galaxy properties (Section 2.4). HST also revolutionized our understanding of the link between galaxies and black hole (BH) evolution (for a review, see Kormendy & Ho 2013).

In parallel to detailed studies of nearby galaxies, deep multi-band photometric galaxy surveys [e.g., GOODS (the Great Observatories Origins Deep Survey), Giavalisco et al. 2004; COSMOS (the Cosmological Evolution Survey), Scoville et al. 2007]; and multiplexed spectroscopic surveys, [e.g., DEEP2 (the Deep Extragalactic Evolutionary Probe 2), Newman et al. 2013b] were delivered. These provided the ability to trace galaxy evolution back in time (for a review, see Conselice 2014).

This review focuses on the progress in our knowledge of the structure of ETGs, and of galaxies in general, brought by the next major technological advance: integral field spectroscopy (IFS). IFS brings the ability to obtain a spectrum at every position on a grid of sky coordinates covering the galaxy image. Effective IFS prototypes appeared on the scene in the 1990s (e.g., Bacon et al. 1995), but it took a decade for them to reach maturity. IFS has recently become a standard asset on all major telescopes and an essential tool of astrophysics research.

This review summarizes the status of our knowledge as another instrumental revolution is about to take place. Currently, the first generation of IFS surveys, which targeted one galaxy at a time [SAURON, de Zeeuw et al. 2002; the DiskMass survey, Bershady et al. 2010; the ATLAS^{3D} survey, Cappellari et al. 2011a; CALIFA (the Calar Alto Legacy Integral Field Area Survey), Sánchez et al. 2012] have been completed, and a new generation of multiplexed IFS surveys, which can observe multiple galaxies simultaneously, is starting [the SAMI (Sydney-AAO Multi-object IFS) survey, Bryant et al. 2015; the MaNGA (Mapping Nearby Galaxies at APO) survey, Bundy et al. 2015].

Early-type galaxy (ETG): elliptical or S0, defined by the lack of spiral arms in optical images

IFS: integral field spectroscopy

2. STRUCTURE FROM PHOTOMETRY

This section gives a concise summary of the main findings on the structure of ETGs before the publication of the first large set of IFS kinematics for ETGs by Emsellem et al. (2004). Most early developments were based on photometry alone. Extensive reviews were written on the classification (Sandage 1975, 2005) and photometry (Kormendy et al. 2009, Kormendy & Bender 2012, Graham 2013) of ETGs. I refer the reader to these publications for a summary of the historic developments, and here I only mention the results and definitions that are essential to understand the findings from two-dimensional spectroscopy, which is the focus of this review.

2.1. Definition of Early-Type Galaxies

The focus of this review is ETGs, although I make clear that one cannot understand ETGs without linking their structure to the general galaxy population. ETGs can be broadly characterized by their old stellar populations, red colors, small amount of cold gas and dust, and lack of spiral arms. These characteristics, however, are not equivalent. In fact, applying different selection criteria for ETGs leads to quite different sets of galaxies (Strateva et al. 2001, Conselice 2006, van den Bergh 2007, Bernardi et al. 2010).

Here I adopt the standard definition of ETGs as the galaxies in the handle of Hubble's (1936) empirical tuning fork diagram. In the revised Hubble classification system (Sandage 1961), the separation between ETGs and spiral galaxies is entirely based on the presence of spiral arms (or extended dust lanes for edge-on galaxies). This separation was adopted unchanged by de Vaucouleurs (1959, 1963) and in the very popular *Third Reference Catalog of Bright Galaxies* (de Vaucouleurs

et al. 1991, hereafter RC3). The same criterion was also used in the RC3 follow-up HyperLeda (Paturel et al. 2003). This nearly universal definition of ETGs is also adopted in this review.

Sérsic index (n): increases with the galaxy concentration

a_4: isophotal shape parameter; $a_4 < 0$ is boxy and $a_4 > 0$ is disky

Disky elliptical E(d): morphological classification intermediate between ellipticals and S0s, for which the disk dominates the central regions, whereas the spheroid dominates at large radii

2.2. Global Galaxy Profiles

For many years, elliptical galaxies were thought to be well described by the de Vaucouleurs (1948) $R^{1/4}$ profile. Caon et al. (1993) discovered that the profiles of many elliptical galaxies require the more general parametrization proposed by Sérsic (1968):

$$I(R) = I_e \exp\left\{-b(n)\left[\left(\frac{R}{R_e}\right)^{1/n} - 1\right]\right\}, \quad (1)$$

where $b(n) \approx 2n - 0.327$ (Capaccioli 1989). R_e is the radius of a circle enclosing half of the projected total galaxy luminosity, and I_e is the surface brightness at R_e. Most importantly, it was found that the Sérsic index n, related to the galaxy concentration (the ratio between the amount of light contained within the central regions and the amount in the outskirts), increases with increasing total galaxy luminosity. This result was confirmed by several authors (e.g., D'Onofrio et al. 1994, Bertin et al. 2002, Graham & Guzmán 2003, Ferrarese et al. 2006, Kormendy et al. 2009).

The reported correlation was interpreted as due to a systematic change in the intrinsic properties of elliptical galaxies with luminosity. However, it can also be partially explained by a systematic variation in the fraction of hidden disks in ellipticals. Most likely, both effects play a role. In fact, the Sérsic index of spiral galaxy and S0 bulges tends to be different from that of massive ellipticals (e.g., Kormendy et al. 2009, Krajnović et al. 2013a), and systematic variations in the Sérsic index of spiral bulges also exist (Andredakis et al. 1995).

2.3. Isophotal Shape

The shape of elliptical isophotes is quantified by finding the best-fitting ellipse, which provides a good first-order approximation, and then measuring the deviations from that ellipse. In the standard algorithm (Carter 1978, Jedrzejewski 1987, Bender et al. 1988, Franx et al. 1989, Peletier et al. 1990), the galaxy surface brightness, sampled along the ellipse, is fitted by a truncated Fourier expansion via linear least squares minimization,

$$\Sigma(\psi) = \Sigma_0 + A_1 \sin(\psi) + B_1 \cos(\psi) + A_2 \sin(2\psi) + B_2 \cos(2\psi), \quad (2)$$

where ψ is the eccentric anomaly, so that the angles between the sampled points are equally spaced when the ellipse is projected onto a circle. Here, Σ is the surface brightness, and Σ_0 is its average value along the ellipse. The best-fitting ellipse is defined as the one for which A_1, B_1, A_2, and B_2 are zero within numerical accuracy. Once the best-fitting ellipse has been determined, the surface brightness residuals along that ellipse are parameterized by the next higher term in the Fourier expansion,

$$\Delta\Sigma(\psi) = A_3 \sin(3\psi) + B_3 \cos(3\psi) + A_4 \sin(4\psi) + B_4 \cos(4\psi). \quad (3)$$

When the profile is properly sampled, the higher-order terms are mathematically orthogonal to the lower-order ones. The amplitude of the axially symmetric fourth Fourier coefficient $a_4 \equiv \sqrt{A_4^2 + B_4^2}$ measures whether an isophote is boxy ($a_4 < 0$) or disky ($a_4 > 0$; **Figure 1**). For physical interpretation, the quantity a_4 is typically divided by $(a\ \mathrm{d}\Sigma/\mathrm{d}a)$, where a is the ellipse semimajor axis, in such a way that a_4 represents fractional deviations of the isophote from the best fitting ellipse.

These studies led to the discovery that disky ellipticals appeared to rotate faster than nondisky or boxy ellipticals (Lauer 1985, Bender 1988). Moreover, radio-loud ellipticals were found to only

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

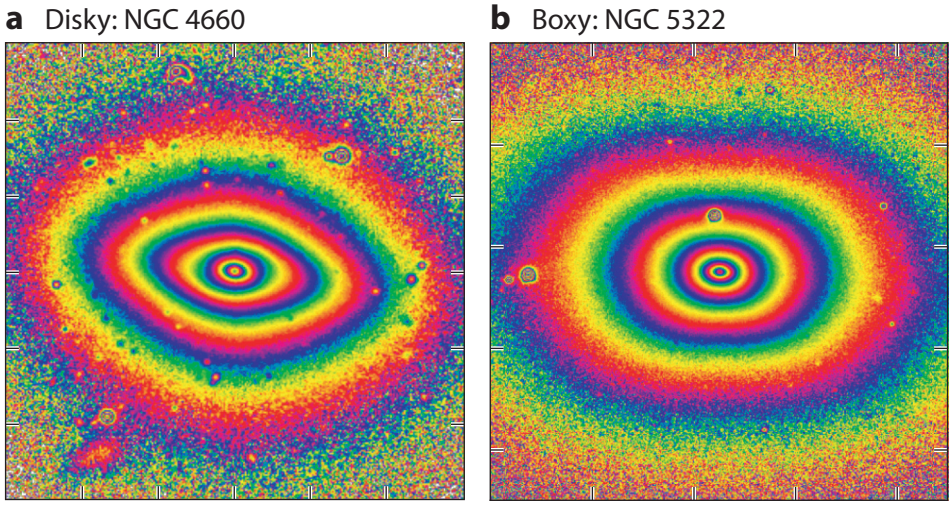


Figure 1

Disky versus boxy isophotes. (*a*) NGC 4660 has disky isophotes with isophotal shape parameter $a_4 \approx 3\%$. (*b*) NGC 5322 has boxy isophotes with $a_4 \approx -1\%$. The photometry is taken from the Sloan Digital Sky Survey. Bender et al. (1988) chose these galaxies as representative of the two classes they described.

be present in ellipticals without disky isophotes, showing that isophotal shape was an intrinsic parameter in galaxy structure (Bender et al. 1989). Ultimately, the realization of the important connection between isophotal shape and other global galaxy properties led to the proposal of a new classification scheme for ellipticals (Kormendy & Bender 1996).

2.4. Nuclear Galaxy Profiles

The launch of HST in 1990 revolutionized the study of inner profiles in ellipticals by firmly establishing that the centers of ellipticals do not possess flat cores (Crane et al. 1993). Additionally, it was found (Ferrarese et al. 1994, Lauer et al. 1995) that the profiles of ellipticals can be broadly separated into two classes, core galaxies and power-law galaxies. The core galaxies show clear breaks in the surface brightness, with a steeper outer profile followed by a more shallow inner one, while the power-law galaxies show modest changes in profile slope (**Figure 2**, Ferrarese et al. 1994, Lauer et al. 1995).

A popular parameterization used to quantify the surface brightness profile shapes is the following double power-law (informally dubbed the Nuker law when it was proposed by Lauer et al. 1995):

$$\Sigma(R) = \Sigma_b \left(\frac{R_b}{R}\right)^{\gamma} \left[\frac{1}{2} + \frac{1}{2}\left(\frac{R}{R_b}\right)^{\alpha}\right]^{(\gamma-\beta)/\alpha}, \quad (4)$$

where γ and β represent the (positive) asymptotic slope for radii smaller and larger (respectively) than the break radius R_b, where the surface brightness is Σ_b and α determines the sharpness of the transition. From this relation, the slope γ' is defined as the analytic value inferred from Equation 4 at HST resolution limit $R_0 \approx 0.''1$. According to this definition, core galaxies are those with $\gamma' \leq 0.3$ (Lauer et al. 1995, 2007b).

Alternative ways of quantifying inner profiles include measuring either extra light or a deficit with respect to a global Sérsic fit to the outer profile (e.g., Ferrarese et al. 2006, Kormendy et al. 2009), or using the core-Sérsic parameterization (Graham et al. 2003, Trujillo et al. 2004b).

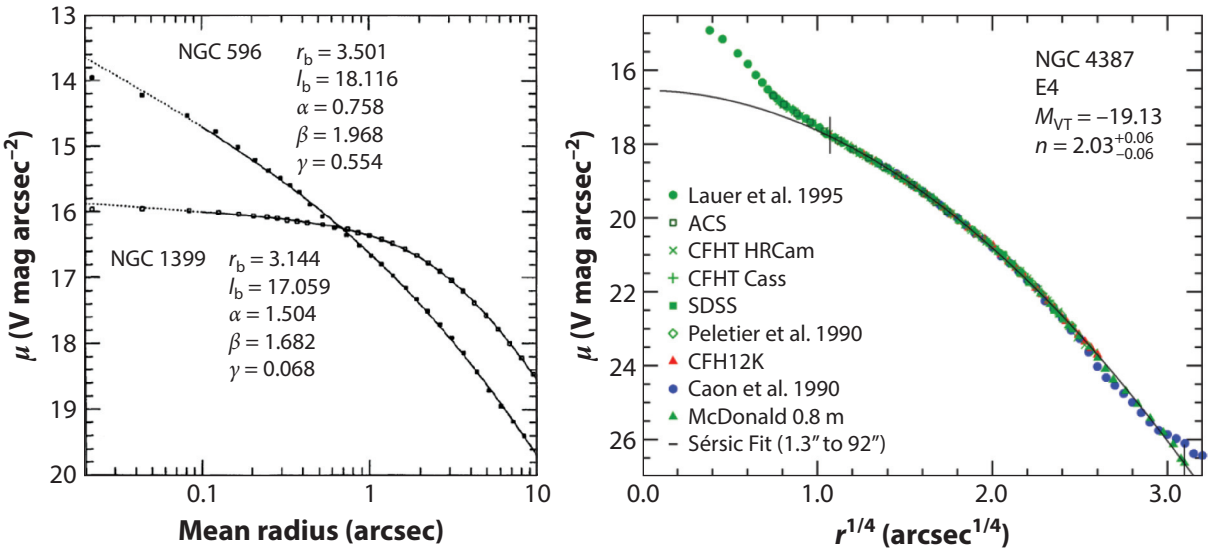


Figure 2

Nuclear profiles of early-type galaxies. (*left*) The break in the surface brightness profile of a core galaxy is contrasted to the smooth, steep rise of a power-law one (from Lauer et al. 1995). Both are fit by Equation 4. (*right*) Extra light (with respect to a Sérsic profile fit) in the inner surface brightness profile. The extra light component is thought to be produced by gas accretion or mergers when cold gas dissipates and falls to the center, producing a starburst (Kormendy et al. 2009). This figure is modified with permission from Kormendy et al. (2009).

Reassuringly, the different profile classifications agree with each other with good accuracy (Krajnović et al. 2013b).

Some studies have found a bimodality, or dichotomy, in the distribution of the profile slopes within the general galaxy population (Lauer et al. 1995; Byun et al. 1996; Lauer et al. 2005, 2007b), whereas others find smooth continuity (Carollo et al. 1997, Ferrarese et al. 2006, Côté et al. 2007). However, the key and remarkable result on which all studies agree is that the inner slope is closely related to the global galaxy properties measured on spatial scales more than one order of magnitude larger (Faber et al. 1997; Lauer et al. 2005, 2007a; Ferrarese et al. 2006; Kormendy et al. 2009). This relationship shows that nuclear slopes provide key physical insights related to galaxy structure. The results described in Section 2 are summarized in the sidebar "Elliptical Galaxy Dichotomy Before Integral Field Spectroscopy."

3. STRUCTURE FROM KINEMATICS

3.1. Visual Classification of Kinematic Maps

The photometric approach to classifying ellipticals described in Section 2 suffers from two limitations: (*a*) measuring nuclear profiles requires subarcsecsecond spatial resolution, and (*b*) deviations from elliptical isophotes are only visible near edge-on orientations. Thus, neither technique is applicable to large galaxy samples at significant distances.

A solution to both limitations is provided by IFS. In fact, IFS observations of stellar kinematics provide the long-sought ability to recognize the presence of stellar disks at virtually any inclination. This is illustrated in **Figure 3**, which shows predictions, using dynamical models (Cappellari 2008), of how two nearly edge-on galaxies, the disky elliptical (as classified by Bender et al. 1994) NGC

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

ELLIPTICAL GALAXY DICHOTOMY BEFORE INTEGRAL FIELD SPECTROSCOPY

Before the two companion SAURON papers introducing a kinematic classification system for elliptical galaxies (Cappellari et al. 2007, Emsellem et al. 2007), two classes among ellipticals were recognized:

Giant ellipticals ($M_V \lesssim -21.5$): (*a*) have Sérsic function outer profiles with $n \gtrsim 4$ (Caon et al. 1993), (*b*) have cores in their nuclear profiles (Ferrarese et al. 1994, Lauer et al. 1995), (*c*) rotate slowly (Illingworth 1977), (*d*) are anisotropic and triaxial (Binney 1978), (*e*) tend to be rounder (Tremblay & Merritt 1996), (*f*) can have boxy isophotes (Bender 1988), (*g*) contain X-ray emitting gas (Bender et al. 1989), and (*h*) have old and α-element enhanced stellar populations (Thomas et al. 2005).

Normal-luminosity ellipticals ($M_V \gtrsim -21.5$): (*a*) have Sérsic profiles with $n \lesssim 3$ (Graham & Guzmán 2003), (*b*) are coreless (Faber et al. 1997), (*c*) rotate rapidly (Davies et al. 1983), (*d*) are nearly isotropic and oblate (Kormendy & Bender 1996), (*e*) can be quite flat, (*f*) can have disky isophotes, (*g*) rarely contain X-ray gas, and (*h*) can have stellar populations that are not young or α-enhanced.

The physical relevance of this possible dichotomy is summarized by Kormendy & Bender (1996), who propose a revision to Hubble's classification of ellipticals based on isophotal shapes, and by Faber et al. (1997) (but see Ferrarese et al. 2006 for a different view). Kormendy & Bender (2012) and Kormendy (2016) provide more recent, post-IFS reviews of the dichotomy.

821 and the S0 galaxy NGC 5308, would appear when seen close to face on. The justification for the adopted models is given in Section 3.4.3. The figure shows that both the centrally concentrated disk of a disky elliptical and the extended disk of an S0 galaxy have clear, observable kinematic signatures at nearly all inclinations. In both cases the velocity fields display extended, ordered rotation, with the kinematic position angle $PA_{\rm kin}$ aligned with the photometric major axis $PA_{\rm phot}$. In contrast, the measurable effects on the isophote shape are hidden in the noise for inclinations $i \lesssim 60°$. This is quantified by Krajnović et al. (2013a, figure 8), who show that even the strong disk of an edge-on S0 galaxy produces nearly elliptical isophotes ($a_4/a_0 \lesssim 2\%$).

The earliest IFS observations of stellar kinematics used a simple but very time-consuming *y*-scanning approach, in which a long slit is moved across the field of view to map the two-dimensional velocity field of the nearest elliptical, Centaurus A (Wilkinson et al. 1986). This was followed by proof-of-concept large-scale observations of the stellar kinematics of individual ETGs with real IFS units such as the TIGER integral field unit (Bacon et al. 1995) on the Canada-France-Hawaii Telescope (Emsellem et al. 1996, 1999) or the MPFS integral field unit (Sil'chenko et al. 1997) on the 6-m telescope of the Special Astrophysical Observatory (Sil'chenko 1999).

A breakthrough came with the introduction of the SAURON (Bacon et al. 2001) IFS due to the dramatic improvement of the instrument data quality with respect to previous prototypes. The SAURON survey (de Zeeuw et al. 2002) was the first project to map the two-dimensional stellar kinematics, ionized gas, and stellar population of a significant sample of 48 nearby ETGs with total absolute magnitudes $M_B < -18$.

A striking feature immediately apparent from a simple visual inspection of the kinematic maps of the sample galaxies (Emsellem et al. 2004) is the qualitative difference between two classes of ETGs: one class of galaxies is consistent with the models of **Figure 3**—namely, with what one would expect for disks seen at various inclinations (**Figure 4*e***)—whereas the other class is clearly inconsistent with simple disks (**Figure 4*a*–*d***). In two companion papers from the survey, this initial insight led to the proposal of a quantitative kinematic classification of ETGs that is virtually independent of inclination effects (Emsellem et al. 2007, Cappellari et al. 2007).

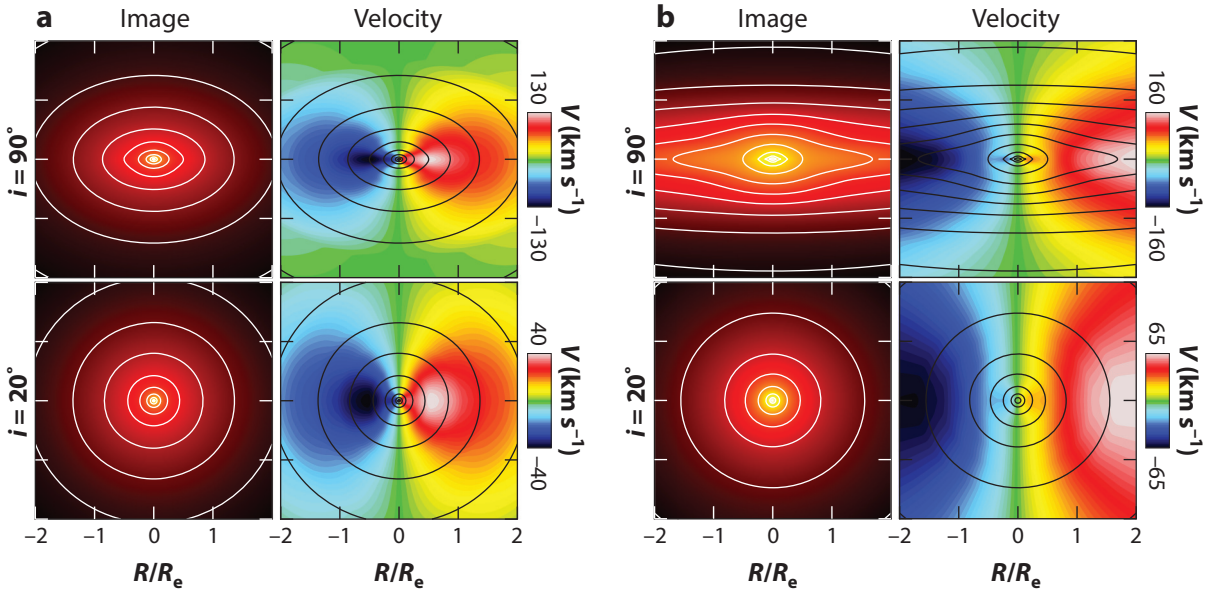


Figure 3

Recognizing face-on stellar disks. (*a*) Photometric models for the disky elliptical galaxy NGC 821, projected at two inclinations. The rainbow-colored panels show the corresponding stellar velocity predicted by the Jeans anisotropic models or Cappellari (2008) for each location. (*b*) As in panel *a*, for the S0 galaxy NGC 5308. The presence of a disk can be visually recognized from the kinematics even near face-on inclination, although the photometric evidence disappears below $i \lesssim 60°$. The input multi-Gaussian expansion parameterization for the surface brightness is taken from Cappellari et al. (2006) and Scott et al. (2009) for panels *a* and *b*, respectively, and the adopted model parameters are from Cappellari (2008). Note the different behavior of the velocity in the two types of galaxies. In disky ellipticals, the stellar disk dominates the surface brightness only out to $R \lesssim R_e$ (the half-light radius), and the velocity sharply drops beyond that radius. In the S0, the disk extends to the edge of the galaxy, and the velocity is still rising at $R \gtrsim 2R_e$. This difference can be predicted by the models from the photometry alone, without the need to invoke differences in the orbital distribution, or anisotropy, which is assumed constant throughout the galaxy. Importantly, in both cases, the rotation is not limited to the disks; however, the spheroids rotate more slowly owing to their rounder shapes.

The SAURON survey was followed by the ATLAS^{3D} project (Cappellari et al. 2011a), which, rather than employing IFS alone, was a multiwavelength survey combined with a theoretical modeling effort. It targeted a complete sample of 260 ETGs, extracted from a volume-limited sample of 871 galaxies brighter than $M_{K_s} < -21.5$, within $D \lesssim 42$ Mpc. The observations of the sample spanned from radio to millimeter and optical wavelengths. The survey includes galaxies with a minimum stellar mass of $M_* \gtrsim 6 \times 10^9\ M_\odot$.

ATLAS^{3D} confirmed the striking visual distinction between the kinematics of regular and nonregular rotators (Krajnović et al. 2011). It additionally defined four subclasses of the nonregular class, as illustrated in **Figure 4**. Nonregular rotators were found to (*a*) either not rotate at all, (*b*) show clear but not bisymmetric or irregular rotation, (*c*) present kinematically decoupled cores (KDCs; these features were previously described by Bender 1988, Jedrzejewski & Schechter 1988, Franx & Illingworth 1988, Franx et al. 1989), or (*d*) have two counter-rotating disks (such as the prototypical S0 NGC 4550 discovered by Rubin et al. 1992, Rix et al. 1992).

KDC: kinematically decoupled core

LOSVD: line-of-sight velocity distribution

3.2. Generalizing Photometry to Kinematics Maps

A galaxy image is only the zeroth moment of the stellar line-of-sight velocity distribution (LOSVD). This suggests that one may use an approach similar to that described in Section 2.3 to

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

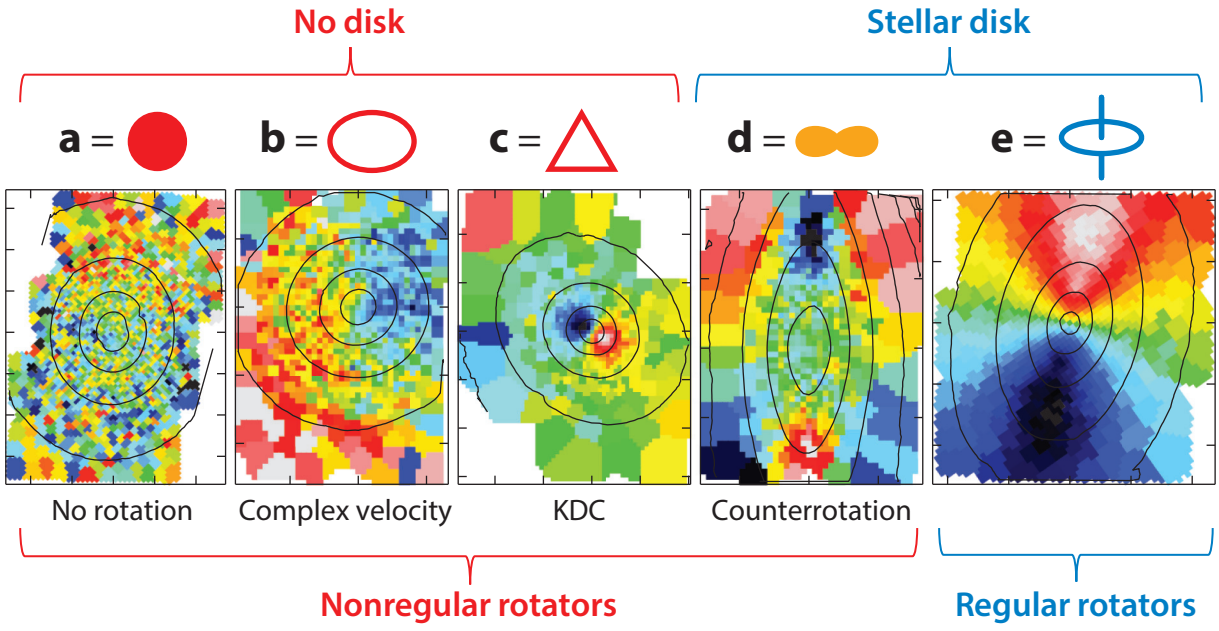


Figure 4

Morphological classifications based on stellar kinematics. The kinematic features in large samples of early-type galaxies can be qualitatively described by five classes: (*a*) no clearly detectable rotation (NGC 4374), (*b*) clear but nonregular rotation (NGC 4552), (*c*) kinematically distinct cores (KDCs) (NGC 5813), (*d*) counter-rotating disks (NGC 4550), and (*e*) regular, extended, disk-like rotation (NGC 2974). The five classes were introduced by Krajnović et al. (2011). The kinematics were taken from Emsellem et al. (2004) and were spatially binned using the Voronoi tessellation method of Cappellari & Copin (2003). The symbols above the maps corresponding to the different morphological classes are used consistently throughout this review.

measure the shape of the higher moments of the LOSVD, and in particular the mean velocity field, which is the most precisely measured quantity. Like in the photometric case, this approach will work as long as one can find a good zeroth-order description of the velocity along the best-fitting ellipse.

In the case of photometry, the zeroth-order approximation along the best-fitting ellipse is a constant. Krajnović et al. (2006, 2008) found that the velocity field of ETGs along the best-fitting ellipse is approximated by a cosine law, $V(\psi) = V_0 + B_1 \cos(\psi)$, with better than 2% accuracy. This is the same form one would expect for the kinematics of an infinitesimally thin disk. It was thus unexpected that this law would be able to describe the stellar kinematics of ETGs. This finding motivated the extension of photometry to velocity fields (and other velocity moments) in a method called kinemetry (Krajnović et al. 2006).

A popular so-called tilted-ring method to perform ellipse fitting of gas velocity fields existed well before kinemetry and was implemented in the ROTCUR program (Begeman 1989). Schoenmakers et al. (1997) used ROTCUR to fit ellipses to the gas velocity field of spiral galaxies. They subsequently measured and interpreted the observed velocity along these ellipses using a Fourier expansion similar to the kinemetry technique. However, the key difference is that ROTCUR defines the best-fitting ellipse as the one that minimizes the squared deviations between the cosine law and the observed velocity samples along the ellipse, whereas kinemetry makes the low-order Fourier coefficient equal to zero. This makes kinemetry more robust and ensures that the low-order Fourier terms cannot affect the higher ones.

Kinemetry: generalizes photometry to kinematics from integral field spectroscopy

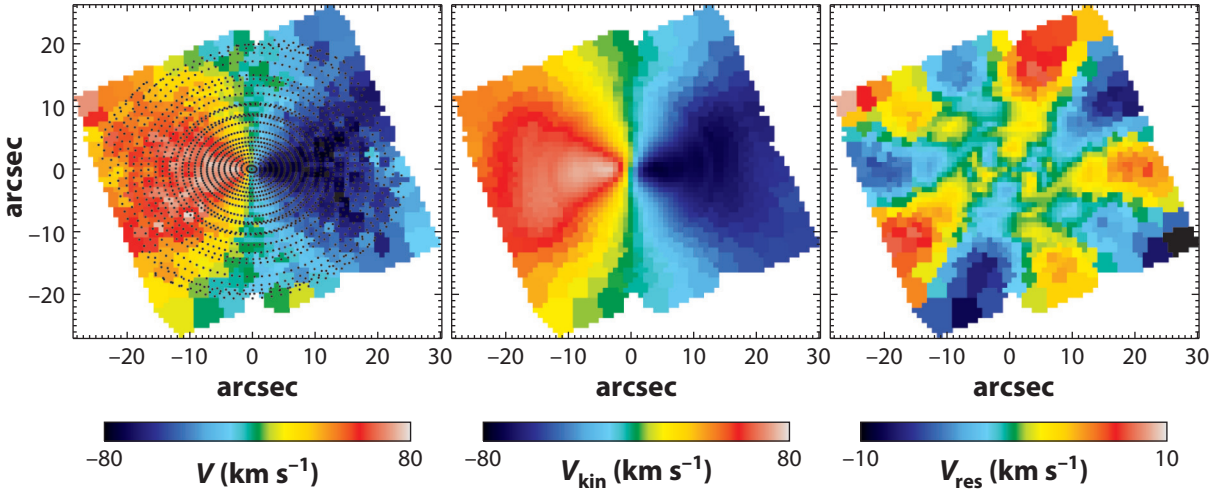


Figure 5

Applying kinemetry to stellar velocity maps. (*left*) Data with best-fitting ellipses overlaid. (*middle*) Reconstructed velocity field adopting a pure cosine law along every ellipse. (*right*) Residuals between the observed and reconstructed velocity field. Note the 5-fold symmetry, which implies a significant contribution from the fifth Fourier term k_5 along the best fitting ellipse, indicative of a secondary kinematic component. Modified with permission from Krajnović et al. (2006).

In practice, kinemetry is designed to be a direct generalization of the Fourier approach used for photometry. In close analogy to Equations 2 and 3, kinemetry samples the galaxy velocity (or other odd moments of the velocity) along an ellipse using the truncated Fourier expansion

$$V(\psi) = V_0 + A_1 \sin(\psi) + B_1 \cos(\psi) + A_2 \sin(2\psi) + B_2 \cos(2\psi) + A_3 \sin(3\psi) + B_3 \cos(3\psi), \quad (5)$$

where ψ is the eccentric anomaly. The best-fitting ellipse is defined as the one for which A_1, A_2, B_2, A_3, B_3 are zero within numerical accuracy, whereas B_1 is allowed to be nonzero. Once the best-fitting ellipse has been determined, the velocity residuals along that ellipse are again parameterized by the next higher term in the Fourier expansion,

$$\Delta V(\psi) = A_4 \sin(4\psi) + B_4 \cos(4\psi) + A_5 \sin(5\psi) + B_5 \cos(5\psi). \quad (6)$$

In this case the deviations of the ellipse from the pure $V(\psi) = V_0 + B_1 \cos(\psi)$ curve are quantified by the $k_5 \equiv \sqrt{A_5^2 + B_5^2}$ term. Like the photometric a_4 parameter, which is sensitive to hidden disks, k_5 is useful for determining the presence of multiple kinematic components. An application of the method to the elliptical galaxy NGC 4473, which is known from dynamical modeling to contain two counter-rotating disks (Cappellari et al. 2007), is illustrated in **Figure 5**.

Kinemetry was used to describe all velocity maps from the ATLAS^{3D} survey. Krajnović et al. (2011) found that the regular rotators have residuals from the cosine law, quantified by the radially averaged ratio $k_5/B_1 \lesssim 0.04$. This ratio measures the fractional amplitude of the fifth Fourier term with respect to the peak velocity amplitude along the same ellipse.

3.3. Intrinsic Shapes of Early-Type Galaxies

Following the first study of the intrinsic shape of elliptical galaxies by Hubble (1926), many papers have investigated the intrinsic shape of elliptical and S0 galaxies by statistical inversion of their apparent shape distribution (e.g., Sandage et al. 1970, Binney & de Vaucouleurs 1981, Fasano &

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

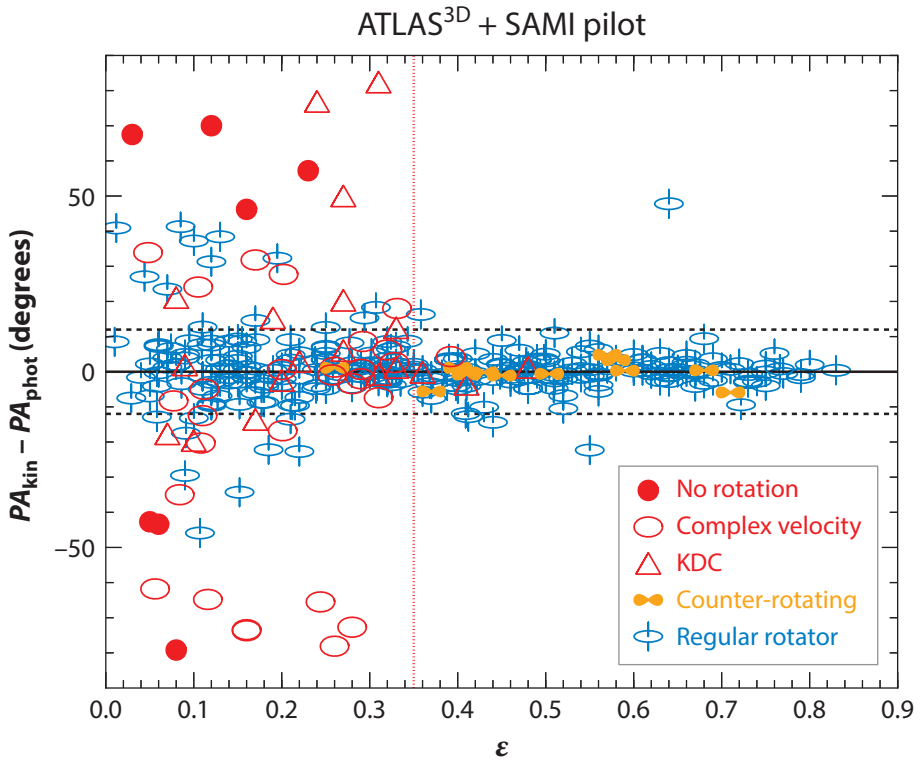


Figure 6

Kinematic misalignment between the photometric major axis PA_{phot}, measured around $R \approx 3R_e$ (where R_e is the half-light radius), and the kinematic major axis PA_{kin}, measured around $R \approx R_e$. The plot includes data for 340 early-type galaxies, of which 260 were taken from Krajnović et al. (2011) and 80 were taken from Fogarty et al. (2015). The symbols represent the types of galaxies illustrated in **Figure 4**. Abbreviations: KDC, kinematically decoupled core; PA, position angle; SAMI, Sydney-AAO Multi-object IFS survey.

Vio 1991, Lambas et al. 1992, Ryden 1992). An intrinsic limitation of these studies is that the recovery of a generally triaxial shape distribution (a two-dimensional function of the two axial ratios) from the distribution of the observed apparent ellipticity (a one-dimensional function) is nonunique. This remains true even when stellar kinematics information is available (Franx et al. 1991).

Luckily, the vast majority of ETGs turn out to be much simpler than purely dynamical considerations would permit them to be. This allows one to measure their shape in spite of degeneracies. In fact, IFS data show that all regular rotators have kinematic axes with position angles PA_{kin} (measured at $R \approx R_e$) essentially aligned with their photometric axes' position angles PA_{phot} (measured at much larger radii of $R \approx 3R_e$) (Cappellari et al. 2007, Krajnović et al. 2011, Fogarty et al. 2015). This is illustrated in **Figure 6** for a combined sample of 340 ETGs. The 1σ rms (biweight) scatter of 4° is almost at the level of the combined measurement errors of PA_{kin} and PA_{phot}. The few deviant objects appear to be either interacting systems or strongly barred.

The only way to observe such a tight alignment for such a large sample of ETGs is if regular rotators are axisymmetric out to their stellar halos at approximately $3R_e$. This is because in a triaxial system, (*a*) the intrinsic symmetry axis need not be aligned with the projected photometric

PA_{kin}: position angle of the axis along which the projected velocities reach the maximum absolute value at a given radius

PA_{phot}: position angle of the axis along which the surface brightness reaches the maximum value at a given radius

major axis (Contopoulos 1956, Stark 1977), and (*b*) the intrinsic angular momentum need not be aligned with the intrinsic symmetry axes but can lie anywhere in the plane containing the major and minor axes (e.g., Statler 1987). This implies that in triaxial galaxies, kinematic misalignments are unavoidable except in very special configurations; the lack of misalignment for a large sample unambiguously implies axisymmetry for the whole class of regular rotators.

The alignment between the kinematics and photometry of regular rotators out to a median radius of $4R_e$ can be seen directly, albeit for a much smaller sample, from the stellar kinematics obtained by the SLUGGS (SAGES Legacy Unifying Globulars and Galaxies) survey (Brodie et al. 2014). The data, presented by Arnold et al. (2014), are reproduced in **Figure 7** together with the SAURON kinematics of the central regions. The kinematics is oriented in such a way that the PA_{phot} from Krajnović et al. (2011) is horizontal. Overlaid are the kinematic axes and errors measured with the procedure fit_kinematic_pa[1] described by Krajnović et al. (2006). Although the SLUGGS data are in some cases rather noisy, the figure shows that in all measurable cases, the kinematic major axis agrees (within the errors) with the photometric one, and this is consistent with the global axisymmetry of the stellar halos of regular rotators.

Because stellar kinematics can separate the class of axisymmetric regular rotators from the rest of the elliptical galaxy population, one can revisit the statistical inversion of the observed shape distribution. This is motivated by the fact that, once a family of objects has been proven to be axisymmetric, the photometric statistical inversion procedure does provide a unique and well-defined solution. A study of the intrinsic shape of regular rotators, performed using the volume-limited ATLAS^{3D} sample, finds that the intrinsic axial ratio of regular rotators (as measured in their outer disks) can be described by a nearly Gaussian distribution with mean axial ratio $\langle q \rangle = 0.25$ and dispersion $\sigma_q = 0.14$ (Weijmans et al. 2014). This distribution is consistent with that of spiral galaxies (e.g., Lambas et al. 1992, Padilla & Strauss 2008). One should note that the flattening measured by these shape inversion studies mainly refers to the outer parts where disks dominate, but ETGs have larger bulges than spirals, as will be discussed later.

The shape inversion remains nonunique for the classes of nonregular rotators, which shows evidence of kinematic misalignment, implying triaxiality. However, the kinematic data show that the intrinsic ratio between the smallest and largest axis of the triaxial ellipsoid, measured around $3R_e$, must be $c/a \gtrsim 0.65$. This is demonstrated by the fact that no kinematic misalignments are observed for the 51 nonregular rotators with $\varepsilon \lesssim 0.35$ (**Figure 6**). Even ignoring the extra information provided by the misalignment, which is not always well defined for nonregular rotators, a stringent limit is placed by the fact that all nonregular rotators have $\varepsilon \lesssim 0.4$ (also see Section 3.6.3). The small ellipticity of slow rotators has also been confirmed using VIMOS (the Visible

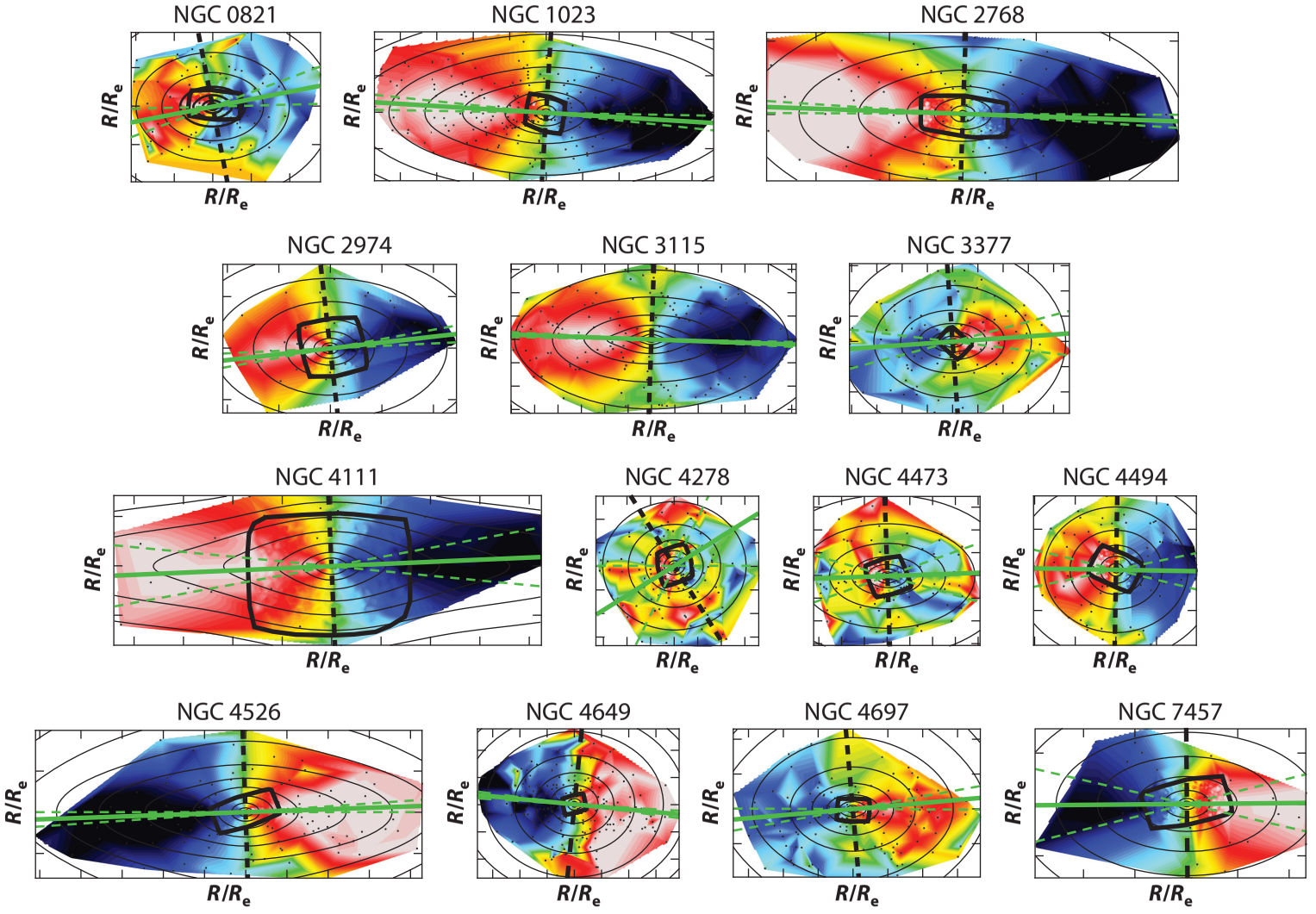
Figure 7

Kinematic misalignment at large radii. The maps show mean stellar velocity fields of regular rotators and one galaxy with counter-rotating stellar disks (NGC 4473) measured by the SLUGGS (SAGES Legacy Unifying Globulars and Galaxies) survey and taken from Arnold et al. (2014). The data are linearly interpolated from the measured positions, which are indicated by the small black dots. The integral field spectroscopic velocities observed by SAURON at smaller radii are enclosed by the solid thick black line (from Emsellem et al. 2004, Cappellari et al. 2011a). The fields are oriented in such a way that the position angle of the photometric axis, PA_{phot}, measured near $3R_e$ (where R_e is the half-light radius) by Krajnović et al. (2011), is horizontal. The solid green lines are the best-fitting global kinematic position angle PA_{kin}, fitted only to the SLUGGS data, and the dashed lines indicate the measurement uncertainties. Isophotes of the galaxy surface brightness are overlaid in 1-magnitude intervals. Tick marks are separated by $1R_e$, and the R_e values are taken from Cappellari et al. (2011a). Adapted with permission from Cappellari et al. (2011a).

[1]Available from **http://purl.org/cappellari/software**.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12



Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Multiobject Spectrograph) kinematic observations of 7 additional slow-rotator brightest cluster galaxies (Jimmy et al. 2013) and using the Mitchell Spectrograph for 11 other massive slow rotators (Raskutti et al. 2014). Overall, of the 73 slow rotators observed so far, all are rounder than $\varepsilon \lesssim 0.4$, showing that nonregular rotators as a class are quite close to spherical and only weakly triaxial.

Stellar distribution function (DF): describes the orbital distribution of stars in a galaxy

3.4. Dynamical Modeling of Stellar Kinematics

3.4.1. Techniques and degeneracies. A complementary way of quantifying the information content of kinematic maps is via dynamical modeling of the stellar kinematics. A key assumption of the models is that the galaxies are in a steady state. The models also generally assume simple spherical, axisymmetric, or triaxial shapes. Under the steady-state assumption, the galaxy dynamics is fully specified by (*a*) the six-dimensional stellar distribution function (DF), which describes the distribution of the positions and velocities of stars in the galaxy, and (*b*) the gravitational potential, or equivalently the total mass distribution, which may include a stellar contribution as well as a dark matter halo and supermassive BHs. This topic was recently reviewed by Courteau et al. (2014). Here, I focus on results specific to IFS observations.

Three major methods have been used in the past three decades: (*a*) the equations of stellar hydrodynamics, first applied to galaxies by Jeans (1922); (*b*) the numerical orbit-superposition method by Schwarzschild (1979); and (*c*) the made-to-measure N-body models by Syer & Tremaine (1996). The first has the advantage that it has predictive power and that one can compute reproducible results nearly to machine precision, whereas the latter two methods have the advantage of generality, which is required for unbiased results. However, these two methods cannot make predictions, and the results they yield depend on the details of how the methods are implemented.

The recovery of the DF and the mass distribution using only line-of-sight quantities is an intrinsically degenerate or ill-posed problem and consequently its solutions is not unique. This is because the DF is a function of the three isolating integrals of motion (Jeans 1915), and one cannot expect to uniquely constrain both the three-dimensional DF and the three-dimensional mass distribution using only a three-dimensional observable, as provided by the LOSVD at every spatial location (e.g., Valluri et al. 2004, section 3).

Moreover, the deprojection of the stellar surface brightness into an intrinsic stellar luminosity density is mathematically nonunique, even when assuming axisymmetry, unless the galaxy is edge-on (Rybicki 1987). This degeneracy increases rapidly at lower inclinations (Gerhard & Binney 1996, Romanowsky & Kochanek 1997, van den Bosch 1997) and represents a fundamental barrier to detailed models of external galaxies. Relaxing the axisymmetry assumption further changes the dimensions of the problem and vastly increases the potential for degeneracies (Gerhard 1996), making the recovery of general triaxial stellar densities impossible without strong assumptions.

Empirical explorations of model degeneracies using IFS data have revealed that in practice, even in an ideal case in which (*a*) one uses simulated noiseless two-dimensional data, (*b*) one artificially removes the deprojection degeneracy, and (*c*) one assumes there is no dark matter and the gravitational potential is produced by the stars alone, basic parameters such as galaxy inclination (or, equivalently, galaxy shape) are virtually unconstrained by the IFS data (**Figure 8**; see also Krajnović et al. 2005, van den Bosch & van de Ven 2009). Tests on the ability of the models to constrain mass profiles have also revealed ample room for degeneracies in the model parameters, even with excellent data (Gerhard et al. 1998, de Lorenzi et al. 2009).

3.4.2. Results from integral field data using general models. With the previous caveats in mind, key results were obtained using dynamical models fitted to stellar kinematics. Schwarzschild (1979) models, generalized to fit kinematic data (Richstone & Tremaine 1988, Rix et al. 1997,

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

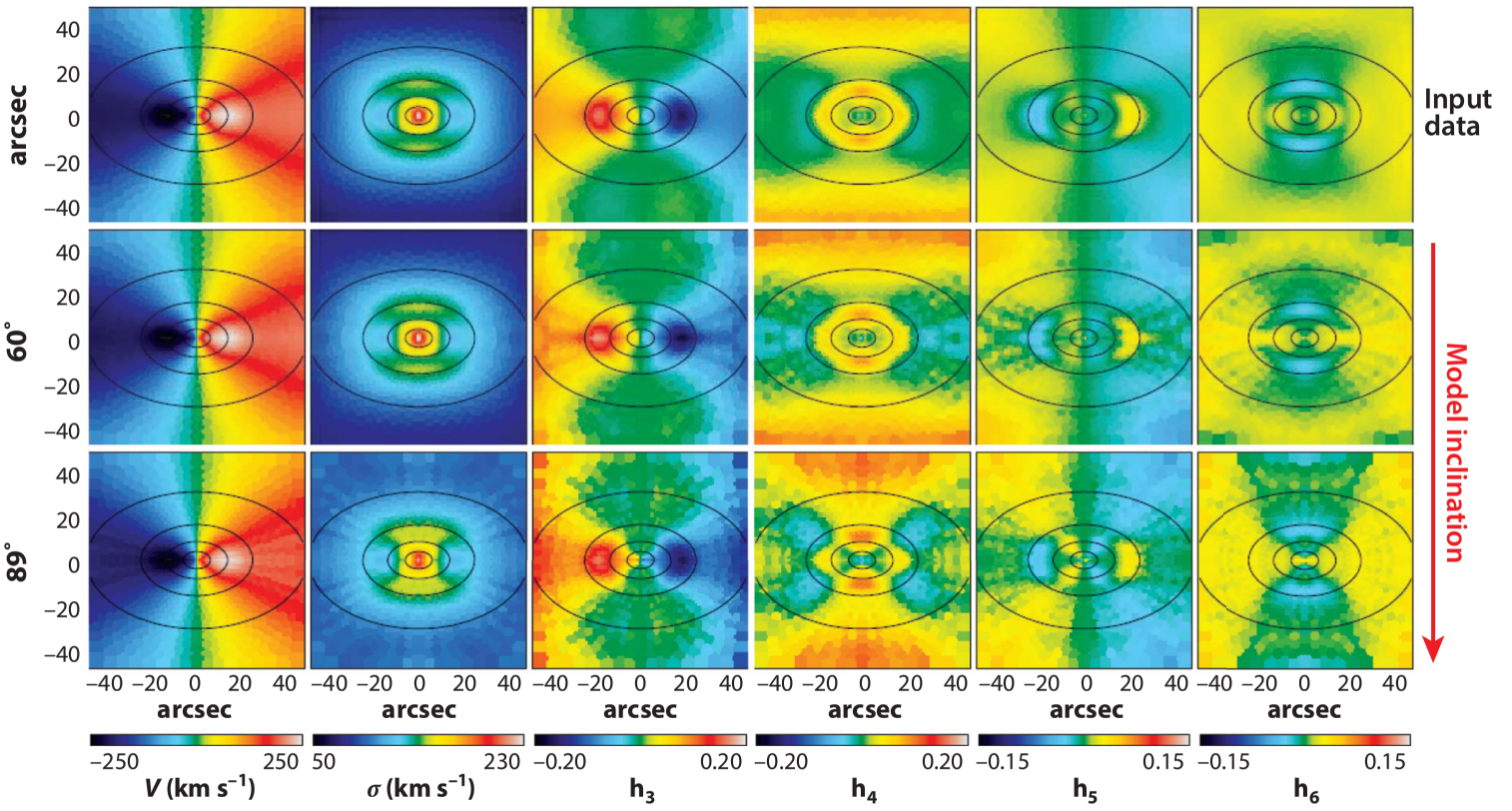


Figure 8

Inclination degeneracy. The top row shows the mean stellar velocity, the velocity dispersion σ, and the higher Gauss-Hermite moments of the velocity, for the input simulated data, an axisymmetric galaxy seen at an inclination $i = 60°$. These data were fit by Schwarzschild models at different inclinations. At every inclination, the model's surface brightness is constrained to agree with the input one. To eliminate the effect of the intrinsic degeneracy in the deprojection, the surface brightness is deprojected in such a way that at the correct inclination, the intrinsic density is the same as the input one. Even in these idealized conditions, the differences in the model fits at different inclinations are at the level of the systematic errors affecting real kinematic observations. Adapted with permission from Krajnović et al. (2005).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

van der Marel et al. 1998) have provided most of the mass determinations of supermassive BHs in galaxies (e.g., van der Marel et al. 1998, Gebhardt et al. 2000b, Cappellari et al. 2002, Valluri et al. 2005, McConnell et al. 2011, van den Bosch et al. 2012, Rusli et al. 2013). These measurements have driven much of our understanding of the connection between supermassive BHs and galaxy evolution. This important topic was reviewed by Kormendy & Ho (2013).

Other significant results are the determination of stellar mass-to-light ratios and total mass profiles. According to these studies (described in Section 4.2.2), dark matter appears to play a minor role within $1R_e$, which implies that at those radii, the shape of the total mass density is close to that of the luminous density (Section 4.2.3). Under this assumption, the availability of the first set of IFS data from the SAURON survey (de Zeeuw et al. 2002), opened up the possibility for a unique inversion of the data cube into the DF. Cappellari et al. (2007) attempt this, using the axisymmetric Schwarzschild (1979) implementation optimized for IFS data described by Cappellari et al. (2006). The dynamical models provide the stellar mass orbiting along the half-million orbits that approximate the whole galaxy. To quantify this large amount of information, three anisotropy parameters were defined. Cappellari et al. (2007) introduced the β_z and γ parameters in close analogy to δ, which is given in equation 4.265 in the classic book by Binney & Tremaine (2008):

$$\beta_z \equiv 1 - \frac{\Pi_{zz}}{\Pi_{RR}}, \quad \gamma \equiv 1 - \frac{\Pi_{\phi\phi}}{\Pi_{RR}}, \quad \delta \equiv 1 - \frac{\Pi_{zz}}{\Pi_{xx}}, \tag{7}$$

where (R, z, ϕ) are the standard cylindrical coordinates, z coincides with the symmetry axis of an axisymmetric galaxy, and x is any direction orthogonal to it. Here,

$$\Pi_{kk} = \int \nu \, \sigma_k^2 \, \mathrm{d}^3\mathbf{x}, \tag{8}$$

where σ_k is the velocity dispersion along the direction k at a given location inside the galaxy, ν is the stellar density, and $\mathbf{x}$ is an integration variable for the volume integral. The numerical integral extends to the region covered by the IFS observations. β_z describes the global shape of the velocity dispersion tensor in the (v_R, v_z) plane. If the anisotropy is spatially constant then $\beta_z = 1-(\sigma_z/\sigma_R)^2$. The anisotropy γ describes the global shape of the velocity dispersion tensor in a plane orthogonal to v_z. If the anisotropy is spatially constant then $\gamma = 1 - (\sigma_\phi/\sigma_R)^2$. For an isotropic system (i.e., a spherical velocity ellipsoid), $\beta_z = \gamma = \delta = 0$. Integrating over the azimuthal angle, one finds that the three anisotropy parameters are related by $\delta = (2\beta_z - \gamma)/(2 - \gamma)$. In the case of $\gamma = 0$, the simple relation $\beta_z = \delta$ applies.

The anisotropy parameters β_z, γ, and δ converge to zero in the case of spherical symmetry and assuming no rotation. For this reason, to quantify the anisotropy of nearly spherical galaxies, a complementary anisotropy parameter is defined, in spherical coordinates:

$$\beta_r \equiv 1 - \frac{\Pi_{tt}}{\Pi_{rr}} = 1 - \frac{\Pi_{\theta\theta} + \Pi_{\phi\phi}}{2\,\Pi_{rr}}, \tag{9}$$

where (r, θ, ϕ) are the standard spherical coordinates. In the spherical limit, assuming the galaxy is nonrotating, $\Pi_{\theta\theta} = \Pi_{\phi\phi}$ by symmetry. The parameter $\beta_r = 0$ for an isotropic galaxy and is positive (negative) when the luminosity-weighted average dispersion along the radial direction is larger (smaller) than the average dispersion along any direction orthogonal to it. If the anisotropy is constant then $\beta_r = 1 - (\sigma_\theta/\sigma_r)^2$.

Calculation of the four anisotropy parameters for 25 axisymmetric galaxies from the SAURON sample showed that on average, (*a*) regular rotators have significant anisotropy δ, (*b*) the velocity ellipsoid is oblate, with $\delta \sim \beta_z$ and $\gamma \sim 0$, and (*c*) $\beta_z \gtrsim 0$ even though no limits are enforced by the models on this parameter (Cappellari et al. 2007). This picture was independently confirmed

using long-slit kinematics of a different galaxy sample and a different modeling code (Thomas et al. 2009).

The two galaxies NGC 4550 and NGC 4473 stand out because of a significant tangential anisotropy ($\gamma < 0$). In both cases, the IFS data shows a characteristic and peculiar enhancement of the stellar velocity dispersion σ along the galaxy's major axis, with two symmetric peaks in σ along the major axis, qualitatively suggesting the presence of a counter-rotating disk [for NGC 4550, this confirms the results by Rix et al. (1992) and Rubin et al. (1992)]. This interpretation was quantitatively confirmed by dynamical models revealing two clearly distinct populations of stars rotating in opposite direction (Cappellari et al. 2007). The strong tangential anisotropy is precisely what one would have expected given the fact that the two counter-rotating stellar disks strongly increase the random motions in the tangential direction. These two well-studied prototypes of counter-rotating disks motivated the definition of a more general class of qualitatively similar galaxies, which Krajnović et al. (2011) term 2σ galaxies because of their distinctive double peaks in σ along the major axis. This population constitutes 4% of the ATLAS^{3D} sample (Krajnović et al. 2011).

2σ galaxies: galaxies that contain two counter-rotating stellar disks

ε: observed ellipticity of a galaxy

$\varepsilon_{\rm intr}$: intrinsic ellipticity of a galaxy

The situation is different for the rounder nonregular rotators. They were found to span a smaller range of anisotropies than the regular rotators, and their anisotropy scatters around zero, implying that on average they are close to isotropic. They cannot be precisely so, given their triaxial shapes, but adopting either cylindrical or spherical coordinates, the global anisotropy parameters indicate that their velocity ellipsoids deviate $\lesssim$10% from spherical ones. This agrees with results obtained using long-slit kinematics of nearly round ellipticals (Gerhard et al. 2001, Gebhardt et al. 2003).

An interesting class of nonregular rotators is that consisting of galaxies with central KDCs. Detailed Schwarzschild dynamical models using SAURON data for NGC 4365 (van den Bosch et al. 2008) reveal that its KDC is not spatially distinct in terms of orbital distribution. The apparent KDC arises because of the superposition of two populations of counter-rotating large-scale tube orbits with mean velocities that cancel each other out over most of the field except near the center. The large spatial extent of the stellar orbits producing the KDC is consistent with the observed homogeneity in the stellar population of the KDC in this galaxy (Davies et al. 2001), and with that of the large KDCs of nonregular rotators in general (McDermid et al. 2006). A similar example of Schwarzschild modeling of a KDC in the nonregular rotator NGC 5813 is presented by Krajnović et al. (2015) using very high-quality MUSE (Multi Unit Spectroscopic Explorer) (Bacon et al. 2010) observations. The study confirmed the "apparent" nature of the KDC in this other nonregular rotator—the seemingly decoupled KDC was not decoupled in terms of the orbital distribution. The similarity in the density distributions of the two counter-rotating orbits, which is needed to produce the observed negligible mean stellar velocity outside the KDC, is reminiscent of the Separatrix crossing mechanism proposed by Evans & Collett (1994). Krajnović et al. (2015) summarize possible KDCs formation mechanisms.

The anisotropy β_z, the z-flattening of the velocity ellipsoid, of regular rotators was found to be related to their intrinsic axial ratios $\varepsilon_{\rm intr}$. This is illustrated in the left panel of **Figure 9**. **Figure 9** shows the intrinsic flattening, deprojected from the observed flattening ε within an isophote enclosing half of the total light using

$$\varepsilon_{\rm intr} = 1 - \sqrt{1 + \varepsilon(\varepsilon - 2)/\sin^2 i}, \tag{10}$$

where $i = 90°$ is edge-on. Regular rotators are found to span a range of β_z at given $\varepsilon_{\rm intr}$, with the range increasing with $\varepsilon_{\rm intr}$. For an oblate velocity ellipsoid ($\sigma_\phi = \sigma_R$), the tensor virial theorem (Binney & Tremaine 2008, section 4.8.3) already sets an upper limit $\beta_z < 1 - 1/\Omega(e)$, corresponding to $V/\sigma = 0$ in Equation 14. However, the observed values do not span the full space allowed

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

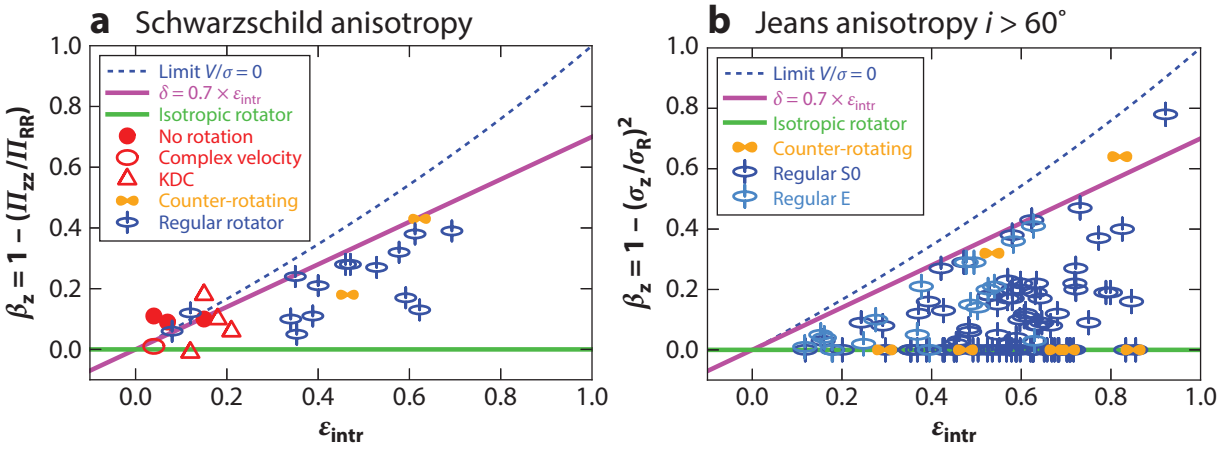


Figure 9

Anisotropy versus intrinsic flattening. (*a*) The anisotropy derived via Schwarzschild models is plotted against the deprojected intrinsic ellipticity of the galaxies around $1\,R_e$ (the half-light radius). Nonregular rotators tend to be nearly isotropic in their central regions, whereas regular rotators span a range of anisotropies but appear bounded by a relation of the form $\beta_z \approx 0.7 \times \varepsilon_{intr}$ (*magenta line*). The green line indicates $\sigma_z = \sigma_R$, and the dashed line is the limit $\beta_z = 1 - 1/\Omega(e)$ (see Equation 15) set by the tensor virial equations for objects with oblate velocity ellipsoids. This plot was adapted with permission from Cappellari et al. (2007). (*b*) As in panel *a*, but for the anisotropy determined via Jeans anisotropic models in Cappellari et al. (2013b). Only galaxies with $i > 60°$ were included, to reduce the effect of the inclination-anisotropy degeneracy. The symbols for galaxy classes are the same as those defined in **Figure 4**.

by virial equilibrium, and in particular they are found to lie approximately below the linear relation (Cappellari et al. 2007)

$$\delta \approx \beta_z = 0.7 \times \varepsilon_{intr}. \tag{11}$$

In the next section I show that these results, obtained from detailed dynamical models of a small set of galaxies, are fully consistent with what one can infer from a much larger sample of galaxies using the very different approach provided by the Jeans (1922) equations.

3.4.3. Results from integral field data using Jeans models. Motivated by the finding that the anisotropy of fast-rotator ETGs is on average best approximated as a flattening of the velocity ellipsoid $\sigma_z < \sigma_R$, Cappellari (2008) developed an accurate and efficient method to solve the axisymmetric Jeans equations allowing for a cylindrically aligned velocity ellipsoid with general axial ratios $\sigma_R \neq \sigma_z \neq \sigma_\phi$. As discussed in detail in the paper, the cylindrical alignment is only an approximation, as it cannot be accurately satisfied in real galaxies. This Jeans anisotropic modeling (JAM) formalism (the results of which are referred to as JAM models) is an anisotropic (three-integral) generalization of the semi-isotropic (two-integral) formalism ($\sigma_R = \sigma_z$) presented by Emsellem et al. (1994). It uses the multi-Gaussian expansion (MGE; Emsellem et al. 1994, Cappellari 2002) to parameterize the observed galaxy surface brightness. The MGE allows for an analytic deprojection of the observed surface brightness (Bendinelli 1991, Monnet et al. 1992). One can employ an arbitrary number of Gaussians to reproduce all features of a galaxy image. In particular, one can describe in detail multiple photometric components, including bulges, disks, inner disks, and general ellipticity variations.

The comparison between the JAM models and state-of-the-art IFS observations of ETGs confirms the expectations from the Schwarzschild models. It shows that, assuming a constant-anisotropy, cylindrically aligned velocity ellipsoid, one can predict the kinematics with

MGE: multi-Gaussian expansion

JAM: Jeans anisotropic modeling

κ: ratio $V_{obs}/V\,(\sigma_\phi = \sigma_R)$ between the observed velocities and those of a model with an oblate velocity ellipsoid

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

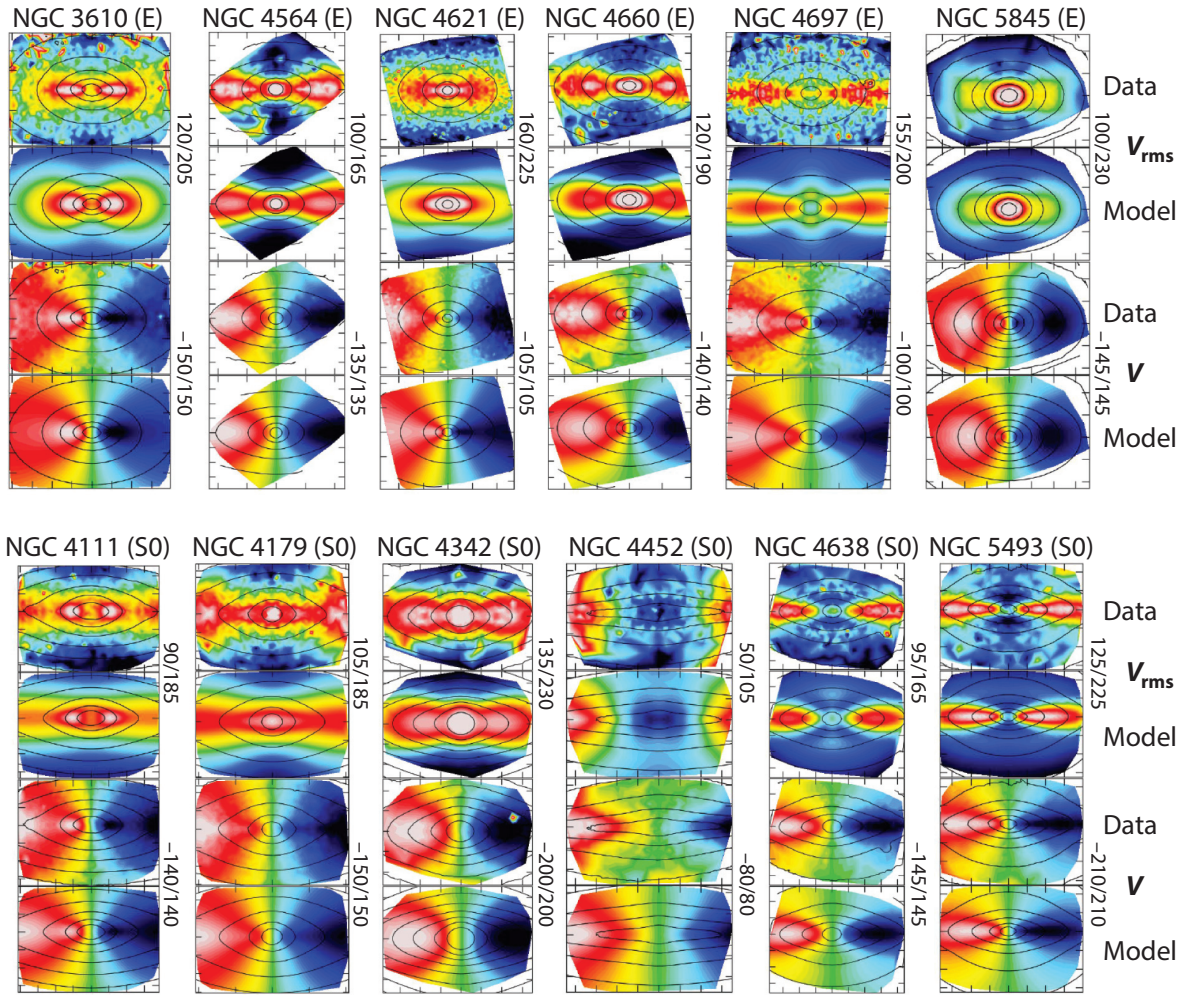


Figure 10

Jeans anisotropy models of regular rotators. The top rows show six ellipticals (E), and the bottom ones show six SOs (as classified by de Vaucouleurs et al. 1991). As expected for regular rotators, most ellipticals have disky isophotes and are better morphologically classified as E(d). In each plot, the top panel shows the symmetrized SAURON stellar $V_{\rm rms} \equiv \sqrt{V^2 + \sigma^2}$, the second panel is the best fitting Jeans anisotropy model to the $V_{\rm rms}$, the third panel is the SAURON mean stellar velocity V, and the bottom panel is the best fit to V obtained by keeping the best-fitting anisotropy β_z, inclination i, and M/L from the previous fit to the $V_{\rm rms}$. Only an overall scaling of V is done here. The SAURON data come from Emsellem et al. (2004) or Cappellari et al. (2011a). The multi-Gaussian expansion surface brightness is from Scott et al. (2013a). Once an accurate description of the surface brightness is given, the observed shape of the kinematics of each galaxy can be predicted with remarkable accuracy by varying the single physical parameter β_z and by choosing an inclination. This illustrates the homogeneity in the dynamics of the regular rotators family. Given that these models assume a mass distribution following the light, the good predictive power suggests this assumption should also be nearly correct.

remarkable accuracy. **Figure 10** shows that the large variety in the observed shapes of the second velocity moment $V_{\rm rms} \equiv \sqrt{V^2 + \sigma^2}$ maps of fast-rotator ETGs are properly captured by the simple JAM models. Once the photometry of the galaxies is given as input, the models are fully specified by the single physical parameter $\beta_z \equiv 1 - (\sigma_z/\sigma_R)^2$ and by the inclination, as well as by the overall mass scaling, which is parameterized via the total mass-to-light ratio (M/L).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Moreover, the distribution of β_z for the regular rotators, as a function of the intrinsic ellipticity $\varepsilon_{\rm intr}$, is consistent with the one observed from the Schwarzschild models (**Figure 9*a*,*b***). It shows that real galaxies have $\beta_z \lesssim 0.7 \times \varepsilon_{\rm intr}$. The JAM models show that this trend is not due to anisotropic disks embedded in isotropic bulges—in fact, the best fits to the kinematics are obtained with a constant anisotropy for both components.

Once the JAM models have been fitted to the $V_{\rm rms}$, the intrinsic second velocity moment $\overline{v_\phi^2}$ in the tangential direction is uniquely defined by the given assumptions. However, to estimate the first velocity moment, the mean stellar velocity V, one needs to make an extra assumption about how the $\overline{v_\phi^2}$ splits into ordered and random motion, as defined by

$$\overline{v_\phi^2} = \overline{v_\phi}^2 + \sigma_\phi^2. \tag{12}$$

The JAM method allows for a general tangential anisotropy $\gamma = 1 - (\sigma_\phi/\sigma_R)^2$. However, this generality is not actually needed to describe the kinematics of real galaxies, at least within approximately $1R_{\rm e}$, where good data have been obtained for many galaxies. In fact, if one makes the simplest assumption of an oblate velocity ellipsoid $\gamma = 0$ (or $\sigma_\phi = \sigma_R$), the shape of the mean stellar velocity field can also be quite accurately predicted without the need to invoke extra parameters (**Figure 10**).

A remarkable finding is that, if one takes the mean projected velocity field $V(\sigma_\phi = \sigma_R)$ predicted by a JAM model with a perfectly oblate velocity ellipsoid as a reference, the overall scaling κ required to best fit the observed velocity $V_{\rm obs}$ is close to $\kappa \approx 1$ with an observed rms scatter of only 7% (**Figure 11**). This small observed scatter also includes the effect of measurement errors in β_z, the inclination, and M/L, which are independently fit to the $V_{\rm rms}$ and used as input to predict the $V(\sigma_\phi = \sigma_R)$. This result confirms that an oblate velocity ellipsoid provides a good approximation to the observed galaxy dynamics within $1R_{\rm e}$.

The class of counter-rotating disks (**Figure 4*d***) necessarily cannot be described by models with oblate velocity dispersion tensors. This is because counter-rotating stars produce strong tangential anisotropy. However, it turns out that these ETGs have similar dynamics to the other regular rotators, once the counter-rotation is taken into account. This is illustrated in **Figure 12**, where the JAM models are used to describe their kinematics. One can see that their $V_{\rm rms}$ is still reasonably well predicted by the models, once the photometry is given, with a similar range of β_z anisotropy.

In this case however, to fit the velocities, the sense of rotation of the stars enclosed within some of the MGE Gaussians was reversed by allowing κ to have different signs for different Gaussians but still remain relatively close to unity. The stellar velocity dispersion was then given by $\sigma = \sqrt{V_{\rm rms}^2 - V^2}$. Using this simple approach, one can naturally describe the observations of the counter-rotating disks. This approach shows that the 2σ galaxies form a physically homogeneous family with the rest of the regular rotators.

However, not all galaxies are well described by JAM models with oblate velocity ellipsoid—this is only true for the ETGs with evidence of stellar disks (**Figure 4*d*,*e***). The situation is dramatically different for the nonregular rotators (**Figure 4*a*–*c***). The $V_{\rm rms}$ of these objects can generally still be well approximated by JAM; however, the shape of the predicted $V(\sigma_\phi = \sigma_R)$ is, even qualitatively, very different from the observed velocity field. A simple way to quantify the difference between the dynamics of regular versus nonregular rotators is to look at the distribution of their measured κ parameters (**Figure 11**). Whereas the regular rotators show a nearly Gaussian distribution with average $\kappa \approx 1$, the nonregular rotators are clearly distinct. Importantly, the kinematics information shows a real dichotomy, not a smooth transition, between these two classes of galaxies, suggesting that they must follow different formation channels in their evolution.

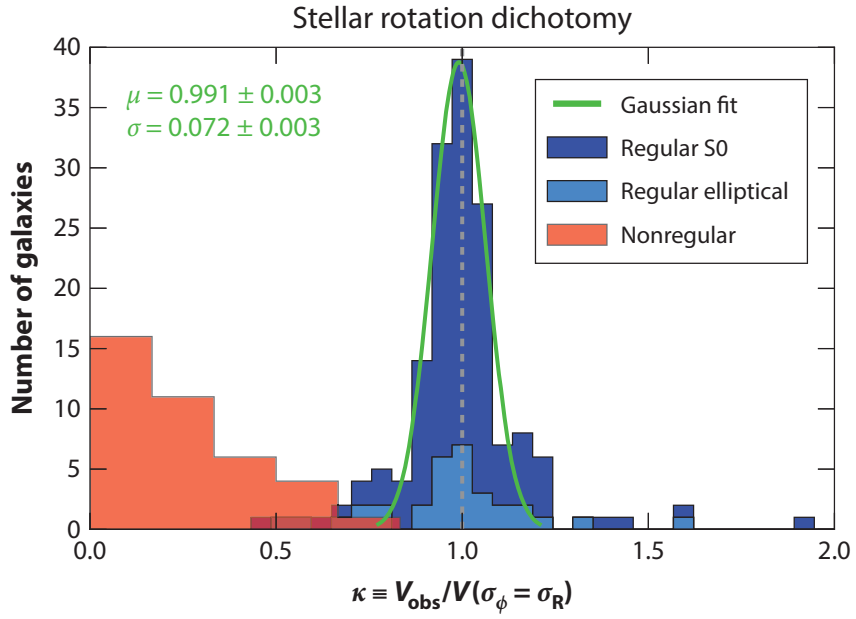


Figure 11

Stellar rotation dichotomy. Histogram of the ratio κ between the observed velocity $V_{\rm obs}$ and the velocity $V(\sigma_\phi = \sigma_R)$ predicted by a Jeans anisotropic model with an oblate velocity ellipsoid. The values are extracted from model A of Cappellari et al. (2013b). More specifically, this ratio is determined by linearly fitting the full Jeans anisotropic model velocity field to the observed one. The green line is a Gaussian fit to the histogram, for the regular rotators only. The distribution peaks with high accuracy at the value $\kappa \approx 1$, which corresponds to an oblate velocity ellipsoid on average. No statistically significant difference is observed (via Kolmogorov–Smirnov test) in the distribution of the elliptical and S0 regular rotators. However, the nonregular rotators have a completely different distribution, which peaks at $k \approx 0$ and has a broad tail. The regular and nonregular rotators are well-separated in this diagram around $k \approx 0.65$, with a transition region due to counter-rotation in disk galaxies. Only nonregular rotators have $\kappa \lesssim 0.5$ and only regular ones have $\kappa \gtrsim 0.75$. This diagram demonstrates a clean dichotomy, rather than a continuity between the two classes of early-type galaxies. The values μ and σ are the location of the peak and the dispersion of the best fitting Gaussian.

3.5. Understanding the $(V/\sigma, \varepsilon)$ Diagram

3.5.1. Before integral field kinematics. Before observations of the stellar kinematics of ETGs became possible, these objects were thought to constitute a class of homogeneous systems with an isotropic velocity dispersion tensor. A revolution was started by the first observations of the stellar kinematics in ETGs (Bertola & Capaccioli 1975), which found much lower velocities than predicted by isotropic models (Illingworth 1977, Schechter & Gunn 1979). To quantify this discrepancy, the now-classic $(V/\sigma, \varepsilon)$ diagram was proposed (Binney 1978). It quantifies the ratio between the ordered rotation and the random motion in a stellar system, as a function of the observed (i.e., apparent) ellipticity ε of a galaxy.

When the observations could be extended to galaxy bulges (Kormendy 1982a,b; Kormendy & Illingworth 1982) and fainter ellipticals ($M_B \gtrsim -21$, Davies et al. 1983), it was found that galaxy bulges and fainter ellipticals were instead more consistent with being isotropic rotators. A separation around $M_B \approx -20.5$ was suggested between (*a*) brighter ellipticals, with slow

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

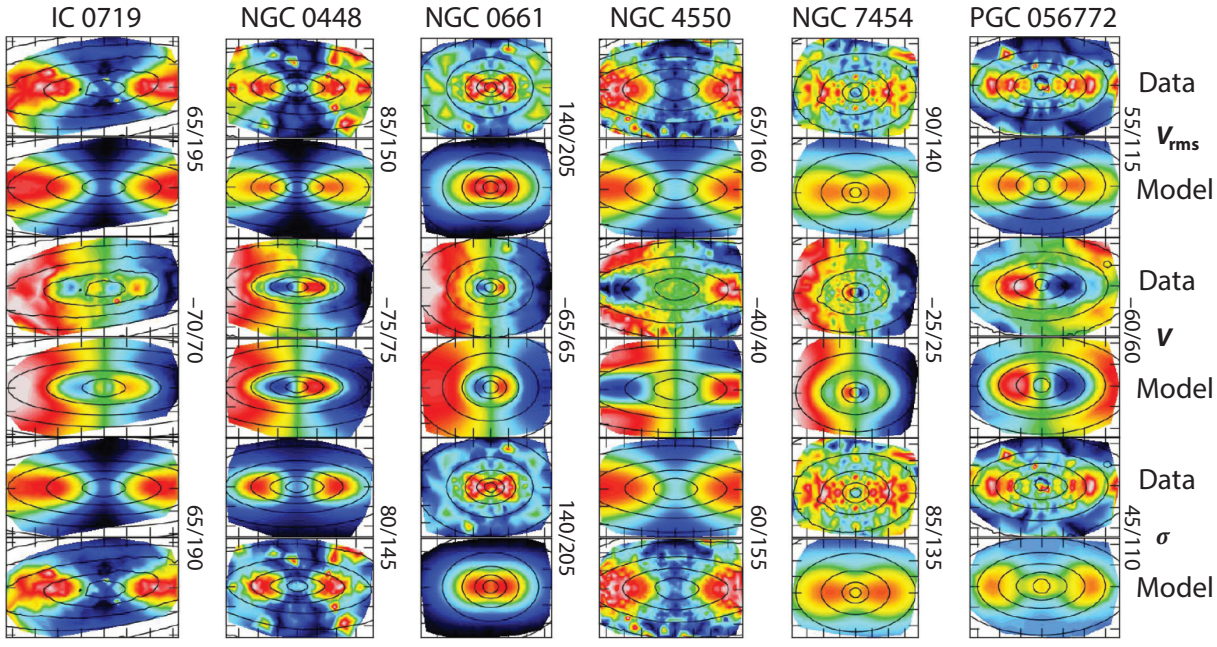


Figure 12

Jeans anisotropic models (JAM models) of counter-rotating disks, which are termed 2σ galaxies by Krajnović et al. (2011). In each plot, the top panel shows the symmetrized SAURON stellar $V_{\rm rms} \equiv \sqrt{V^2 + \sigma^2}$, the second panel is the best fitting JAM model to the $V_{\rm rms}$, the third panel is the SAURON mean stellar velocity V, and the fourth panel is the best fit to V obtained by keeping the best-fitting anisotropy β_z, inclination i, and M/L from the previous fit to the $V_{\rm rms}$. The penultimate panel is the velocity dispersion σ, which is characterized by two maxima along the major axis on opposite sides of the nucleus. The bottom panel is the JAM model $\sigma = \sqrt{V_{\rm rms}^2 - V^2}$. The SAURON data come from Cappellari et al. (2011a), except for NGC 4550, which comes from Emsellem et al. (2004). The multi-Gaussian expansion (MGE) surface brightness comes from Scott et al. (2013a). The only difference from **Figure 10** is that the fit to V allows for the opposite sense of rotation for different Gaussians in the MGE. This reversal leaves the model $V_{\rm rms}$ rigorously unchanged. According to the models, the amount of mass that is affected by the counter-rotation ranges from approximately 10% for NGC 661 to approximately 50% for NGC 4550. This plot illustrates the physical continuity between the regular rotators and the counter-rotating disks.

rotation and triaxial shapes, and (*b*) fainter ellipticals with faster rotation. The former were thought to be significantly anisotropic and likely triaxial, whereas the latter were interpreted as nearly isotropic stellar systems flattened by rotation. The subsequent discovery of a connection between elliptical galaxies' stellar rotation, isophotal shape (Section 2.3), and nuclear profile slopes (Section 2.4) significantly strengthened the case for two different types of elliptical (Kormendy & Bender 1996, Faber et al. 1997).

The theoretical interpretation of the $(V/\sigma, \varepsilon)$ diagram is based on the tensor virial theorem, which relates the kinetic and potential energy tensors in a stellar system (Binney & Tremaine 2008, section 4.8.3). The tensors are formally integrated over the full extent of the galaxies. However, for decades, V/σ could only to be measured in galaxies by using long-slit kinematics, which provides only a crude approximation of the global galaxy kinematics.

3.5.2. Formalism for integral field kinematics. The advent of IFS motivated a more rigorous and robust formulation of the theoretical diagram in which the V and σ are luminosity-weighted quantities integrated over the full extent of the system (Binney 2005). The first group that applied

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Binney's revised $(V/\sigma, \varepsilon)$ formalism to integral field unit data performed the luminosity-weighting within an ellipse that encloses half of the projected total galaxy light (Cappellari et al. 2007). In this case, the observed quantity becomes

$$\frac{\langle V^2 \rangle}{\langle \sigma^2 \rangle} \approx \left(\frac{V}{\sigma}\right)_e^2 \equiv \frac{\sum_{n=1}^{N} F_n V_n^2}{\sum_{n=1}^{N} F_n \sigma_n^2}, \tag{13}$$

i: galaxy inclination, with $i = 90°$ being edge-on

where V_n and σ_n are the mean stellar velocity and dispersion within a given spatial bin, F_n is the flux enclosed within that bin, and the sum is performed over all bins falling within the half-light ellipse. One can verify using theoretical models that, by limiting the sum to $1R_e$, the measured values are no more than $\Delta(V/\sigma) \lesssim 0.1$ lower than the theoretical ones extended to infinite radii (Cappellari et al. 2007, Emsellem et al. 2011).

As shown by Binney (2005), for oblate galaxies with different anisotropies δ (Equation 7) and intrinsic ellipticity $\varepsilon_{\rm intr}$, the theoretical prediction for an edge-on view is

$$\frac{\langle V^2 \rangle}{\langle \sigma^2 \rangle} = \frac{(1-\delta)\Omega(e) - 1}{\alpha(1-\delta)\Omega(e) + 1}, \tag{14}$$

with

$$\Omega(e) = \frac{0.5[(\arcsin e)/\sqrt{1-e^2} - e]}{e - (\arcsin e)\sqrt{1-e^2}}, \qquad e = \sqrt{1-(1-\varepsilon_{\rm intr})^2}, \tag{15}$$

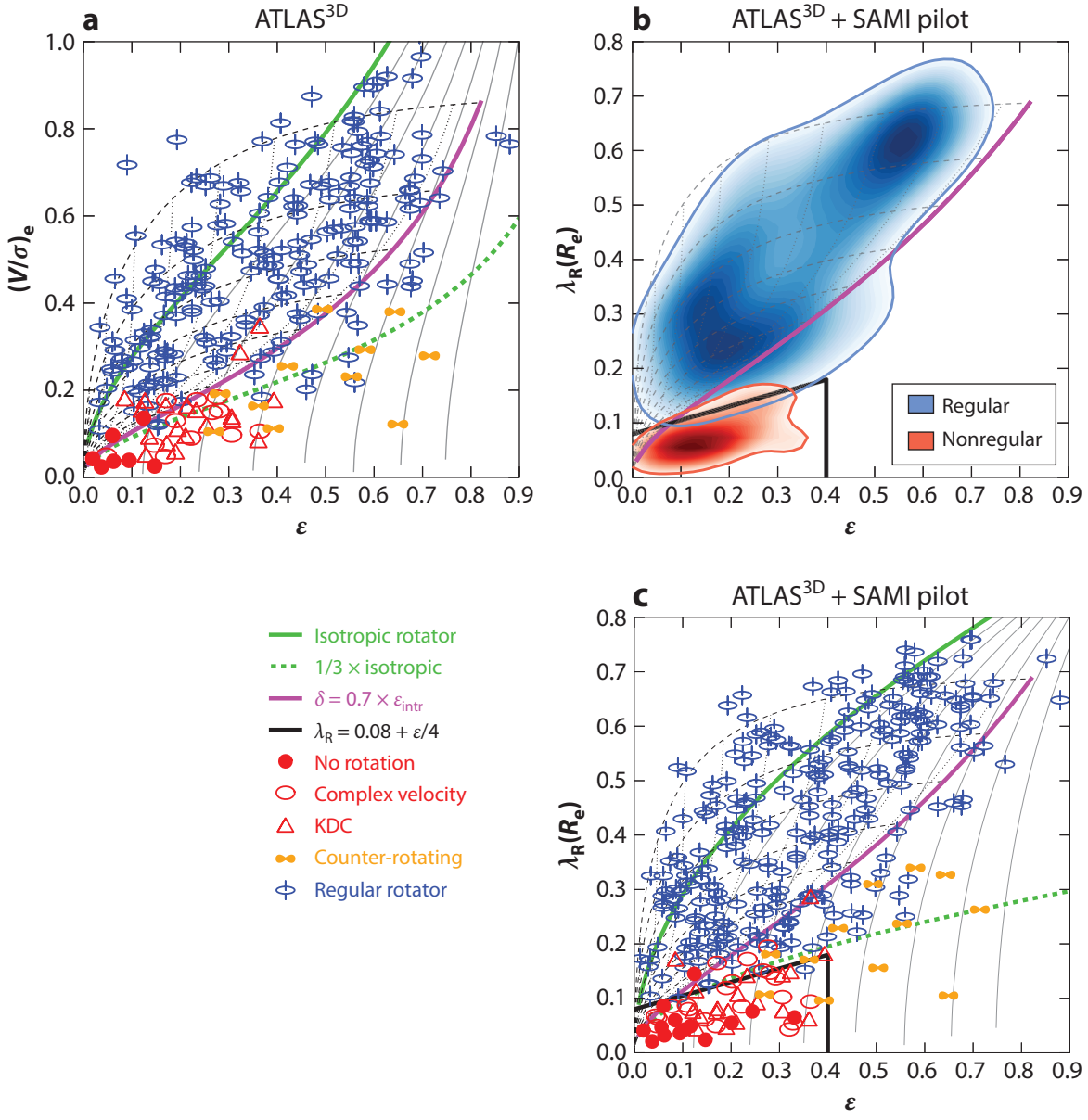
and a parameter α that depends on the shape (but not the amplitude) of the galaxy's intrinsic rotation curve, and the galaxy's radial luminosity profile. A fixed value $\alpha \approx 0.15$ was found to provide a good representation of real galaxies (Cappellari et al. 2007) and is adopted in what follows. The edge-on isotropic line and the corresponding edge-on relations for different anisotropies are shown in the $(V/\sigma, \varepsilon)$ diagram of **Figure 13*a***. Given the edge-on ($i = 90°$) values of $(V/\sigma, \varepsilon_{\rm intr})$, their projections at different inclinations are given by

$$\left(\frac{V}{\sigma}\right)_e^{\rm obs} = \left(\frac{V}{\sigma}\right)_e \frac{\sin i}{\sqrt{1-\delta\cos^2 i}}, \qquad \varepsilon = 1 - \sqrt{1 + \varepsilon_{\rm intr}(\varepsilon_{\rm intr} - 2)\sin^2 i}. \tag{16}$$

3.5.3. Results from integral field data. The first application of the formalism in Section 3.5.2, to 66 galaxies with SAURON IFS, revealed that galaxies with the kinematic morphology of nonregular rotators (**Figure 4*a–d***) tend to lie well below the isotropic line in the $(V/\sigma, \varepsilon)$ diagram (Cappellari et al. 2007). With the exception of the special class of counter-rotating disks (**Figure 4*d***), these slow-rotating ETGs are generally bright (Emsellem et al. 2007) and consistent with the triaxial class of elliptical galaxies found by the previous long-slit studies. However, they were found to be only weakly triaxial and close to spherical and isotropic within $1R_e$ (Cappellari et al. 2007). The intrinsic ellipticity of the slow-rotating galaxies was found to be $\varepsilon \lesssim 0.4$. The only significantly flattened slow-rotating galaxies are the counter-rotating disks (**Figure 4*d***). The regular rotators (**Figure 4*e***) show a completely different distribution on the $(V/\sigma, \varepsilon)$ diagram. They are broadly consistent with a population of randomly oriented axisymmetric galaxies with oblate velocity ellipsoids ($\gamma \approx 0$), satisfying the anisotropy condition $\beta_z \lesssim 0.7 \times \varepsilon_{\rm intr}$ of Equation 11. This equation was suggested by the detailed dynamical models of Section 3.4.2 and 3.4.3. In fact, on the $(V/\sigma, \varepsilon)$ diagram, the regular rotators are broadly distributed within the envelope defined by the edge-on relation of Equation 11 and its projection at different inclinations (Cappellari et al. 2007). These relations are shown in **Figure 13**. The anisotropy of regular rotators within $1R_e$ was found to span a larger range than that of the nonregular rotators. The former are generally not close to isotropic but often fall close to the edge-on isotropic line owing to projection effects. This explain previous observations suggesting the fast-rotating galaxies are more isotropic than

the slow-rotating ones. In practice, starting from a range of ε values, the edge-on relation (the magenta line in **Figure 13**) can be plotted on the $(V/\sigma, \varepsilon)$ diagram using Equations 14 and 15, with the different projections obtained from Equation 16.

Also shown in **Figure 13** is the relation $(V/\sigma)^* = 1/3$, where $(V/\sigma)^* \equiv (V/\sigma)/(V/\sigma)_{\rm iso}$ is the ratio between V/σ and the corresponding theoretical value for an isotropic galaxy with the same ε. This parameter was called the anisotropy parameter (e.g., Kormendy & Illingworth 1982; Davies et al. 1983; Bender 1988; Bender et al. 1992, 1994; Naab & Burkert 2003), although is clear from **Figure 13** that galaxies with the same $(V/\sigma)^*$ can span the full range of anisotropies. Nonetheless, $(V/\sigma)^*$ is still a very useful tool to quantify the global dynamics of ETGs. In fact, Cappellari



Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

et al. (2007) note that a value of $(V/\sigma)^* \approx 0.4$ approximately separates the fast/regular and slow/nonregular rotator classes indicated by the IFS kinematics (Emsellem et al. 2007, Cappellari et al. 2007). This fact confirms and explains the early results, based on long-slit spectroscopy, reporting a connection between $(V/\sigma)^*$ and galaxy properties (e.g., Davies et al. 1983, Kormendy & Bender 1996).

These results about the $(V/\sigma, \varepsilon)$ diagram from the SAURON survey (de Zeeuw et al. 2002) were confirmed and strengthened by the volume-limited sample of 260 galaxies from the ATLAS^{3D} survey (Cappellari et al. 2011a), which also provided a reliable census of the different classes of ETGs. The results were presented by Emsellem et al. (2011) and are reproduced on top of the theoretical relations in **Figure 13**. The figure shows that the edge-on relation of Equation 11 (magenta line) and the envelope produced by its values at different inclinations still broadly describe the location of regular rotators on the $(V/\sigma, \varepsilon)$ diagram. Again, a fixed $(V/\sigma)^*$ value provides a rough separation of the two main kinematic classes, and the larger sample allows one to more accurately define the dividing line around $(V/\sigma)^* \approx 1/3$.

The ATLAS^{3D} survey discovered 11 counter-rotating disks (**Figure 12**), which constitute 4% of the volume-limited ETG sample (Krajnović et al. 2011). They can appear quite flat, as expected because of their disk-like nature. However, unlike the regular rotators, they all lie below the magenta line in the $(V/\sigma, \varepsilon)$ diagram of **Figure 13** because they do not have oblate velocity ellipsoids but are instead dominated by tangential anisotropy ($\sigma_\phi > \sigma_R$ or $\gamma < 0$). Below the magenta line, there are also some galaxies that are classified as regular rotators by Krajnović et al. (2011). These are likely transition objects containing enough counter-rotating stars to produce a detectable decrease of their global rotation, but not enough to produce clear evidence for counter-rotation in the σ field. There is, in fact, no reason to expect a sharp transition between the regular rotators and the counter-rotating disks—the classification of a galaxy will simply depend on the amount of externally acquired counter-rotating gas (Bois et al. 2011).

3.6. Quantitative Kinematic Classification

3.6.1. Early-type galaxies. It is useful to define a quantitative measure of the visual distinction between the two classes of regular and nonregular rotators illustrated in **Figure 4**. A physically motivated and clean separation can be obtained using dynamical models (**Figure 11**), and this approach can be used when maximum accuracy is desired, but a more economical approach is also possible. One possible approach is to use the dividing line $(V/\sigma)^* \approx 1/3$ on the $(V/\sigma, \varepsilon)$

Figure 13

The $(V/\sigma, \varepsilon)$ and $(\lambda_{R_e}, \varepsilon)$ diagrams. (*a*) The symbols are the $(V/\sigma, \varepsilon)$ values, where V/σ is the ratio between the ordered stellar rotation V and the random motions σ, and ε is the galaxy observed ellipticity, for 260 early-type galaxies (ETGs) from Emsellem et al. (2011), using the symbols introduced in **Figure 4**. The solid green line is the prediction for an edge-on isotropic rotator, and the thin lines are separated by $\Delta\delta = 0.1$ in anisotropy, using Equation 14 (Binney 2005). The magenta line is the edge-on relation of Equation 11 (Cappellari et al. 2007), and the dotted lines indicate how this relation transforms at different, equally spaced inclinations using Equation 16. The distribution of regular rotators is broadly consistent with a family of axisymmetric systems following the anisotropy trends in **Figure 9**, seen at random orientations. The nonregular rotators tend to lie well below the magenta line. These two classes can be approximately separated by the line $(V/\sigma)^* = 1/3$ (*dashed green line*). (*b*) $(\lambda_{R_e}, \varepsilon)$ values, a quantity related to the stellar specific angular momentum, using kernel density estimates for the regular/nonregular distributions. For each class, the thick solid lines enclose 80% of the total probability. The lines are the same as in panel *a*, with the projected V/σ values transformed into λ_{R_e} using Equation 18 (Emsellem et al. 2007). The solid black line is the best empirical separation between regular and nonregular rotators (Equation 19), and it nearly overlaps with the $(V/\sigma)^* = 1/3$ line on this diagram. (*c*) Same as panel *b*, except the symbols are the $(\lambda_{R_e}, \varepsilon)$ values for 340 ETGs from Emsellem et al. (2011) and Fogarty et al. (2015). Other abbreviations: KDC, kinematically decoupled core; SAMI, Sydney-AAO Multi-object IFS survey.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

ε_e: luminosity-weighted ellipticity within the half-light isophote

diagram (Section 3.5.3). However, using the V/σ quantity has one major limitation: It does not take into account the spatial distribution in the kinematic maps. For this reason, a galaxy with a KDC such as NGC 5813 (**Figure 4*c***) can overlap on the $(V/\sigma, \varepsilon)$ diagram with inclined regular rotators (e.g., NGC 3379), which have a dramatically different kinematic appearance, as noted by Emsellem et al. (2007). This was the motivation to develop a new physical parameter that retains the useful characteristics of the classic V/σ quantity but also includes spatial information. A natural replacement for V is the magnitude of the luminosity-weighted averaged projected angular momentum $\langle \mathbf{L} \rangle = \langle \mathbf{R} \times \mathbf{V} \rangle$. To remove the need to determine vector directions and make a more easily computable quantity, $\langle \mathbf{L} \rangle$ was replaced by the surrogate $\langle R|V| \rangle$, where R is the projected distance from the galaxy center. When this proxy for the angular momentum is made dimensionless and normalized by a quantity such as $V_{\rm rms} \equiv \sqrt{V^2 + \sigma^2}$, which according to the scalar virial theorem is proportional to mass (Binney & Tremaine 2008), then one obtains the parameter (Emsellem et al. 2007)

$$\lambda_R \equiv \frac{\langle R|V| \rangle}{\langle R\sqrt{V^2+\sigma^2} \rangle} = \frac{\sum_{n=1}^{N} F_n R_n |V_n|}{\sum_{n=1}^{N} F_n R_n \sqrt{V_n^2+\sigma_n^2}}, \tag{17}$$

where the values of F_n are the fluxes within the N spatial bins where mean stellar velocities V_n and velocity dispersion σ_n are measured. The summation is extended out to a certain finite radius $R_{\rm max}$ within a galaxy isophote, and R_n is the mean radius of the bin.

The distribution of the $(\lambda_{R_e}, \varepsilon)$ values for ETGs from ATLAS^{3D} (from Emsellem et al. 2011) and the SAMI Pilot Survey (from Fogarty et al. 2015) is shown in **Figure 13** (data from CALIFA are not included here because kinematic classes are not yet available). Also overlaid are the same lines shown in the $(V/\sigma, \varepsilon)$ diagram, computed with the analytic expressions in Equations 14–16, as well as the anisotropy-shape relation of Equation 11. To calculate those lines, the projected locations on the $(V/\sigma, \varepsilon)$ diagram were simply converted into the corresponding ones for the $(\lambda_{R_e}, \varepsilon)$ diagram using the empirical calibration (Emsellem et al. 2007, 2011)

$$\lambda_{R_e} \approx \frac{k\,(V/\sigma)_e}{\sqrt{1+k^2(V/\sigma)_e^2}} \qquad \text{with} \qquad k = 1.1. \tag{18}$$

Given the close relation between V/σ and λ_{R_e}, broadly speaking, the $(\lambda_{R_e}, \varepsilon)$ diagram shows the same information as the $(V/\sigma, \varepsilon)$ diagram. Regular rotators are still generally described by the envelope of the magenta line in **Figure 13** and its projections, whereas nonregular rotators are not. The difference between the two diagrams is that on the $(\lambda_{R_e}, \varepsilon)$ diagram, the regular and nonregular rotators of Krajnović et al. (2011) are significantly better separated (Emsellem et al. 2011).

More recently, the SAMI Pilot sample of Fogarty et al. (2015) and the CALIFA sample of Falcón-Barroso et al. (2015) increased the number of observed slow rotators by 2.5×. They strongly confirm the ATLAS^{3D} finding that genuine disk-less slow rotators are all rounder than $\varepsilon_e < 0.4$. As shown in Section 3.4.3, counter-rotating disks are structurally equivalent to regular rotators and should not be classified as slow rotators. The new data motivate a refinement to the $\lambda_{R_e} = 0.31\sqrt{\varepsilon_e}$ fast/slow rotator division of Emsellem et al. (2011) as follows, to reduce the risk of missing very round nonregular rotators (see **Figure 13**):

$$\lambda_{R_e} < 0.08 + \varepsilon_e/4 \qquad \text{with} \qquad \varepsilon_e < 0.4. \tag{19}$$

The explicit inclusion of the roundness criterion in the classification allows one to identify ETGs with disks, or fast rotators, without the need for kinematic observations when they are sufficiently flat in projection.

The separation between fast and slow rotators was defined by Emsellem et al. (2007, 2011) for λ_{R_e} measured within the half-light isophote. For this reason a galaxy cannot change its class when more extended data become available. In this classification scheme, as in most others, the scale within which the classification is performed is important. The same is true, for example, for Hubble's classification: a nuclear dust disk in an elliptical galaxy may resemble a spiral galaxy from HST observations alone (e.g., Young et al. 2008), but this does not make us classify the galaxy as a spiral. Similarly, an elliptical that turns out to have extended spiral arms in very deep observations (e.g., Duc et al. 2015), should not be classified as a spiral.

I stress that the classification based on the $(\lambda_{R_e}, \varepsilon)$ diagram only constitutes an automatic and objective proxy for the more detailed classification based on the kinematic morphology of **Figure 4**. For reliable results, it is important to verify that the two agree. This implies that, when the data are of such a low quality that no visual classification is possible, one should avoid using the $(\lambda_{R_e}, \varepsilon)$ diagram alone to kinematically classify ETGs, as this may lead to meaningless results. This is because systematic effects on the velocity fields, such as spurious fluctuations, produce systematic biases in λ_{R_e}, as discussed by Emsellem et al. (2007).

3.6.2. Connection between dynamics and nuclear surface brightness profiles. As discussed in Section 2, the slow/fast rotator classes are expected to trace a similar phenomenon as the core/coreless classes—they are both expected to distinguish galaxies assembled mainly via gas poor mergers from the rest of the ETGs. This hypothesis was tested by Emsellem et al. (2011) using a sample of 56 galaxies with measured central light deficits from Kormendy et al. (2009) and Glass et al. (2011). They concluded that there is indeed a close connection between being a slow rotator and having a central deficit in the surface brightness, with respect to a Sérsic (1968) profile extrapolated from larger radii. However, the two classifications were not found to agree in all cases.

Lauer (2012) revisited the question using a subset of 63 galaxies in common between the ATLAS^{3D} sample and the Nuker sample of Lauer et al. (2007b). He concluded that the two indicators generally agree, confirming they are both able to recognize the relics of gas-poor, or dry, mergers. Krajnović et al. (2013b) doubled the size of both samples to 122 galaxies thanks to new HST photometric determinations. They used a core versus coreless definition based on a Nuker profile fit (Equation 4), but they showed that different criteria differ only at the ∼3% level. The main figure from that paper is adapted here in **Figure 14**. In agreement with the previous studies, one can see that the general agreement is very good: Nearly all round and slow-rotating ETGs do indeed have a core, and the vast majority of the regular rotators have a steep profile as expected. However, core and coreless ETGs do not overlap perfectly with the regular versus nonregular rotator classes, and for the same reason they do not separate precisely at the slow/fast rotator divide.

To understand, at least in part, the small disagreements between the core/coreless and slow/fast classifications, it is instructive to consider a few individual examples. Three of the coreless slow rotators are counter-rotating disks (**Figure 4*d***). As these objects have disks and must have formed via gas accretion, the lack of a core makes physical sense and further emphasizes the fact, discussed in Section 3.4.3, that counter-rotating disks are essentially misclassified slow rotators, which instead form a continuous sequence with the rest of the fast rotators. One of the counter-rotating disks, however, has a core. This galaxy is the well-studied elliptical galaxy NGC 4473. The cored nature of its inner profile was studied in detail by Pinkney et al. (2003), who suggested that it may be due

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

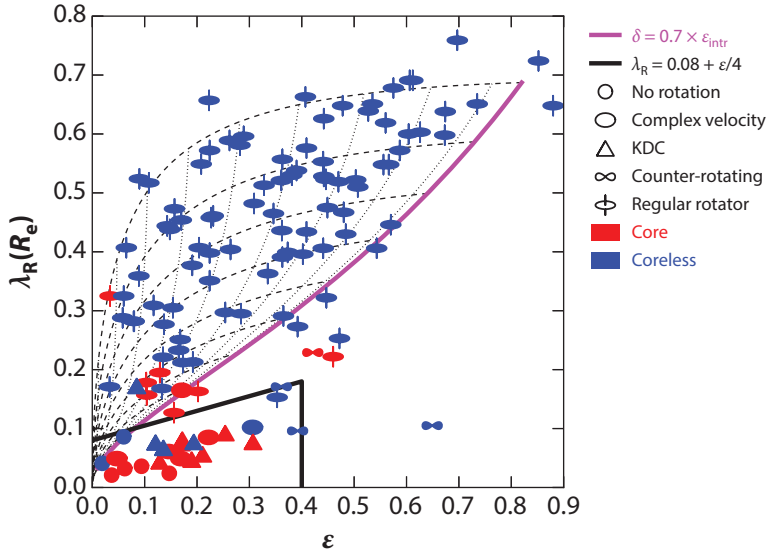


Figure 14

Cusps and cores on the (λ_{R_e}, ε) diagram. The values and symbols are the same as in **Figure 13** but are shown only for the galaxies for which a core or coreless classification exists (taken from Krajnović et al. 2013b). The magenta line is the same $\delta = 0.7 \times \varepsilon_{intr}$ (Cappellari et al. 2007) as in **Figure 13**, and the black line is the fast/slow rotator divide of Equation 19. There is a broad agreement between the slow rotators and the core galaxies but not a full overlap. Other abbreviations: KDC, kinematically decoupled core.

to a recent merger that flattened a preexisting cusp. The dynamics of NGC 4473 was modeled in detail by Cappellari & McDermid (2005) and Cappellari et al. (2007). It was found to consists of two counter-rotating stellar components with an approximate mass ratio of 1:3. This galaxy was proposed as a prototype of the counter-rotating class. The large-scale stellar counter-rotation and the lemniscate (which motivates the symbol used in this review to represent the kinematic class), or figure-eight, or 2σ nature of this galaxy is beautifully illustrated by the combination of the SAURON and SLUGGS kinematics in Foster et al. (2013). In summary, this galaxy does not differ from the other counter-rotating disks, and in this case the core classification does not imply that the galaxy assembled by dry mergers like the core slow rotators.

Overall, as these few examples already illustrate, it is not surprising that the core/coreless and slow/fast classifications do not agree in 100% of cases, although they agree most of the time. A galaxy that formed predominantly via gas accretion and that originally possessed a steep inner profile may sometimes acquire a small gas-poor satellite which, in favorable conditions, can destroy the inner cusp and nuclear disk (Sarzi et al. 2015), producing a core galaxy. The fact that most fast rotators are coreless indicates that this event is not a common occurrence. The small disagreement between the two classifications shows that, like in any classification, intermediate cases may exist and one should consider different characteristics of a galaxy before making strong inferences about the formation mechanism of any individual object. The safest choice in this case is to consider as dry merger remnants only galaxies that both are classified as slow rotators and have an inner core. Unfortunately this information is not available for many galaxies. However, the broad agreement indicates that on a statistical basis, either method is robust and allows one to draw conclusions about the overall galaxy population.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

3.6.3. Angular momentum across the Hubble sequence. In the $(V/\sigma, \varepsilon)$ and $(\lambda_{R_e}, \varepsilon)$ diagrams, I use a single ε per galaxy. However, this quantity is not constant within a galaxy. In fact, fast rotators are galaxies with a bulge and a disk. The flattest ones tend to be the more disk-dominated ones. This means that the fast rotators near the top of the $(\lambda_{R_e}, \varepsilon)$ diagram are expected to be, on average, the ETGs with small bulge fractions. This trend of decreasing λ_{R_e} with increasing bulge fraction, or concentration, was shown by Krajnović et al. (2013a). It has a large scatter because of projection effects, which make the observed ellipticity a poor estimator of bulge fraction, and the observed λ_{R_e} an equally poor estimator of the intrinsic angular momentum. In this picture one would expect spiral galaxies, which on average have smaller bulge fractions than S0s (e.g., Simien & de Vaucouleurs 1986, de Jong 1996, Graham 2001, Graham & Worley 2008), to overlap with the fast-rotator ETGs with the largest λ_{R_e} values. This was shown by Falcón-Barroso et al. (2015) (see also Querejeta et al. 2015) in a work that utilizes IFS kinematics of spiral galaxies from the CALIFA survey (Sánchez et al. 2012). The trend between λ_{R_e} and galaxy concentration was also confirmed using data from the SAMI Pilot Survey by Fogarty et al. (2015). An exception to the trend are the latest-type spiral galaxies (Sd), which appear to have quite low λ_{R_e}. An explanation for this fact will need to await a more detailed analysis on the kinematics of those objects, which is not yet available.

In **Figure 15** I illustrate the trend between galaxy morphology and λ_{R_e} by combining data from the three largest IFS survey for which homogeneously measured parameters were published, for a total sample of 666 galaxies of all morphological types: ATLAS^{3D} (Cappellari et al. 2011a),

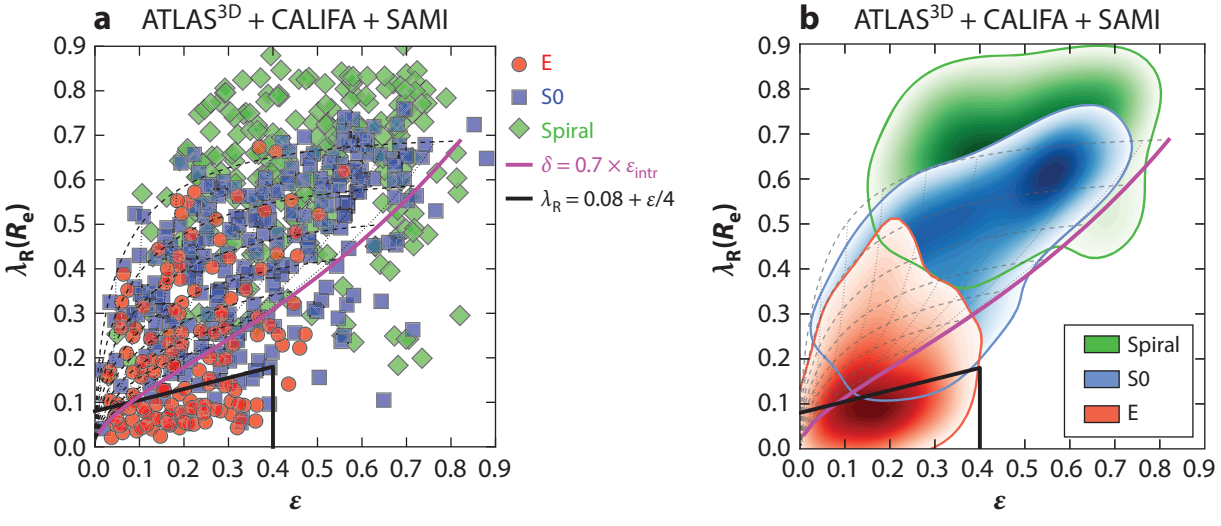


Figure 15

Galaxy morphology on the $(\lambda_{R_e}, \varepsilon)$ diagram (where λ_{R_e} is a quantity related to the stellar specific angular momentum). (*a*) 666 $(\lambda_{R_e}, \varepsilon)$ values, including 260 early-type galaxies (ETGs) in ATLAS^{3D} (from Emsellem et al. 2011), 300 mostly spiral galaxies in CALIFA (the Calar Alto Legacy Integral Field Area Survey) (from Falcón-Barroso et al. 2015), and 106 galaxies, mostly ETGs, in SAMI (the Sydney-AAO Multi-object IFS survey) (from Fogarty et al. 2015). The magenta line is the same $\delta = 0.7 \times \varepsilon_{\rm intr}$ relation (Cappellari et al. 2007) as in **Figure 13**, and the black line is the fast/slow rotator divide (Equation 19). The CALIFA sample, by design, lacks very round and very flat spiral galaxies, and this explains their apparently different distribution in ε from the ETGs. In this extended sample, which has 2.5× more slow rotators than ATLAS^{3D}, all secure slow rotators (i.e., excluding counter-rotating disks, cuspy ETGs, and spirals) have $\varepsilon \lesssim 0.4$. (*b*) A kernel density estimate of the distributions of elliptical (E) galaxies, S0s, and spirals in the left panel. For each class, the thick solid line encloses 80% of the probability. The region of slow rotators encloses almost exclusively Es.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

CALIFA (Sánchez et al. 2012), and the SAMI Pilot (Fogarty et al. 2014). I take (*a*) 260 values for the ETGs of the ATLAS^{3D} survey from Emsellem et al. (2011), (*b*) 300 values for all morphological types, but mostly spiral a galaxies of the CALIFA survey from Falcón-Barroso et al. (2015) and Falcón-Barroso et al. in preparation, and (*c*) 106 values for all morphological types, but mostly ETGs from SAMI from Fogarty et al. (2015). The distribution of the spiral galaxies (unlike that of the ETGs), which mostly come from CALIFA, is not randomly oriented by design (Walcher et al. 2014), and this explains the lack of very flat or very round spiral galaxies. Unlike the fast rotator ETGs, these galaxies should not be approximately distributed like the envelope of the magenta line shown in **Figures 13–15**.

An interesting result of **Figure 15** is that the area of this diagram populated only by ellipticals approximately traces the region defining the slow rotator class (Equation 19). This shows that, even from images alone, the ellipticals classified as slow rotators look different from S0 galaxies and cannot be mistaken for S0s. The same is not true for the ellipticals classified as fast rotators in the same range of ellipticity, which can be classified as either elliptical or S0. This result further confirms the adopted criterion to separate the two classes of fast and slow rotators.

Overall, the plot clearly shows that the classic and still widely used distinction between elliptical and S0 (Hubble 1936, Sandage 1961, de Vaucouleurs et al. 1991) has little physical meaning, in fact many inclined fast rotators are classified as ellipticals. This is mainly due to the obvious difficulty of recognizing an inclined S0 from a genuinely elliptical galaxy. The fact that ellipticals always have $\varepsilon \lesssim 0.5$ is essentially a matter of definition. For this reason, flat fast rotators are always classified as S0. The misclassification of ellipticals is very significant. In the volume-limited ATLAS^{3D} sample, up to 66% (=2/3) of ellipticals turn out to be fast rotators, namely an inclined axisymmetric galaxy with a disk (Emsellem et al. 2011). Similarly, in the combined sample of **Figure 15**, which has twice as many elliptical galaxies, 60% are actually fast rotators. Most misclassifications can be corrected using other photometric indicators, such as isophotal shape for edge on cases, and nuclear slopes more generally (Kormendy & Bender 1996), but these are currently not applicable to large surveys or at high redshift.

3.7. Two-Dimensional Stellar Population

IFS provides spatially resolved stellar population information together with stellar kinematics. Using profiles of stellar population parameters averaged over elliptical isophotes, Kuntschner et al. (2010) found a trend of increasing metallicity gradient with increasing galaxy mass for low-mass fast rotators with $M_* \lesssim 3 \times 10^{10}$, whereas the slopes decrease above that critical mass so that the most massive systems have quite shallow logarithmic gradients. This finding agrees with long-slit observations by Spolaor et al. (2009). The majority of the population was found to have negligible age trends. This is also true for slow rotators with KDCs, which have a population indistinguishable from the rest of the galaxy, in agreement with the SAURON observations of NGC 4365 by Davies et al. (2001) and the OASIS high-resolution observations by McDermid et al. (2006).

It had been suggested that the local escape velocity V_{esc}, derived from dynamical models in which mass follows light, may be a better parameter to study stellar population gradients. The key advantage is that using V_{esc} instead of radius, one finds local relations within a galaxy in quantitative agreement with global ones among different galaxies. In particular, Franx & Illingworth (1990) found a good correlation between V_{esc} and galaxy color, and Davies et al. (1993) and Carollo & Danziger (1994) found good global and local correlation of V_{esc} with the Mg absorption line strength, for two samples of eight and five ETGs respectively. Emsellem et al. (1996) first investigated the Mg–V_{esc} correlation using IFS data.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

The significance of the Mg-V_{esc} correlation was demonstrated by Scott et al. (2009) for the 48 ETGs of the SAURON survey and by Scott et al. (2013a) for the full ATLAS^{3D} sample of 260 ETGs. The dramatic improvement of these two studies was due not only to the large number of objects, but also to the fact that IFS data allowed one to obtain very high-quality radial gradients of the stellar population and accurate dynamical models. These studies showed that the Mg-V_{esc} correlation is truly universal in passive ETGs and constitutes an ideal benchmark for numerical simulations trying to describe in detail the local variations of stellar population in ETGs.

The IFS data not only provided accurate radial gradients, but they also provide fully resolved spatial distributions. The two-dimensional distribution revealed an increase of metallicity in fast rotators that is specifically associated with the disk component rather than being uniformly distributed within the spheroidal or bulge component (Kuntschner et al. 2010, Scott et al. 2013a). The association of the high metallicity with the disk is indicated by the fact that all inclined fast rotators have contours of equal metallicity that are flatter than the galaxy isophotes and generally consistent with the expected flattening of the disk component. This effect is visualized in **Figure 16**. The figure also shows that the disks do not stand out in the age maps.

In recent times, thanks to the availability of high-quality stellar population models (see Conroy 2013 for a review) with high spectral resolution based on large empirical spectral libraries (Bruzual & Charlot 2003, Maraston & Strömbäck 2011, Conroy & van Dokkum 2012a, Vazdekis et al. 2012), the study of stellar populations from integral field data has moved to the full-spectrum fitting approach, using fitting codes such as PPXF (Cappellari & Emsellem 2004), STARLIGHT (Cid Fernandes et al. 2005), STECKMAP (Ocvirk et al. 2006), and VESPA (Tojeiro et al. 2007). Recent results from the CALIFA (González Delgado et al. 2015), MASSIVE (Greene et al. 2015), and MaNGA surveys (Wilkinson et al. 2015) illustrate the power of IFS studies of stellar populations. However, a detailed overview of these stellar population studies goes beyond the scope of this review.

4. LOCAL SCALING RELATIONS

Dynamical scaling relations of ETGs relate the size, luminosity (L), and stellar kinematics of galaxies. Sizes are typically described by the half-light radius (R_e), whereas kinematics is generally quantified by the stellar velocity dispersion (σ) within a given aperture, which in this review is assumed no larger than R_e. Given that luminosity and size depend on distance, whereas kinematics does not, one of the first key applications of galaxy scaling relations was to infer galaxy distances (Dressler et al. 1987, Djorgovski & Davis 1987).

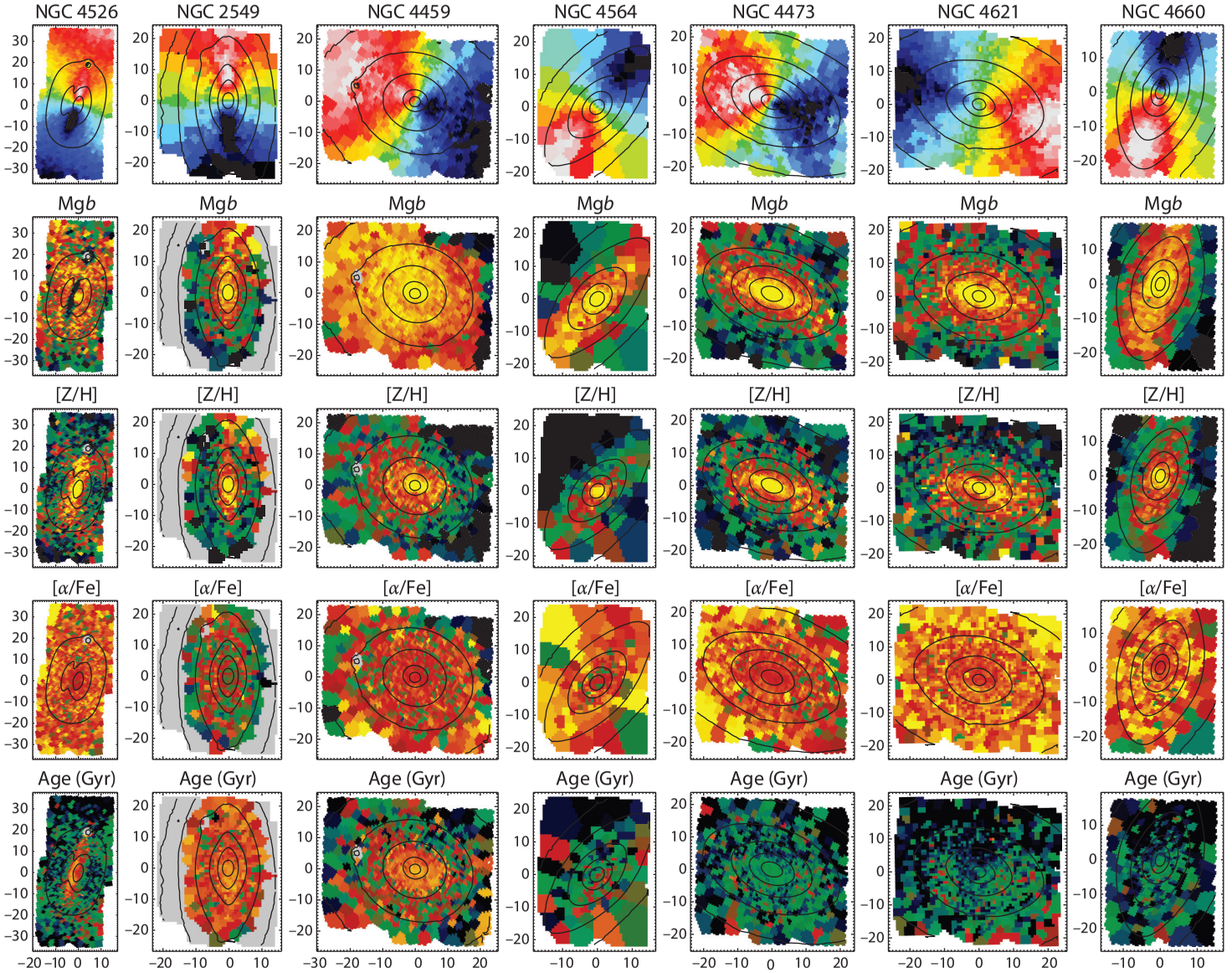
Nowadays, dynamical scaling relations are a key tool for studying galaxy formation. The main reasons are that (*a*) they provide a statistical description for easily measurable properties of galaxies as a function of time (redshift), which can be directly compared with numerical simulations (e.g., Boylan-Kolchin et al. 2006, Robertson et al. 2006, Oser et al. 2012, Porter et al. 2014) and (*b*) scaling parameters are actually expected to evolve very differently depending on the galaxy formation mechanism (e.g., Naab et al. 2009, Hopkins et al. 2010).

In this section I review recent developments in scaling relations, focusing specifically on the advancements in our understanding of scaling relations made possible by the use of IFS. Kormendy et al. (2009) and Graham (2013) present two contrasting overviews focused on results derived via photometry or long-slit spectroscopy.

4.1. Classic Scaling Relations

4.1.1. Scaling relations and the fundamental plane. The first dynamical scaling relation to be discovered was the one between luminosity (L) and stellar velocity dispersion (σ) (Faber &

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12



Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Jackson 1976). The observed relation had the form $L \propto \sigma^4$, and the authors pointed out it also suggests a trend between the mass-to-light ratio (M/L) and galaxy luminosity.

Soon thereafter, a correlation between galaxy surface brightness Σ and galaxy size was also found (Kormendy 1977). When one defines the surface brightness as the mean value within R_e, then $\Sigma_e = L/(2\pi R_e^2)$. Thus, the Kormendy relation describes a correlation between galaxy radius and luminosity. The latter form has the advantage that it does not explicitly include galaxy size on both axes of the correlation, reducing the covariance between the measured values. The L-R_e relation (e.g., Shen et al. 2003) has recently become quite popular to study galaxy evolution as a function of redshift (e.g., van der Wel et al. 2014), given that it does not involve any kinematic determination and for this reason is much more economical to observe than the Faber-Jackson relation.

Fundamental plane (FP): the distribution of (L, σ, R_e) galaxy parameters

Virial relation: $M \propto \sigma^2 R_e$

Thanks to larger systematic surveys of ETGs, it was later discovered that the Faber-Jackson and the Kormendy relations are just two special projections of a plane described by galaxies in ($\log L$, $\log \sigma$, $\log R_e$) coordinates (Djorgovski & Davis 1987, Dressler et al. 1987). This plane was aptly named the fundamental plane (FP; see Kormendy & Djorgovski 1989 for a review of the initial results). It was found to hold for all ETGs, including S0s and elliptical galaxies, with a scatter smaller than 20% in R_e (e.g., Jorgensen et al. 1996).

The existence of the FP was interpreted as due to galaxies satisfying virial equilibrium $M \propto \sigma^2 R_e$, with M the galaxy mass (Faber et al. 1987). However, the exponents of the FP were found to deviate significantly from the virial predictions, as confirmed by all of the numerous subsequent studies (e.g., Hudson et al. 1997, Scodeggio et al. 1998, Pahre et al. 1998, Colless et al. 2001, Bernardi et al. 2003). In particular, a recent determination of the plane, with σ_e measured from IFS within R_e, gives $L \propto \sigma_e^{1.25} R_e^{0.96}$ (figure 12 in Cappellari et al. 2013b). This deviation of the FP from the virial predictions is called the tilt of the FP.

4.1.2. Why the fundamental plane deviates from the virial relation. The variation of the (M/L) of stellar populations was immediately recognized as a potential source for the tilt of the FP (Faber et al. 1987). Systematic changes in the galaxy population were already known, with galaxies becoming older and more metal rich with increasing M or σ (e.g., Thomas et al. 2005, McDermid et al. 2015). This variation can potentially explain a major part of the FP tilt and scatter, predicting larger (M/L) as a function of σ by a factor of a few, as observed (depending on the photometric band) over the full range of galaxy masses (Prugniel & Simien 1996, Forbes et al. 1998).

The surface brightness profiles of ETGs also display systematic variations as a function of their luminosity, with profiles becoming more concentrated in more luminous objects (Section 2.2). At a fixed mass, a steeper profile implies a larger σ within the central regions (Ciotti 1991) where the kinematics information is observed. The amount of σ variation is again, in principle, sufficient to explain a major part of the FP tilt (Ciotti et al. 1996, Graham & Colless 1997, Prugniel & Simien 1997, Bertin et al. 2002, Trujillo et al. 2004a).

A third potential cause for the FP tilt is the fraction of dark matter within the region where kinematics is observed. The dark matter fraction is expected to increase systematically with mass for the range of interest of FP studies (e.g., Moster et al. 2010, 2013; Behroozi et al. 2010; Guo

Figure 16

Metallicity-enhanced disks in fast rotators. The top panels show the mean stellar velocity of seven fast-rotator early-type galaxies. Subsequent panels show the Mg*b* line strength, the metallicity [Z/H], the elemental abundance [α/Fe], and age. Strong, nearly edge-on disks show an enhancement in the measured Mg*b* or the inferred metallicity, which is flatter than the isophotes, whereas the age distribution is flat and featureless. Exceptions are NGC 4526 and NGC 4459, which contain young star forming disks and are also rich in molecular gas (Young et al. 2008). Adapted with permission from Kuntschner et al. (2010).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

et al. 2010; Leauthaud et al. 2012). This increase can cause variations in the observed total (M/L) of an amount again sufficient to produce a significant fraction of the measured tilt (Renzini & Ciotti 1993, Borriello et al. 2003, Tortora et al. 2012).

Mass plane (MP): the distribution of (M, σ, R_e) galaxy parameters

M_*^{dyn}**:** dynamically determined total galaxy stellar mass

4.2. Understanding Scaling Relations via Integral Field Spectroscopy

4.2.1. The mass plane follows the virial relation. Using semi-isotropic Jeans models and stellar kinematics of 37 galaxies from long-slit observations, it was found that the M/L trend with galaxy mass remains nearly unchanged when one includes the effects of galaxy nonhomology (van der Marel 1991, Magorrian et al. 1998). The model accuracy was improved by fitting models to IFS stellar kinematics. Using a sample of 25 galaxies and both the Schwarzschild and Jeans approaches, Cappellari et al. (2006) found that the (M/L)-σ relation is extremely tight and "can be included in the remarkable series of tight correlations between σ e and other galaxy global observables" (p. 1126). The relation was found to account for the entire scatter and tilt of the FP. Independent confirmation of these facts came from strong lensing studies (Bolton et al. 2007, Auger et al. 2010a).

This result was dramatically strengthened by two larger studies using integral field stellar kinematics: the modeling of the 260 ETGs of the ATLAS^{3D} sample (Cappellari et al. 2013b) and a similar study using 106 galaxies, of generally larger masses, from the SAMI Pilot Survey (Scott et al. 2015). Both studies employed the JAM method, which was shown using simulation to produce unbiased M/L estimates (Lablanche et al. 2012, Li et al. 2016). These modeling studies showed that when luminosity is replaced with mass in the FP (L, σ, R_e), the coefficients of the resulting mass plane (M_{JAM}, σ, R_e) match the virial predictions $M \propto \sigma^2 R_e$ within the errors. However, the studies also pointed out the significant dependence of the plane coefficients on the technique used to measure them. This sensitivity explains the apparent contradictions between some past studies of the FP tilt.

As demonstrated in detail by Cappellari et al. (2013b), the mass M_{JAM} that appears in the mass plane represents the most accurate empirical approximation of a galaxy's total stellar mass M_*^{dyn}:

$$M_{JAM} = L \times (M/L)_{JAM} \approx 2 \times M_{1/2} \approx M_*^{dyn}, \tag{20}$$

where L is the total galaxy luminosity and $M_{1/2}$ is the total mass contained within a sphere enclosing half of the total galaxy luminosity.

In **Figure 17** I show the combined (M/L)-σ values taken from both the ATLAS^{3D} survey (Cappellari et al. 2013b) and the SAMI Pilot Survey, in three different clusters (Scott et al. 2015). In both cases the values are computed in the Sloan Digital Sky Survey (SDSS) r-band (AB mag system). Here, the published SAMI (M/L) values were divided by a factor $(1+z)^3$ to account for both the $(1+z)^4$ contribution from cosmological surface brightness dimming and the $(1+z)^{-1}$ contribution owing to the frequency decrease (using AB mag) (e.g., Hogg 1999), at the redshift $z \approx 0.06$ of the observations. This correction was not included in the original paper (Scott et al. 2015). The resulting correlation, for a combined sample of 366 galaxies, has an observed scatter of $\Delta = 0.11$ dex (29%), from which can be inferred an intrinsic scatter of 20%. The scatter further decreases for the subsample of 69 galaxies with $\sigma_e > 200$ km s^{-1}, for which the observed scatter becomes $\Delta = 0.078$ dex (20%). The best-fitting relation was obtained with the Python version of the program LTS_LINEFIT[2] described by Cappellari et al. (2013b), which combines the least trimmed squares robust technique of Rousseeuw & Van Driessen (2006) into a least-squares

[2]Available from **http://purl.org/cappellari/software.**

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

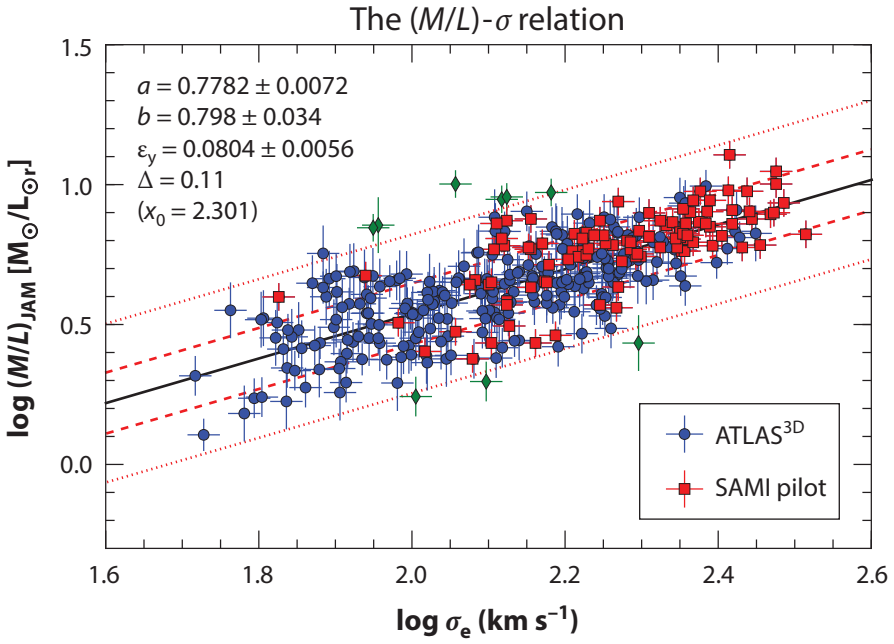


Figure 17

The (M/L)-σ relation. The dynamical total r-band mass-to-light (M/L) ratios and the stellar velocity dispersion within the half-light radius σ_e values for 366 early-type galaxies. The figure shows 260 ATLAS^{3D} early-type galaxies from Cappellari et al. (2013b) and 106 from the SAMI Pilot Survey of Scott et al. (2015), with an extra correction for surface brightness dimming. In both cases the (M/L) values were determined from Jeans anisotropic modeling (JAM) of the IFS stellar kinematics. The two complementary samples agree extremely well. They define a tight relation of the form $(M/L)_r \propto \sigma^{0.80}$, consistent with earlier findings, with an observed scatter of 29%, decreasing to 20% for $\sigma_e > 200$ km s^{-1}. The best-fitting values of the linear relation $y = a + b\,(x - x_0)$, the observed rms scatter Δ in dex, and the inferred intrinsic one ε_y, are shown in the top left corner.

fitting algorithm that allows for errors in both variables and intrinsic scatter. It is given by

$$\left(\frac{M}{L_r}\right)_{\rm JAM} = (6.00 \pm 0.10) \times \left(\frac{\sigma_e}{200\,{\rm km\ s^{-1}}}\right)^{0.80\pm0.03}. \qquad (21)$$

This relation and its scatter are consistent, within the errors, with the one for the ATLAS^{3D} subsample alone (Cappellari et al. 2013b), whereas the slope is also consistent with the one from Cappellari et al. (2006), albeit with much smaller errors. However, both the scatter and the slope in the relation weakly depend on galaxy properties; the relation is shallower and has smaller scatter for galaxies in cluster and for slow rotators (Cappellari et al. 2013b), in agreement with previous IFS studies of the FP (Falcón-Barroso et al. 2011).

The (M/L_r) in the r-band can be can be accurately converted to a different band B using the color $(B-r)_{\rm gal}$ of the galaxies and of the Sun $(B-r)_\odot$ as follows

$$M/L_B = (M/L_r) \times 10^{0.4[(B-r)_{\rm gal} - (B-r)_\odot]}. \qquad (22)$$

I used the B (Vega system) magnitudes from RC3, and I measured $r - K_s$ colors from matched apertures extracted from the 2MASS and SDSS images, which are available for all ATLAS^{3D} galaxies, to fit a tight color-σ relation. This was used, together with the sun colors from Blanton

& Roweis (2007), to provide the $(M/L) - \sigma$ relation in the B and K_s bands:

$$\left(\frac{M}{L_B}\right)_{\rm JAM} = (7.46 \pm 0.12) \times \left(\frac{\sigma_e}{200\,{\rm km\ s^{-1}}}\right)^{0.87\pm0.03} \quad (23)$$

and

$$\left(\frac{M}{L_{K_s}}\right)_{\rm JAM} = (1.27 \pm 0.02) \times \left(\frac{\sigma_e}{200\,{\rm km\ s^{-1}}}\right)^{0.53\pm0.03}. \quad (24)$$

The normalization of Equation 23 agrees within the relative errors with the one derived by van der Marel & van Dokkum (2007), but with a slightly more shallow slope.

$(M_*/L)_{\rm dyn}$: stellar mass-to-light ratio measured via dynamical models

$(M_*/L)_{\rm pop}$: stellar mass-to-light ratio measured via stellar population

$f_{\rm DM}(R_{\rm e})$: fraction of dark matter within a sphere of radius $R_{\rm e}$

IMF: stellar initial mass function

A natural question is how much of the observed trend in (M/L) can be explained by variations in the stellar population, and to first order of its age and metallicity. The study by Gerhard et al. (2001), using detailed dynamical models, found that the dynamically derived $(M_*/L)_{\rm dyn}$ was related to the $(M_*/L)_{\rm pop}$ inferred from stellar population models. This confirmed that at least part of the FP tilt is due to stellar population variations. It agrees with the fact that the scatter around the FP is also linked to stellar population variations (Graves et al. 2009, Falcón-Barroso et al. 2011, Magoulas et al. 2012, Springob et al. 2012). However, even after improving the accuracy of the models using IFS kinematics, the relation between dynamically derived total (M/L) and the stellar population $(M_*/L)_{\rm pop}$ still showed significant systematic deviations (Cappellari et al. 2006). Deviations between accurate determinations of the stellar and total masses were also found by combining the strong lensing technique with VIMOS integral field observations (e.g., Barnabè et al. 2011) or using multiple long-slit observations (Thomas et al. 2011). These could only be explained by either dark matter or initial mass function (IMF) variations among galaxies.

4.2.2. Little dark matter and nonuniversal stellar initial mass function. Using long-slit data and general models for two samples of approximately 20 ETGs, different studies appeared to agree that dark matter represents a minor fraction of the total mass within a sphere of radius $r \sim R_{\rm e}$ (Gerhard et al. 2001, Thomas et al. 2011). The dark matter content within $R_{\rm e}$ was better quantified with the modeling of the IFS kinematics for the 260 ETGs of the ATLAS$^{\rm 3D}$ sample (Cappellari et al. 2013b). A median dark matter fraction $f_{\rm DM}(r = R_{\rm e})$ as low as 13% was measured for the full sample (**Figure 18**).

Thanks to the large sample and two-dimensional stellar kinematics, the ATLAS$^{\rm 3D}$ study was able to show that the systematic trend in the discrepancy between $(M_*/L)_{\rm dyn}$ and $(M_*/L)_{\rm pop}$ could not be explained by a variation in the $f_{\rm DM}$. The constraining power of the IFS data allowed the JAM dynamical models to explore, with a Bayesian approach, a range in dark matter inner slopes including a flat inner dark matter profile, a standard NFW profile (Navarro, Frenk & White 1996), and a contracted halo (Gnedin et al. 2004) as special cases. Even with this freedom, the models were unable to reproduce the photometry and IFS kinematics by varying the dark matter fraction alone (for an illustration, see Cappellari et al. 2012). The only remaining option for the discrepancy was a systematic variation of the IMF (Cappellari et al. 2012). This was inferred to vary in mass normalization over the full galaxy mass range from Milky-Way type (Kroupa 2001, Chabrier 2003) to heavier-than-Salpeter (1955) type, with a dependence on galaxy velocity dispersion (Cappellari et al. 2012, 2013a). The IMF variation contrasted with previous evidence of a universal IMF (Bastian et al. 2010). However, the IMF for the most massive galaxies in the sample appeared consistent with previous indications of a heavy IMF in massive ETGs from either stellar population (van Dokkum & Conroy 2010) or strong gravitational lensing (Auger et al. 2010b). A recent update to the IMF-σ relation by Cappellari et al. (2013a) was obtained by combining $(M_*/L)_{\rm dyn}$ determinations from ATLAS$^{\rm 3D}$ with fully consistent redeterminations of $(M_*/L)_{\rm dyn}$

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

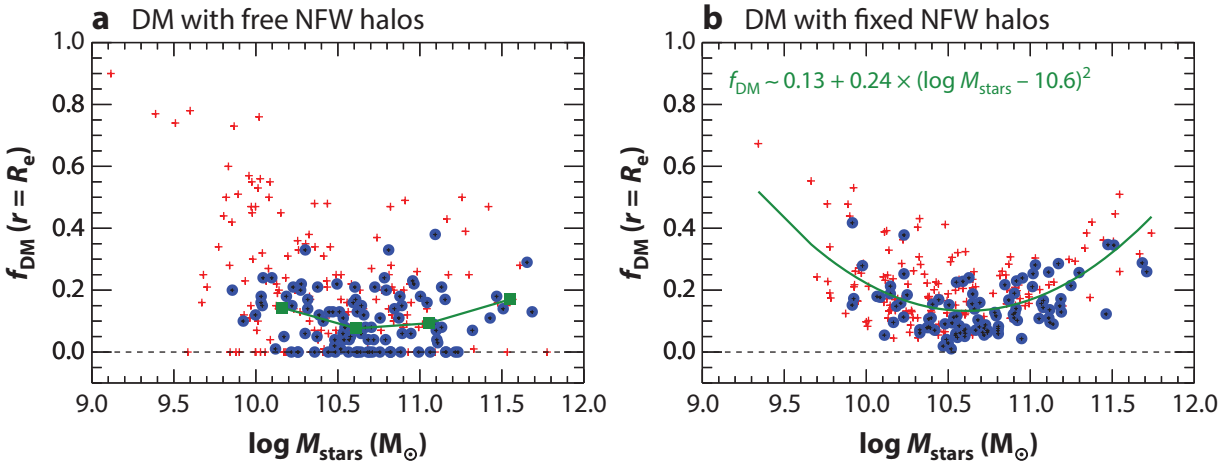


Figure 18

Dark matter (DM) from stellar dynamics. Either (*a*) leaving the halo normalization as a free parameter in the models, or (*b*) constraining it to the value predicted by ΛCDM, assuming a Navarro et al. (1996) (NFW) dark halo profile, the inferred dark matter fraction f_{DM} within R_e (the half-light radius) must be small for the models to be able to accurately fit the photometry and the integral field spectroscopy kinematics. Blue circles have higher-quality data, whereas red crosses have lower-quality data. Adapted with permission from Cappellari et al. (2013b).

from SLACS (Sloan lens ACS) (Bolton et al. 2006), combining strong lensing masses with JAM models. This relation is shown in **Figure 19** (from Posacki et al. 2015), which illustrates the good agreement between the two approaches and the need for a heavy IMF at the largest σ values.

The trend in IMF was found to be most closely related to the bulge mass fraction, as inferred from the galaxy dynamics, rather than to the galaxy mass alone. The heaviest IMF was generally measured for the densest fast-rotator galaxies and not for the most massive galaxies (Cappellari et al. 2013a; see also **Figure 22*i*** in Section 4.3.1). This result seems to have been confirmed so far by individual studies of three dense ETGs using long-slit kinematics (Läsker et al. 2013) and IFS data (Yildrm et al. 2015).

A large number of papers have appeared in recent years on the nonuniversality of the IMF in ETGs. Most of the studies used IMF-sensitive spectral features in combination with stellar population models (e.g., Conroy & van Dokkum 2012b, van Dokkum & Conroy 2012, Smith et al. 2012a, Spiniello et al. 2012, Ferreras et al. 2013, La Barbera et al. 2015, Martín-Navarro et al. 2015b, Spiniello et al. 2015). Others used constraints provided by scaling relations (Dutton et al. 2013) or approximate dynamical models in combination with stellar population information (Conroy et al. 2013, Tortora et al. 2013). There is an encouraging consensus about the need for systematic IMF variations among different galaxies. However, there are also clear tensions between different results, with stellar population models being unable to accurately predict the $(M_*/L)_{dyn}$ observed by dynamical or lensing approaches (Smith 2014), and possible contrast on the parameters driving the IMF variation (e.g., McDermid et al. 2014, Martín-Navarro et al. 2015b).

Initial studies at $z \sim 1$ found a Salpeter-like mass normalization of the IMF for $M_* \gtrsim 10^{11}\ M_\odot$ (Shetty & Cappellari 2014, Sonnenfeld et al. 2015). This is consistent with the IMF in the centers of massive nearby ETGs, assuming passive evolution. Similar consistency was found for the IMF at $z \sim 1$ from spectral features (Martín-Navarro et al. 2015a).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

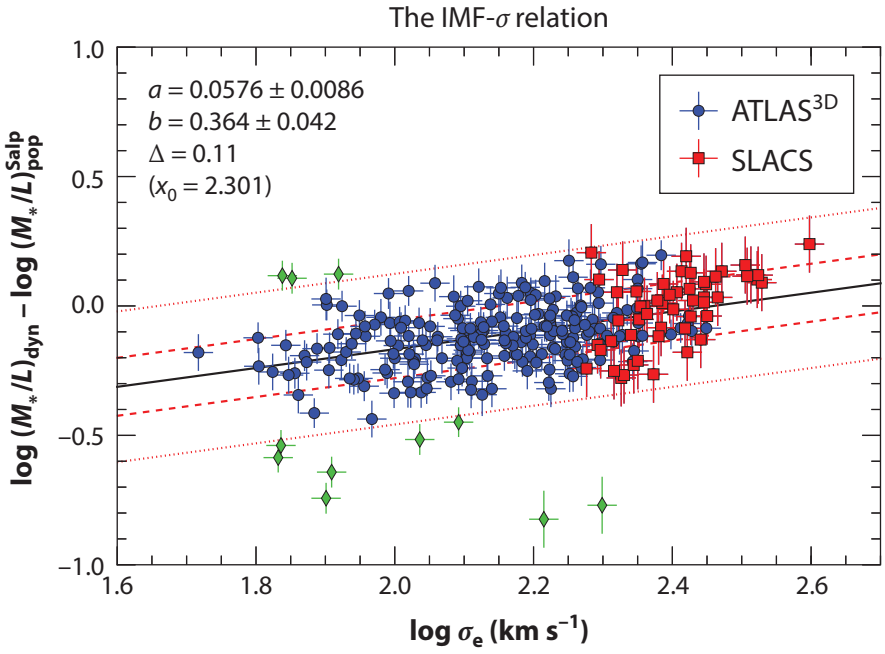


Figure 19

The IMF-σ (stellar initial mass function versus velocity dispersion) relation. The y-axis shows the ratio of the stellar mass-to-light ratio $(M_*/L)_{\rm dyn}$ inferred from mass dynamical modeling or strong lensing to the $(M_*/L)^{\rm Salp}_{\rm pop}$ inferred from full-spectrum fitting using stellar population models. A reference Salpeter (1955) IMF is assumed. The dynamical models of the integral field ATLAS$^{\rm 3D}$ data are taken from Cappellari et al. (2013b), and the combined lensing and dynamics models are based on the SLACS (Sloan Lens ACS survey) lens models of Treu et al. (2010), but were redone using axisymmetric Jeans anisotropic models for maximum consistency between the two sets of determinations. The two sets of data consistently indicate a mass excess that cannot be ascribed to dark matter and can be interpreted as a variation of the IMF (adapted from Posacki et al. 2015). The best-fitting values of the linear relation $y = a + b(x - x_0)$ and the observed rms scatter Δ in dex are shown in the top left corner.

IFS has the potential to revolutionize this field once more. In particular, the ongoing MaNGA survey (Bundy et al. 2015) combines a large wavelength range, which provides access to the most important IMF-sensitive spectral features (Conroy & van Dokkum 2012a), with good quality IFS data for a sample of 10,000 galaxies. The survey data can be used to try to reach a consensus, if possible, between different approaches, and clarify our understanding of IMF trends in galaxies. Progress in this field can also be expected from MUSE observations, thanks to the impressive spectral data quality that the instrument can provide.

4.2.3. Nearly isothermal mass profiles to four half-light radii. In spiral galaxies, circular velocities could be measured decades ago using ionized (Rubin et al. 1980) and neutral gas (Bosma 1978). Observations indicated flat circular rotation curves and consequently nearly isothermal $\rho_{\rm tot} \propto r^{-2}$ profiles (see review by Courteau et al. 2014).

Mass profiles of ETGs are much more difficult to infer, as one generally has to use expensive observations of the stellar kinematics and more complex dynamical models. For this reason, until recently, most studies did not extend to radii much larger than $R_{\rm e}$. Long-slit observations of two different samples of ≈20 ETGs revealed rotation curves to be nearly flat with nearly isothermal

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

mass profiles, as in spiral galaxies, within the median radius $r \approx 2R_e$ where kinematics was available (Gerhard et al. 2001, Thomas et al. 2011). Fully consistent results were independently found using strong gravitational lensing. In particular, the SLACS survey (Bolton et al. 2006) found an isothermal slope, with small scatter, for the total galaxy density profile of 73 ETGs, at a typical radius of $r \approx R_e/2$ (Koopmans et al. 2009, Auger et al. 2010a).

For comparison, at group and cluster scales, mass profiles derived using X-ray modeling indicate a trend in the total density profiles varying systematically from $\rho_{tot} \propto r^{-2}$ for the smallest systems to $\rho_{tot} \propto r^{-1}$ for the largest ones (e.g., Humphrey & Buote 2010). A broadly consistent systematic trend was obtained by combining strong and weak lensing for massive galaxy clusters (Newman et al. 2013a, 2015). The stacked weak-lensing approach by Gavazzi et al. (2007) indicated, on average, isothermal profiles out to very large radii.

Numerous studies of individual ETGs exist. These studies used observations of extended HI disks (Weijmans et al. 2008), deep observations of the stellar kinematics at sparse locations (Weijmans et al. 2009, Forestell & Gebhardt 2010, Murphy et al. 2011), discrete stellar kinematics measurements within individual globular clusters (Deason et al. 2012, Napolitano et al. 2014), and the kinematics of planetary nebulae (Romanowsky et al. 2003; de Lorenzi et al. 2008, 2009; Napolitano et al. 2011; Morganti et al. 2013). But a consistent picture did not emerge because of the inhomogeneity of the targets and observational techniques (for a review, see Gerhard 2013).

Only recently did large-scale two-dimensional stellar kinematic data start to become available for statistically significant samples of ETGs (Brodie et al. 2014, Raskutti et al. 2014). Cappellari et al. (2015) combined the two-dimensional stellar kinematics of 14 fast-rotator ETGs out to a median radius of $r \approx 4R_e$ for the sample from the SLUGGS survey (Brodie et al. 2014), with IFS within $\lesssim 1R_e$ from ATLAS^{3D} (Cappellari et al. 2011a). The data were modeled using the JAM approach. The study did not try to separate the luminous and dark matter components, as this generally requires making restrictive assumptions, but only focused on the total density, using a quite general parameterization. The resulting total density profiles were found to be well described by a nearly isothermal power law $\rho_{tot}(r) \propto r^{-\gamma}$ from $R_e/10$ to at least $4R_e$, the largest average deviation being 11%. The average logarithmic slope is $\langle\gamma\rangle = 2.19 \pm 0.03$ with an observed rms scatter of just $\sigma_\gamma = 0.11$. This scatter out to large radii, where dark matter dominates, is as small as that previously reported by lensing studies at $r \approx R_e/2$, where the stars dominate.

The apparent profile universality seems part of a more general trend in the total density slopes. In fact, the slopes within $r \lesssim R_e$, where models using IFS were constructed for large numbers of galaxies, already indicate (**Figure 22*c***) that the total mass profiles become more shallow for galaxies with $\sigma_e \lesssim 100$ km s^{-1}, whereas the profile remains universal with high accuracy for larger σ_e, with just a small decrease in the absolute slope above the critical mass $M_{crit} \approx 2 \times 10^{11}$ $M_\odot$. However, larger samples of homogeneously modeled galaxies are needed to understand whether these central trend persist at larger radii where dark matter dominates. Profile slopes provide important constraints to galaxy formation models (e.g., Remus et al. 2013, Dutton & Treu 2014).

4.3. The Mass-Size and Mass-σ Distributions

4.3.1. Results from integral field spectroscopy. In **Figure 20** I show two projections of the MP. As expected from the thinness of the plane, both projections provide essentially the same information, simply transformed into a different coordinate system. The key novelties of this plot, made possible by the use of IFS data, are that (*a*) the stellar σ values are not central values but are instead integrated within an ellipse enclosing half of the total galaxy light, and thus more closely related to the galaxy mass appearing in the virial relation, and (*b*) the x-axis is not the commonly used luminosity or the stellar mass determined from stellar population models, but a dynamically

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

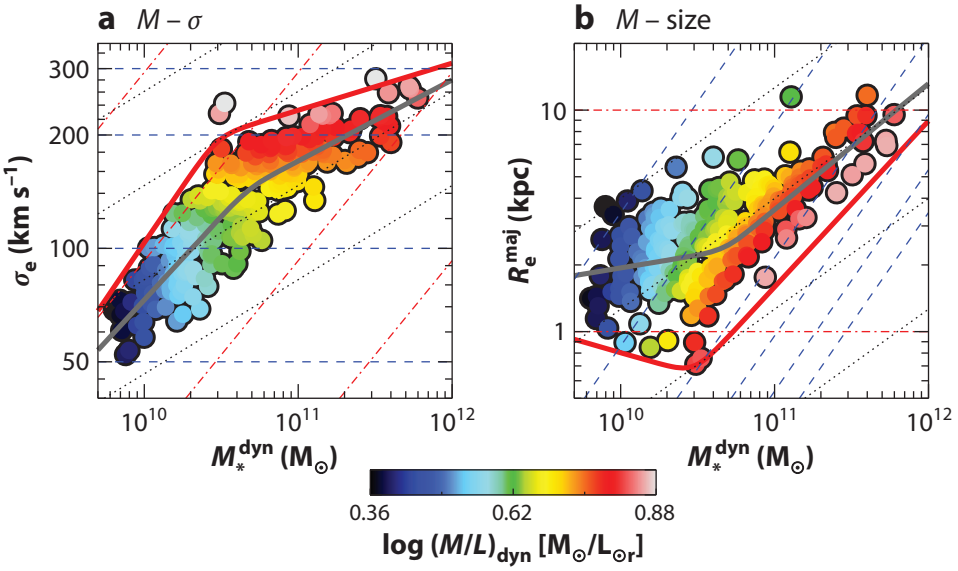


Figure 20

Break in scaling relations. The projections of the mass plane in (*a*) ($M_*^{\rm dyn}$, $\sigma_{\rm e}$) and (*b*) ($M_*^{\rm dyn}$, $R_{\rm e}^{\rm maj}$) coordinates. $M_*^{\rm dyn}$ is the stellar mass inferred from dynamical models, $\sigma_{\rm e}$ is the stellar velocity dispersion and $R_{\rm e}^{\rm maj}$ is the semimajor axis of the half-light isophote. Overlaid are lines of constant $\sigma_{\rm e} = 50$, 100, 200, 300, 400, 500 km s^{-1} (*dashed blue*), constant $R_{\rm e}^{\rm maj} = 0.1$, 1, 10, 100 kpc (*dot-dashed red*), and constant $\Sigma_{\rm e} = 10^8$, 10^9, 10^{10}, 10^{11} M$_\odot$ kpc^{-2} (*dotted black*) transformed in different panels using the virial relation. In each panel, the galaxies are colored according to the (LOESS smoothed) $\log(M/L)_{\rm JAM}$ from Jeans anisotropic models, as shown in the color bar at the bottom. In both panels, the thick red line shows the same zone of exclusion relation given by Equation 25. The gray line is the same $M_*^{\rm dyn} - \sigma_{\rm e}$ relation with trends $M_*^{\rm dyn} \propto \sigma_{\rm e}^{4.6}$ for $\sigma_{\rm e} \gg 140$ km s^{-1} and $M_*^{\rm dyn} \propto \sigma_{\rm e}^{2.3}$ for $\sigma_{\rm e} \ll 140$ km s^{-1}. Adapted with permission from Cappellari et al. (2013a).

determined stellar mass $M_{\rm JAM} \approx M_*^{\rm dyn}$ (Equation 20), which includes possible variations in the IMF. The accurate masses and σ allow one to infer the following results (Cappellari et al. 2013a):

1. The distribution of galaxy properties on both projections is characterized by the same zone of exclusion (ZOE), which can be accurately converted from one projection to the other with the virial relation. The ZOE can be roughly approximated by a double power-law, with a break that defines the minimum radius or maximum density of the ETG population, at a characteristic mass $M_b \approx 3 \times 10^{10}$ M$_\odot$:

$$R_{\rm e}^{\rm maj} = R_b \left(\frac{M_*^{\rm dyn}}{M_b}\right)^{\gamma} \left[\frac{1}{2} + \frac{1}{2}\left(\frac{M_*^{\rm dyn}}{M_b}\right)^{\alpha}\right]^{(\beta-\gamma)/\alpha}, \qquad (25)$$

   with $R_b = 0.7$ kpc, $\alpha = 8$, $\beta = 0.75$, and $\gamma = -0.20$. The relation has an asymptotic trend $R_{\rm e}^{\rm maj} \propto (M_*^{\rm dyn})^{0.75}$ above M_b, and a sharp transition into $R_{\rm e}^{\rm maj} \propto (M_*^{\rm dyn})^{-0.20}$ below this break.
2. The ZOE produces a clear bend in both the mean $M - \sigma$ and $M - R_{\rm e}$ relations, with trends of $M_*^{\rm dyn} \propto \sigma_{\rm e}^{2.3}$ [correspondingly, $R_{\rm e}^{\rm maj} \propto (M_*^{\rm dyn})^{0.12}$] at small masses and $M_*^{\rm dyn} \propto \sigma_{\rm e}^{4.7}$ [correspondingly, $R_{\rm e}^{\rm maj} \propto (M_*^{\rm dyn})^{0.57}$] at large masses.

$\rho_{\rm tot}$: total mass density of luminous plus dark matter

ZOE: zone of exclusion

$M_{\rm crit}$: critical mass of approximately 2×10^{11} M$_\odot$, above which passive slow rotators with cores dominate

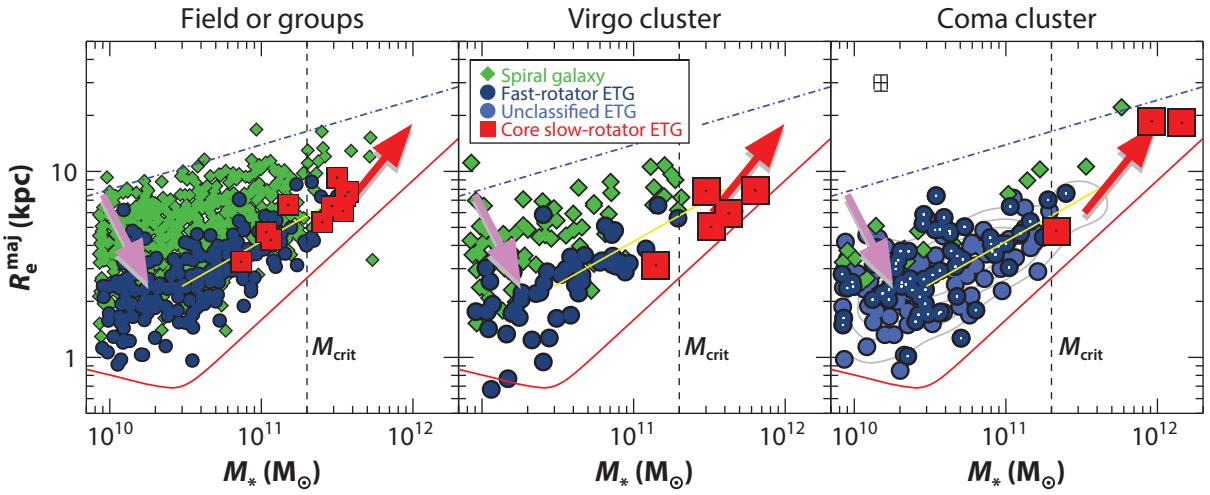


Figure 21

Mass-size versus environment. (*left*) The (M_*, R_e) distribution (where M_* is the stellar mass and R_e is the half-light radius) for a sample of galaxies located in either the field or small groups. (*center*) A sample in the moderately dense Virgo cluster. (*right*) A sample in the Coma cluster, which is one of the densest environments in the Universe. The plots visualize the two main channels for the formation of early-type galaxies (ETGs). The magenta arrow qualitatively indicates the evolutionary channel starting from star-forming spiral galaxies and producing fast rotators via gas accretion and bulge growth followed by quenching. The red arrow shows the dry-merging and halo-quenching channel, which produces massive slow rotators with central cores in their surface brightness profiles. The left and right panels are adapted with permission from Cappellari (2013).

3. A second characteristic mass at $M_{\rm crit} \approx 2 \times 10^{11}\,{\rm M}_\odot$ separates the axisymmetric fast rotators with disks and the spiral galaxies, at lower masses, from the rounder slow rotators with inner cores in their stellar surface brightness, the robustly determined dry merger relics (Section 3.6.2), at larger masses (**Figure 21**). The near absence of spiral galaxies above $M_{\rm crit}$ produces a sudden drop in the overall specific star formation of galaxies (Brinchmann et al. 2004, Salim et al. 2007). This steep decrease in the fraction of star-forming systems is well approximated by a simple model in which the quenching rate is proportional to the galaxy star formation rate (Peng et al. 2010).
4. Below $M_{\rm crit}$, galaxy properties closely follow lines of constant velocity dispersion (**Figure 22**). These lines trace (**Figure 22*b***) equal mass concentration or (**Figure 22*c***) equal steepness of the dynamically determined mass density profile, or bulge mass fraction. Remarkably, this trend is true for nearly every galaxy parameter related to their stellar population, such as (**Figure 22*d***) Hβ and (**Figure 22*e***) Mg*b* line strength, (**Figure 22*f***) optical color, (**Figure 22*g***) molecular gas fraction, (**Figure 22*h***) $(M_*/L)_{\rm dyn}$, (**Figure 22*i***) IMF mass normalization, (**Figure 22*j***) age, (**Figure 22*k***) metallicity and (**Figure 22*l***) α-element enhancement. This shows that galaxy formation models need to reproduce this crucial dependence on the central mass density if they want to describe reality.
5. The sequence of ETG properties, as a function of bulge fraction, merges smoothly with the sequence of spiral galaxy properties, which lie above the ETGs on the (M, R_e) diagram (**Figure 21**; Cappellari et al. 2011a, 2013a; Kormendy & Bender 2012). Late-type spiral galaxies (Sc) lie near the top of the diagram at any given mass and do not overlap with the ETGs, whereas early-type spirals (Sa), which have a smaller R_e, largely overlap with fast rotators with small σ_e. The densest and most spheroid dominated fast rotators (including

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

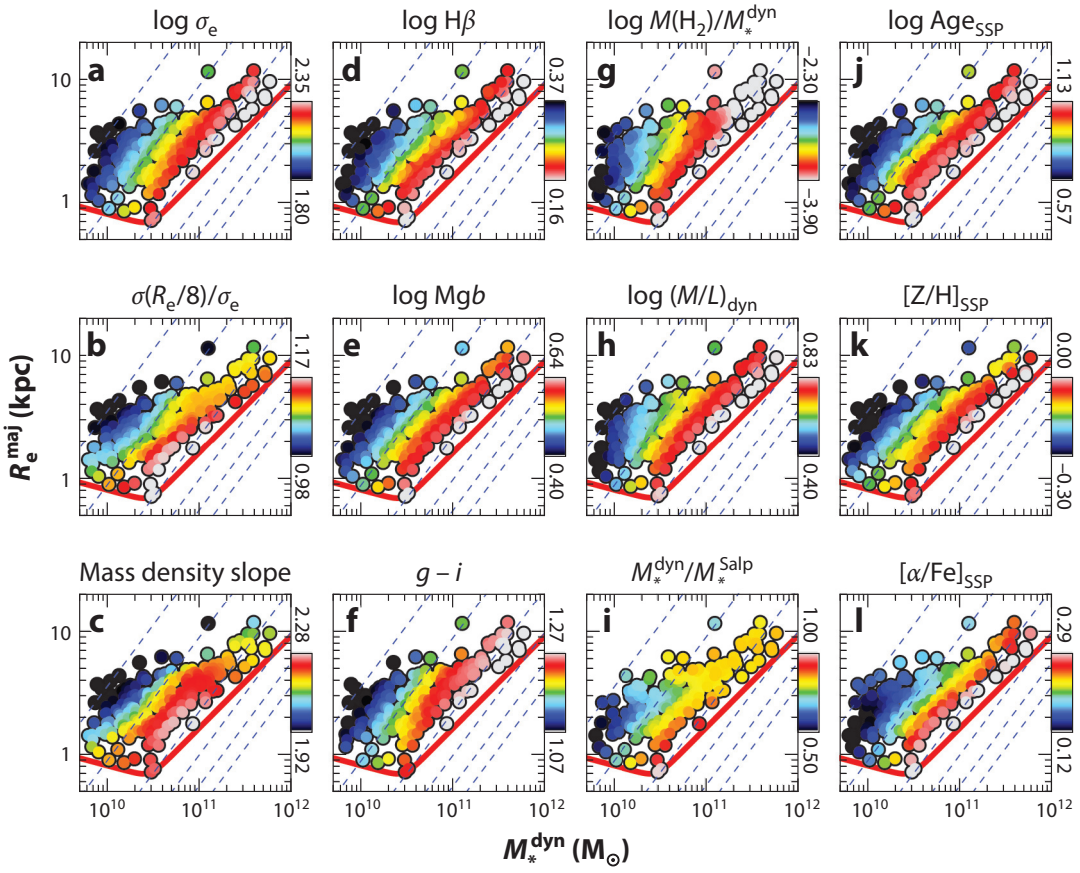


Figure 22

Early-type galaxy properties on the mass-size diagram. (*a*) Effective velocity dispersion σ_e (from Cappellari et al. 2013b); (*b*) ratio between the central dispersion within $R_e/8$ (where R_e is the half-light radius) and σ_e, which is related to the steepens of the mass profile (from Cappellari et al. 2013a); (*c*) average logarithmic slope $\langle\gamma\rangle_r = \Delta \log \rho_{tot}/\Delta \log r$ in the interval 0.1–1 R_e, measured from the B set of models of Cappellari et al. (2013b), which had NFW (Navarro et al. 1996) dark halos; (*d*) Hβ line strength index within $R_e/2$; (*e*) Mg*b* line strength index within $R_e/2$ from McDermid et al. (2015); (*f*) $g-i$ color from SDSS (from Aihara et al. 2011); (*g*) ratio of the molecular hydrogen H_2 from Young et al. (2011) and $M_*^{dyn} \equiv M_{JAM}$ from Cappellari et al. (2013b); (*h*) total (luminous plus dark) dynamical $(M/L)_{dyn} \equiv (M/L)_{JAM}$ from Cappellari et al. (2013b); (*i*) ratio $\alpha \equiv (M*/L)_{dyn}/(M*/L)_{pop}$ between the stellar M/L inferred from dynamical modeling, with the dark matter removed, and the same quantity inferred from full-spectrum fitting using stellar population models, assuming as a reference a Salpeter (1955) IMF (from Cappellari et al. 2013a); (*j*) mean SSP (single stellar population) age; (*k*) metallicity [Z/H]; and (*l*) elemental abundance [α/Fe] within $R_e/2$ (from McDermid et al. 2015). Panels *b*, *d*, and *f–i* adapted with permission from Cappellari et al. (2013a); and panels *j–l* adapted with permission from McDermid et al. (2015). Panels *a*, *c*, and *e* are original works.

many disky ellipticals; Kormendy & Bender 1996) occupy the bottom of the diagram and do not overlap with the spiral galaxies.

6. At stellar masses below the ATLAS^{3D} selection limits, the sequence of spiral galaxies and bulgeless fast rotators continues smoothly with a sequence of dwarf spheroidals (see figure 20 of Kormendy & Bender 2012 or figure 9 of Cappellari et al. 2013a). Interestingly, the stellar mass $M_\star \approx 2 \times 10^9\ M_\odot$, where there is a sharp bend in the M-size relation of dwarf galaxies and the sequence of increasing bulge fraction starts, corresponds to the threshold

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

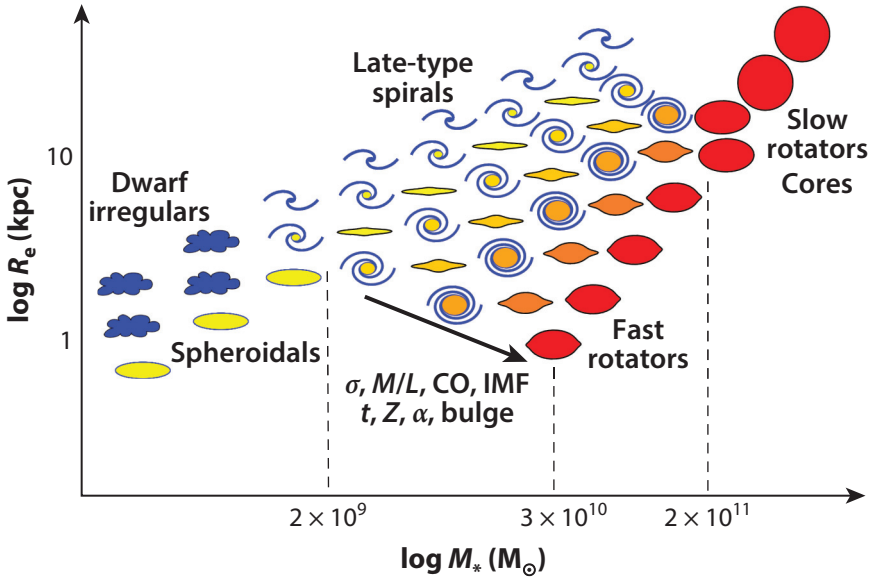


Figure 23

Galaxy properties on the mass-size diagram, where R_e is the half-light radius and M_* is the stellar mass. All early-type galaxy (ETG) properties related to the stellar population or gas content tend to vary along lines of nearly constant σ_e(effective velocity dispersion), which traces the bulge mass fraction or the steepness of the mass density profile. This sequence of ETG properties merges smoothly with the one of spiral galaxies, with little overlap between late-type spirals (Sc–Irr) and ETGs, a significant overlap between early spirals (Sa–Sb) and fast-rotator ETGs with low mass-to-light ratios (M/L), and no overlap between spirals and fast rotators with high (M/L). Three characteristic masses are emphasized in this diagram: (*a*) below $M_\star \approx 2 \times 10^9$ $M_\odot$, there are no regular ETGs and the mass-size lower boundary is increasing; (*b*) $M_\star \approx 3 \times 10^{10}$ $M_\odot$ is the mass at which ETGs reach their minimum size (or maximum stellar density) before a sudden change in slope to $R_e \propto M^{0.75}$ at larger masses; (*c*) below $M_\star \approx 2 \times 10^{11}$ $M_\odot$, ETGs are dominated by flat fast rotators showing evidence for disks. Above this mass there are no spirals and the population is dominated by round or weakly triaxial slow rotators with flat (core or deficit) central surface brightness profiles. Adapted with permission from Cappellari et al. (2013a).

for quenching of field galaxies discovered by Geha et al. (2012). They found that below that mass, only the cluster or group environment can strip galaxies of their gas. Bulge growth cannot happen below that mass: Star formation cannot be quenched by internal processes, only by environmental ones.

All of these empirical observations are graphically summarized in **Figure 23**. These results were interpreted by Cappellari et al. (2013a) as due to the buildup of ETGs happening via two separate routes. The first is in situ star formation, which involves growth via gas accretion or minor gas-rich mergers. The gas sinks toward the center and builds a bulge, which is associated with the quenching of star formation and disk fading (e.g., Cheung et al. 2012, Fang et al. 2013). This moves galaxies from left to right, causing it to cross lines of constant σ_e. The second route is external accretion, which involves dry merging and increases R_e by moving galaxies along lines of roughly constant σ_e (or steeper) but leaves the stellar population unchanged (e.g., Bezanson et al. 2009, Naab et al. 2009, Hopkins et al. 2010).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

4.3.2. Connection to previous results. The observed break in the $M-\sigma$ relation is consistent with previous reports of a change in the slope of the $L-\sigma$ relation of elliptical galaxies at low luminosities, where the relation was reported to become $L \propto \sigma^2$. An initial "marginally significant difference" in slope was noted by Davies et al. (1983, p. 50). But it took a couple of decades for statistically significant results to emerge (Matković & Guzmán 2005, de Rijcke et al. 2005, Lauer et al. 2007a, Forbes et al. 2008, Tortora et al. 2009). However, there is still some debate in the literature about the location and interpretation of the break. A break in the $M_* - \sigma$ relation at $M_* \approx 2\times10^{11}$ $M_\odot$ was reported by large SDSS studies, but no break was observed at lower masses (Hyde & Bernardi 2009, Bernardi et al. 2011). In the review by Graham (2013), the break in the $L-\sigma$ relation is also reported at $M_B \approx -20.5$, which corresponds to the same $M_* \approx 2\times10^{11}$ $M_\odot$.

Thanks to the quantitative consistency between the (M, σ) and the (M, R_e) projections and the link to other observables, **Figure 20** demonstrates that the break in the $L-\sigma$ relation reported in the earliest studies actually reflects the break in the ZOE around $M_* \approx 3\times10^{10}$ $M_\odot$, and not around $M_* \approx 2\times10^{11}$ $M_\odot$, where core slow rotators start dominating. In fact, the break in the $M_* - \sigma$ relation persists even when all slow rotators with cores are removed. However, it is also clear from **Figure 21** that core slow rotators approximately follow the relation defined by fast rotators with $M_* \gtrsim 3\times10^{10}$ $M_\odot$. However, the relation for core slow rotators is slightly steeper, producing a second break at $M_* \approx 2\times10^{11}$ $M_\odot$ as reported by SDSS. A more detailed picture above this interesting mass regime is expected from the MASSIVE IFS survey (Ma et al. 2014), but no kinematics data have been published yet.

A proper understanding of the break in the $M_* - \sigma$ relations is important for studies of supermassive BH scaling relations. The existence of a break implies that the $M_{\rm bulge} - M_{\rm BH}$ (Marconi & Hunt 2003, Häring & Rix 2004) and $\sigma - M_{\rm BH}$ (Gebhardt et al. 2000a, Ferrarese & Merritt 2000) cannot both be equally good predictors of BH masses (Lauer et al. 2007a, Graham 2012, Scott et al. 2013b). The break provides insight into the mechanism by which BHs grow. This very interesting aspect was beautifully reviewed by Kormendy & Ho (2013), and it will not be further addressed here.

The sequence defined by the most passive (old, high-metallicity, heavy IMF, CO-poor, etc.) and bulge-dominated (large σ_e) galaxies in the (M, R_e) diagram is another view of the Kormendy (1977) relation between galaxy luminosity and effective surface brightness. However, here the relation uses mass in place of luminosity. It is the end point of a continuous sequence that starts from the bulgeless spirals and ends up at the ZOE. The existence of the ZOE at large stellar densities was noted before (Bender et al. 1992, Burstein et al. 1997). Importantly, the power-law $M - R_e$ relation defined by the most passive ETGs (those with oldest ages in **Figure 22**) stops around the break in the ZOE at the characteristic mass $M_* \approx 3\times10^{10}$ $M_\odot$. This mass was recognized by Kauffmann et al. (2003) as the divide between two distinct families of galaxies: star forming and disk-like below this mass, and passive and spheroidal above. **Figures 22** and **23** confirm and explain this result, but they also illustrate that mass, unlike σ_e, is actually not a good predictor of galaxy properties (also see Cappellari et al. 2006, Franx et al. 2008, Graves et al. 2009).

Unlike the bend in the $L-\sigma$ relation, the corresponding one in the more easily observable $L - R_e$ relation has been known for a long time. It was noted by Binggeli et al. (1984, their figure 7) and further illustrated by several authors (e.g., Kormendy 1985, Graham & Guzmán 2003, Kormendy et al. 2009, Misgeld & Hilker 2011). Kormendy (1985) and Kormendy et al. (2009) see dwarf spheroidals as distinct from elliptical galaxies, constituting a separate family of gas-stripped dwarf spirals and irregulars, whereas Graham & Guzmán (2003) and Graham & Worley (2008) interpret the curvature in the $L - R_e$ relation as due to a smooth variation of the Sérsic (1968) index with luminosity (e.g., Young & Currie 1994, Graham & Guzmán 2003) in an otherwise homogeneous class of elliptical galaxies. Kormendy & Bender (2012) and

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

SCALING RELATIONS FOR BULGES AND DISKS

In the scaling relations presented in this review (**Figures 20–23**), I treat galaxies as single entities and do not try to show separate scaling relations for bulges and disks. I prefer a global approach, as it leads to more reproducible quantities. But this should not be interpreted as implying that the bulge/disk distinction is not physically meaningful. Instead, the popular photometric bulge/disk decomposition is a complementary way to quantify the relation between galaxy bulges and other galaxy properties, which is not covered in this review. These works reach conclusions that are remarkably consistent with the overall picture I infer from IFS observations, strengthening both conclusions (see Kormendy & Bender 2012, Kormendy 2016, for reviews). In the near future, I envision an approach in which photometric and IFS kinematic information are combined to bring bulge/disk decompositions to the next level. This will (*a*) reduce the inclination degeneracies that affect photometry alone and (*b*) allow researchers to quantify the kinematically determined bulge/disk contributions.

Graham (2013) illustrate two different views on this subject. The ATLAS^{3D} results agree with the former interpretation, with the key difference that the class of fast-rotator ETGs is now bridging the previous apparent gap between genuine ellipticals and dwarf spheroidals (Cappellari et al. 2013a).

The important link between bulge growth and quenching has also been observed using photometric data alone, both for local galaxies (Fang et al. 2013) and as a function of redshift (Bell et al. 2012, Cheung et al. 2012). The advantage of using dynamically determined masses and density slopes is that it allows one to exclude population gradients (e.g., disk fading) as the only driver of the observed trend (e.g., Carollo et al. 2013). The connection between the results presented in this section and those from photometric decomposition is discussed in the sidebar "Scaling Relations for Bulges and Disks."

4.4. Parallel Sequencing of Early-Type Galaxies and Spirals

The fast rotators are morphologically classified as either elliptical or S0 galaxies. All ETGs with photometric evidence for disks, including the S0s and all galaxies classified as E(d) (disky elliptical) by Bender et al. (1994), either belong to the fast-rotator class or contain counter-rotating disks. The reverse is not true, because disks cannot be seen in the photometry at low inclination. However, dynamical models show that all apparently round and nondisky fast rotators are inclined disks and are intrinsically still quite flat (Cappellari et al. 2013a). This is also confirmed by statistical inversion of the observed shape distribution of the fast rotator class (Weijmans et al. 2014) (Section 3.3). This indicates that all fast rotators have disks and would be classified as either S0 or E(d) if they could be observed edge-on. The fast rotators span the same range of bulge fractions as spiral galaxies (Krajnović et al. 2013a), like S0 galaxies (Laurikainen et al. 2010, Kormendy & Bender 2012), and include some extremely flat disks. They form a parallel sequence to spiral galaxies on scaling relations (**Figure 23**).

The slow rotators (or, equivalently, the nonregular rotators) are clearly different. They are never intrinsically flatter than $\varepsilon \approx 0.4$ (Section 3.6.3). As a class, they are inconsistent with being axisymmetric, as evidenced by kinematic twists (Section 3.3). They have cores or light deficits in their central surface brightness (Section 3.6.2), and their velocity fields are qualitatively very different from simple JAM models.

All these arguments led Cappellari et al. (2011b) to propose a revision of the classic Hubble (1936) tuning fork diagram. The new ATLAS^{3D} comb diagram combines elements from two

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

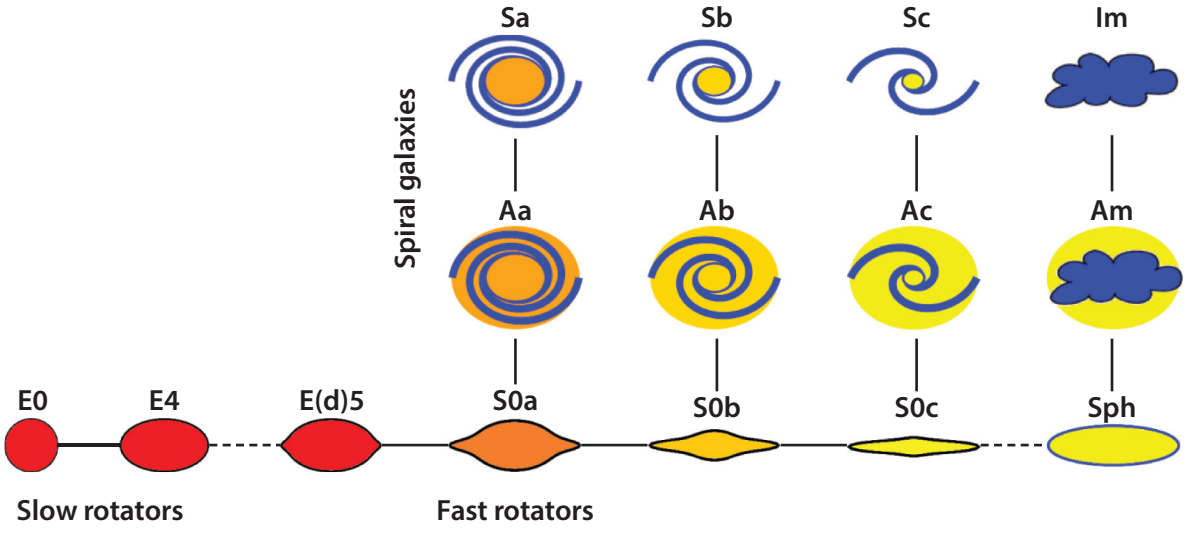


Figure 24

The comb morphology diagram, using the same symbols as **Figure 23**. If all fast rotators could be seen edge-on, they would appear morphologically with a range of spheroid fractions from thin S0s to flat ellipticals with disky isophotes [E(d) galaxies, Kormendy & Bender 1996], as illustrated here. The fast-rotator early-type galaxies (ETGs) form a parallel sequence to spiral galaxies, as already emphasized for the subset of S0 galaxies by van den Bergh (1976), who proposed the above distinction of SOs into S0a–S0c. Fast rotators are intrinsically flatter than $\varepsilon \gtrsim 0.4$ and span the same full range of shapes as spiral galaxies, including very thin disks. However, very few Sa galaxies have spheroids as large as those of E(d) galaxies, indicating that bulges must grow in the transformation. The slow rotators are always rounder than $\varepsilon \lesssim 0.4$ and have central cores in their surface brightness. The right side of the diagram includes spheroidal (Sph) galaxies, following Kormendy & Bender (2012). These are bulgeless dwarf galaxies, but they are significantly rounder than S0c disks. The black solid lines connecting the galaxy images indicate an empirical continuity, and the dashed lines emphasize the dichotomy between the fast- and slow-rotator ETGs, as well as between S0c and Sph galaxies. Adapted with permission from Cappellari et al. (2011b), with the addition of Sph galaxies from Kormendy & Bender (2012).

previously proposed revisions to Hubble's diagram with the new findings from IFS observations: (*a*) the parallelism between spiral galaxies and S0s in the revised classification diagram by van den Bergh (1976), (*b*) the link between S0s and disky ellipticals in the diagram proposed by Kormendy & Bender (1996), and (*c*) the distinction between slow and fast rotators, and the link of the latter with S0s, indicated by the IFS observations.

In the revised diagram (**Figure 24**), the ETGs are moved from the handle of the tuning fork to a sequence parallel to that of spiral galaxies. Contrary to previous diagrams, the proposed shape is not a symmetric fork or trident, but rather an asymmetric comb. This is to emphasize the fact that many of the galaxy properties (e.g., stellar age and gas content) are shared by the fast and slow rotator families of ETGs, and mostly vary along the sequence going from ETGs to spirals. This sequence includes the anemic spirals of van den Bergh (1976) to point out that the distinction between ETGs and spirals is not well defined and includes transition objects of uncertain classification. In some cases faint spiral arms become visible in ETGs with very deep optical (Duc et al. 2011, 2015) or HI observations (Serra et al. 2012).

The need for revising Hubble's tuning fork diagram, and the parallelism between S0 and spiral galaxies was also pointed out, based on photometric arguments, by Kormendy & Bender (2012). The main conceptual difference is that the ATLAS^{3D} diagram emphasizes that spiral galaxies form a parallel sequence to the entire class of kinematically classified fast rotators, rather than just to the subset of S0 galaxies. Moreover, Kormendy & Bender (2012) extended the S0 sequence to

include bulgeless spheroidal (Sph) galaxies. Cappellari et al. (2013a) reached a similar conclusion regarding Sph galaxies. I now also include Sph in **Figure 24**. The similarity of the analysis and conclusions in the two independent studies is a demonstration of the robustness and relevance of the proposed parallelism.

$T - \Sigma$: relation between the optical morphology and environmental density

$kT - \Sigma$: relation between the kinematic classification and environmental density

5. ENVIRONMENTAL TRENDS

Galaxy environmental trends represent a vast subject with a long history. Here I only briefly mention some of the earliest works and again focus on what IFS specifically adds to this topic. For more general reviews, the reader is referred to Blanton & Moustakas (2009) and Kormendy & Bender (2012) for normal galaxies and to Boselli & Gavazzi (2014), for emphasis on the faint end of the galaxy luminosity distribution.

5.1. Kinematic Morphology-Density Relation

Large optical surveys of galaxy clusters in the 1970s discovered that that galaxy morphology significantly depends on environment (Oemler 1974, Davis & Geller 1976, Melnick & Sargent 1977). The classic work by Dressler (1980) revealed a nearly universal morphology-density $T - \Sigma$ relation (where T is the classic galaxy morphology derived from optical images and Σ is the galaxy number density), in which the fraction of spiral galaxies systematically decreases with increasing projected galaxy number density. This general trend was subsequently confirmed with samples of ever-increasing size, reaching a peak with the Galaxy Zoo sample of 10^5 galaxies (Bamford et al. 2009, Skibba et al. 2009). The trend was found to hold over extended density ranges (Postman & Geller 1984, Giovanelli et al. 1986).

The $T - \Sigma$ relation is partly driven by the fact that galaxies become more massive in dense environments (Kauffmann et al. 2004), and galaxy mass drives galaxy properties. However, the $T - \Sigma$ relation is also clearly detectable at fixed mass (left panel of **Figure 25**, Bamford et al. 2009, Smith et al. 2012b), and this shows that purely environmental effects must play a role. In fact, Peng et al. (2010) convincingly argued that the effects of mass and environment are separable and they act independently in transforming galaxies, and this naturally explains the existence of the Schechter (1976) luminosity function.

Studies of the $T - \Sigma$ relation later extended to larger redshift, thanks to the resolving power of HST, which allowed for morphological classification to be performed initially to $z \lesssim 0.5$ (e.g., Dressler et al. 1997, Fasano et al. 2000, Treu et al. 2003, Wilman et al. 2009) and later out to $z \lesssim 1$ (e.g., Stanford et al. 1998, van Dokkum et al. 2000, Smith et al. 2005, Postman et al. 2005, Cooper et al. 2006, Capak et al. 2007, Poggianti et al. 2008). These studies have found that the fraction of S0 galaxies in clusters decreases with redshift, whereas the fraction of spiral galaxies correspondingly increases. The trend is much reduced in the field (Dressler et al. 1997, Fasano et al. 2000, Postman et al. 2005, Smith et al. 2005). These results indicate that spiral galaxies become passive and transform into S0s because of cluster environmental effects (Boselli & Gavazzi 2006). The morphological evolution is less significant at the largest galaxy masses, which indicates that these massive ellipticals were already passive from larger redshift and remain so during their evolution (Stanford et al. 1998, Postman et al. 2005, Tasca et al. 2009), while most of the observed evolution consists of a transformation of spirals into S0s (Smith et al. 2005, Moran et al. 2007).

Motivated by the results reviewed in Section 3, Cappellari et al. (2011b) introduced the kinematic morphology-density relation $kT - \Sigma$, which uses the physically robust fast/slow rotator classification to replace the strongly inclination-dependent S0/E morphological classification. The

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

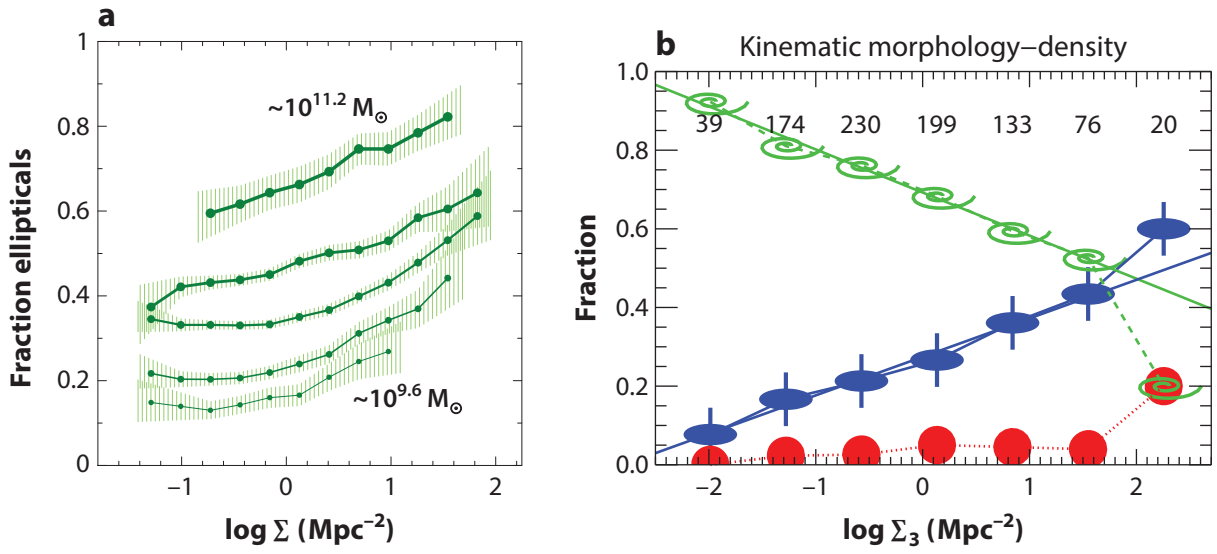


Figure 25

Photometric versus kinematic morphology-density relation. (*a*) The $T - \Sigma$ relation, where T is the classic galaxy morphology derived from optical images and Σ is the local galaxy number density. The ellipticals fraction is plotted against Σ for galaxies in selected narrow bins of stellar mass. Adapted with permission from Bamford et al. (2009). (*b*) The $kT - \Sigma$ relation for fast rotators (*blue ellipses*), slow rotators (*red circles*), and spiral galaxies (*green spirals*), where kT is the kinematic morphology. The numbers above the symbols indicate the total number of galaxies in each density bin. This panel was adapted from Cappellari et al. (2011b) by including the $\varepsilon < 0.4$ criterion for slow rotators (Equation 19). A key difference between the environmental trends using morphological or kinematic classifications is that in the former case, even at the lowest densities, there is always a fraction $\gtrsim$10% of ETGs with stellar disks misclassified as ellipticals galaxies; however, in the latter case, the slow rotators are virtually absent in the lowest density environments, with a slow rotator fraction of $\lesssim$2%.

key differences (**Figure 25**, right panel) between the kinematic and morphological relations are the following:

1. The genuinely spheroidal systems, the slow rotators that would be classified as E from any direction, are essentially absent in the field. Considering the lowest three density bins out of a statistically significant sample of 443 galaxies, only 2% are slow rotators. Slow rotators only play a significant role at the largest environmental density, which corresponds to the center of cluster and groups. This contrasts with the fraction $\gtrsim$10% of (misclassified) ellipticals, which is inferred from morphological classification even at the lowest densities (e.g., Dressler 1980, left panel of **Figure 25**). Given the small-number statistics of slow rotators, it is revealing to see their actual distribution within clusters. **Figure 26** visualizes the clear tendency for slow rotators to be found either at the cluster center or within local overdensities, owing to the infalling subcluster being assembled into the main cluster. This tendency has been seen in most clusters for which IFS data exist and where well-defined overdensities are present (Cappellari et al. 2011b, Cappellari 2013, D'Eugenio et al. 2013, Houghton et al. 2013, Fogarty et al. 2014, Scott et al. 2014). It explains why the $T - \Sigma$ relation seems to work better than the $T - R$ relation, which uses the cluster-centric radius (Dressler et al. 1997).
2. Focusing only on the ETGs subset, the ratio $f(SR)/f(FR)$ between slow and fast rotators is not a simple or monotonic function of environment. Instead, the ratio is nearly constant

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

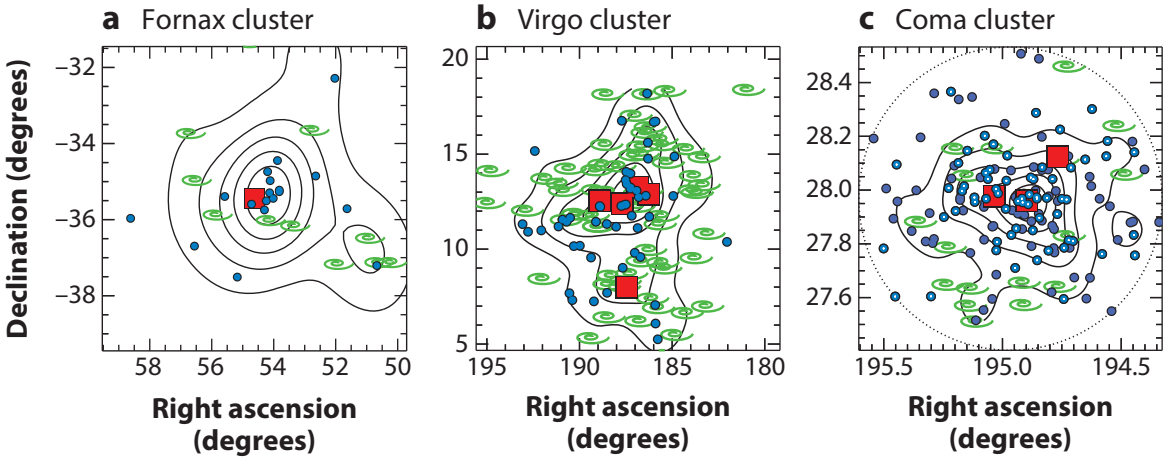


Figure 26

Core slow rotators in clusters. (*a*) Distribution of galaxies in the Fornax cluster of galaxies (Abell S0373). The red squares are the slow rotators with cores, the blue circles are the rest of the early-type galaxies, and the green spirals are the spiral galaxies. A kernel density estimate for the galaxy distribution is overlaid with linearly spaced contours. This panel was adapted with permission from Scott et al. (2014), with the main difference that the slow rotator NGC 1427 is not plotted in red because it is coreless (Lauer et al. 2007b). The only core slow rotator, NGC 1399, sits near the peak of the cluster density. (*b*) Same as panel *a*, for the Virgo cluster. Four core slow rotators are found very near the center of the main cluster A, and one (M49) is at the center the infalling subcluster B to the south of the center. (*c*) Same as panel *a*, for the Coma cluster (Abell 1656). The cluster is dominated by two massive core slow rotators near the peak density, and another lies along a slight overdensity. Panels *b* and *c* adapted with permission from Cappellari (2013). The samples in all three clusters are selected to be accurately complete down to $M_{K_s} < -21.5$, the same luminosity as the ATLAS^{3D} sample.

outside clusters, whereas inside clusters it becomes a strong function of the local galaxy number density (Cappellari et al. 2011b, D'Eugenio et al. 2013, Houghton et al. 2013, Fogarty et al. 2014). This shows that in the field, slow rotators are produced by random events at an extremely low rate, whereas clusters are the natural place for the presence of slow rotators. They must form via a different process, which must be linked to the formation of the clusters themselves.

The study of the $kT - \Sigma$ relation has just recently started. The Virgo (Cappellari et al. 2011b) and Fornax (Scott et al. 2014) clusters are the only two that have currently been mapped by IFS observations with good completeness. With the currently small and generally incomplete samples, a few misclassified slow rotators (e.g., counter-rotating disks, recent mergers, or galaxies with uncertain kinematics) can still significantly affect the conclusions. Moreover, ideally one would like to be able to recognize the genuine dry merger relics: slow rotators with cores. But this is currently only possible with HST, and not much farther than the Coma cluster (100 Mpc). The situation is going to be revolutionized in the near future by IFS observations, thanks to the two large SAMI (Bryant et al. 2015) and MaNGA (Bundy et al. 2015) IFS surveys, which are expected to bring the study of the local $kT - \Sigma$ relation up to the level of the $T - \Sigma$ one, in terms of number statistics and completeness.

5.2. Mass-Size Versus Environment

Recently, the $M - R_e$ relation (Section 4.3) has become one of the most popular observational constraints on galaxy formation models. This is because the variation of galaxy sizes during the

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

hierarchical mass assembly of galaxies depends sensitively on the details of the process by which the assembly occurs (e.g., Nipoti et al. 2003, Bezanson et al. 2009, Naab et al. 2009, Hopkins et al. 2010). For example, using simple virial equilibrium arguments (Hausman & Ostriker 1978), one can show that during a gas-free merger of identical spherical galaxies on parabolic orbits, which is the most likely situation (Khochfar & Burkert 2006), the radius grows proportionally to the mass, and the galaxy stellar velocity dispersion (integrated over the whole galaxy) remains unchanged (White 1983, Barnes 1992). However, when the mass growth happens via very small gas-poor satellites, then R is predicted to increase as the square of the mass increase, whereas the σ decreases as the square root of the mass increase (Bezanson et al. 2009, Naab et al. 2009). These predictions properly capture the trends actually observed in detailed numerical simulations (Barnes 1992, Hernquist et al. 1993, Nipoti et al. 2003, Boylan-Kolchin et al. 2006, Naab et al. 2009).

In contrast, when gas is present, it dissipates energy, falls toward the center, and forms stars in a compact starburst (Mihos & Hernquist 1994). Owing to the shrinkage caused by the gas infall, the size of the remnant will increase more slowly than in the collisionless case, by an amount that is related to the amount of dissipation (e.g., Dekel & Cox 2006, Khochfar & Silk 2006). The galaxy center will be denser and will contain a steep inner profile (cusp or extra light) as observed in many fast rotators (Kormendy et al. 2009).

To test these predictions one would ideally like to be able to follow the redshift evolution of fast rotators and core slow rotators on the (M, R_e) plane. This is unfortunately not yet feasible with the current generation of instruments. However, examining the environmental dependence is a good proxy for studying the redshift evolution. This is because at high redshift, the abundance of massive halos declines and fewer galaxies live in clusters.

The environmental evolution of fast and slow rotators on the (M, R_e) plane was investigated by Cappellari (2013), using two extreme environments for which IFS data were available. The low-density sample was taken from the field/group environment of the ATLAS^{3D} sample, defined as not belonging to the Virgo cluster. The high-density sample used IFS data by Houghton et al. (2013) on the Coma cluster, which is the densest environment for which resolved spectroscopic observations can currently be taken. It has one of the largest, and carefully determined, dark halo virial masses ($M_{200} \approx 1.4 \times 10^{15}$ $M_\odot$; Łokas & Mamon 2003) that is expected to be found in the whole Universe (e.g., Springel et al. 2005). Both samples were carefully selected to be nearly 100% complete to $M_{K_s} < -21.5$ ($M_* \gtrsim 6 \times 10^9$ $M_\odot$) and have fully homogeneous size and luminosity determinations from 2MASS (Skrutskie et al. 2006).

The study found that the mass-size distribution in the dense environment differs from the field/group one in two ways: (*a*) Spiral galaxies are replaced by fast-rotator ETGs, which follow the same mass-size relation and have the same mass distribution as in the field sample, and (*b*) the core slow-rotator ETGs are segregated in mass from the fast rotators, with their size increasing proportionally to their mass. A transition between the two processes appears around the stellar mass $M_{\rm crit} \approx 2 \times 10^{11}$ $M_\odot$. This is illustrated in **Figure 21**, which also includes the distribution of core slow rotators and fast rotators in the Virgo cluster, which has lower density than Coma and a halo virial mass more than three times smaller of $M_{200} \approx 4 \times 10^{14}$ $M_\odot$ (McLaughlin 1999). The distribution of galaxies on the Virgo (M, R_e) diagram is intermediate between that on the field/groups and Coma diagrams. The core slow rotators have slightly larger masses than the field/groups ones and 4/5 lie above $M_{\rm crit}$.

The environmental dependency of the distribution of spiral galaxies, fast rotators, and core slow rotators was interpreted as direct evidence for the two-channel buildup of ETGs inferred from the distribution of galaxy properties in Section 4.3. One channel, the bulge-growth and quenching route, is illustrated by the fact that spiral galaxies are gradually replaced by fast rotators, with

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

smaller sizes and larger σ_e. The latter have the same properties as those in the field but are simply more numerous in the cluster. The other channel, the merger-growth route, is illustrated by the mass increase of the core slow rotators, with M_* increasing proportionally to R_e, broadly following the dry-merging model prediction.

The impressive universality of the mass-size relation for $M_* \lesssim M_{crit}$ in these two extreme environments is in excellent agreement with other studies which used larger samples but lack IFS kinematic information. Differences between the sizes of ETGs in different environments are consistently found to be $\lesssim$10%, which is the level of possible systematic effects in the size determinations or sample selections (Maltby et al. 2010, Huertas-Company et al. 2013, Fernández Lorenzo et al. 2013, Poggianti et al. 2013, Cebrián & Trujillo 2014).

The only significant difference between the low-density and high-density environments was found to be a slight asymmetry of the mass-size distribution. The peak is at the same location in the two environments, but there is a tail toward ETGs with larger sizes in clusters (Cappellari 2013, Delaye et al. 2014). In Coma, where galaxies are well resolved, this appear to be due to the well-known excess of passive disks in clusters (van den Bergh 1976, Wolf et al. 2009, Masters et al. 2010).

6. REDSHIFT EVOLUTION

Only the line emission from the gas-rich star-forming galaxies can be currently spatially resolved with IFS at significant redshift (e.g., Wisnioski et al. 2015, Stott et al. 2016). This is because at $z \gtrsim 1.3$, most key spectral features are redshifted to the near infrared, where the strong contribution of the OH atmospheric lines dramatically degrades the data quality. IFS observations of the stellar continuum in ETGs are essentially nonexistent at these redshifts. For this reason this section will only include an brief overview of some of the key finding that closely relate to what has been discovered with IFS observations of nearby galaxies. A comprehensive review of the redshift evolution of galaxy structure was recently provided by Conselice (2014), and the evolution of stellar population with redshift was reviewed by Renzini (2006).

6.1. Mass-Size Evolution

The stellar population of nearby ETGs, the scatter of the color-magnitude relation in clusters, and the redshift evolution of their FP clearly indicate that ETGs formed their stars at $z \gtrsim 2$ (Thomas et al. 2005; see review by Renzini 2006). When samples of spectroscopically confirmed passive galaxies beyond $z \gtrsim 1.4$ were finally discovered, they appeared to have large masses of $M_* \gtrsim 10^{11}$ $M_\odot$, comparable to some of the most massive local ETGs (Cimatti et al. 2004, 2006; Glazebrook et al. 2004). However, their sizes were found to be significantly smaller than their local counterparts (Daddi et al. 2005; Trujillo et al. 2006a,b, 2007; Zirm et al. 2007; Toft et al. 2007; Longhetti et al. 2007; Cimatti et al. 2008; van Dokkum et al. 2008).

Initial concerns about the possible effect of unresolved nuclear emission from active galactic nuclei (AGNs), strong stellar population gradients, or inaccurate mass estimates were cleared out via spectroscopic determinations of the galaxies' velocity dispersions, which showed values broadly consistent (but with nonnegligible size bias) with the virial predictions inferred from the given masses and radii (Cappellari et al. 2009; Cenarro & Trujillo 2009; Onodera et al. 2010; van de Sande et al. 2011, 2013; Toft et al. 2012; Bezanson et al. 2013; Belli et al. 2014).

It now appears well established that the mean size of ETGs grows significantly from $z \sim 3$ to the present day, whereas spiral galaxies grow at a smaller rate. The precise amount of this evolution depends sensitively on how ETGs are defined, given the strong dependence of galaxy size on their

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

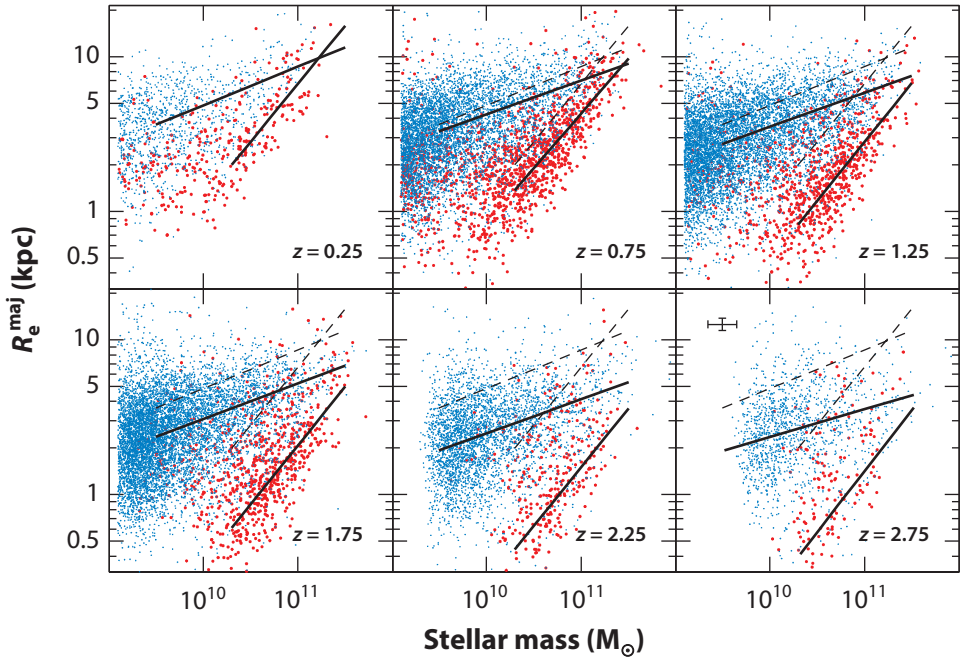


Figure 27

Evolution of the mass-size distribution. (M_*, R_e) distribution of color-selected star-forming and quiescent galaxies. The lines indicate model fits to the passive and star-forming galaxies. The dashed lines, which are identical in each panel, represent the model fits to the galaxies in the lowest redshift bin. The solid lines represent fits to the higher-redshift samples. Strong evolution in the intercept of the mass-size relation is seen for passive galaxies and moderate evolution for the star forming ones. Adapted with permission from van der Wel et al. (2014).

properties such as color, star formation, Sérsic index, or morphology (**Figure 22**, Newman et al. 2012, Cappellari et al. 2013a, Poggianti et al. 2013). The largest and most homogeneous study is currently the analysis of CANDELS (the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey) with extensive multiband HST data (Grogin et al. 2011, Koekemoer et al. 2011) by van der Wel et al. (2014), reproduced in **Figure 27**. The slope of the $M_* - R_e$ relation for ETGs with $M_* \gtrsim 3 \times 10^{10}$ is found to be consistent with the slope $M_* \propto R_e^{0.75}$ of the ZOE (Equation 25) over the whole redshift range $0 \lesssim z \lesssim 3$. Also apparent at all redshifts is break in the overall mass-size, owing to the minimum size of ETGs around a few 10^{10} $M_\odot$, observed locally (see **Figure 23**).

The evolution of the distribution of galaxies on the (M_*, R_e) plane was interpreted by van Dokkum et al. (2015) using a simple statistical model that tries to capture the dominant mode of growth. They concluded that the population of ETG progenitors likely follows two main simple evolutionary tracks in the (M_*, R_e) plane. The first is a shallow $\Delta \log R_e \sim 0.3 \Delta \log M_*$ growth dominated by gas accretion. Along this track, the originally gas-rich and star-forming galaxies become denser, increasing their σ, until they reach a threshold at which their stars are quenched. The second is a steeper track dominated by (mainly dry) mergers, where galaxy size increases proportionally to mass. An independent analysis of the CANDELS data reaches quite

similar conclusions and also emphasizes the need for two similar formation channels for ETGs (Huertas-Company et al. 2015).

Although the detailed mechanism is still actively debated, a consensus is emerging, based on high-redshift observations, that galaxy quenching is linked to the growth of the galaxies' central density, quantified using either photometrically predicted σ (Franx et al. 2008, Bell et al. 2012, van Dokkum et al. 2015) or the stellar density Σ_1 within an aperture of radius $R = 1$ kpc (Cheung et al. 2012, Barro et al. 2015). Importantly, central density was found to be a much better predictor of quenching than other galaxy parameters, in particular galaxy mass. These findings are fully consistent with the link between bulge growth and quenching from IFS observations of local ETGs, reviewed in Section 4.3.

The two evolutionary channels described by van Dokkum et al. (2015) and Huertas-Company et al. (2015) from direct observations of the (M_*, R_e) redshift evolution and galaxy morphology were also proposed to explain galaxy properties on the (M_*, R_e) plane using IFS observations in Section 4.3 and to understand the observed environmental dependency in Section 5.2. The remarkable agreement between these two sets of independent observations, based either on the fossil record in nearby galaxies or on redshift evolution, strengthens the conclusions of both complementary approaches.

6.2. Profile Evolution

Using observations at different redshifts, one can directly trace the variation of the Sérsic indices as a function of time. This analysis was performed for the progenitors of ETGs with present-day masses $M_* \approx 3 \times 10^{11}$ $M_\odot$ by van Dokkum et al. (2010), followed by a similar analysis of progenitors with masses $M_* \approx 5 \times 10^{10}$ $M_\odot$, by van Dokkum et al. (2013). The first set of galaxies have present-day masses above the critical mass $M_{crit} \approx 2 \times 10^{11}$ $M_\odot$, dominated by the core slow rotators, whereas the second set have present-day masses near the break $M_b \approx 3 \times 10^{10}$ where spiral galaxies and fast rotators overlap and slow rotators are absent.

The results, summarized in **Figure 28**, indicate that the progenitors of slow rotators grow rapidly in size, whereas those of fast rotators remain almost unchanged from $z \sim 2$. Similarly, the progenitors of slow rotators change little in mass from $z \sim 2$, whereas the progenitors of fast rotators still grow significantly (Muzzin et al. 2013). This mass and size growth difference is accompanied by a strikingly different behavior of the Sérsic indices: Both classes of galaxies start with disk-like $n \sim 1$, but the slow rotators progenitors rapidly increase their concentration and reach $n \gtrsim 4$ near $z \sim 0$, whereas the fast rotators progenitors maintain a more constant Sérsic index, with $z \sim 0$ values around 2–3, which is consistent with the typical value for the fast rotators in the ATLAS^{3D} sample (Krajnović et al. 2013a).

7. IMPLICATIONS FOR GALAXY FORMATION

The aim of this review is to provide an overview of the empirical signatures of galaxy formation in ETGs, as they were obtained mainly by IFS observations. Here I sketch ideas on galaxy formation driven by these observations. A detailed review of the theoretical models of the formation of ETGs is given by Somerville & Davé (2015) and T. Naab & J.P. Ostriker (manuscript in preparation). Theoretical studies trying to specifically address the formation of the fast- and slow-rotator ETGs classes revealed by IFS observations were presented by Bois et al. (2011), Khochfar et al. (2011), and Naab et al. (2014).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

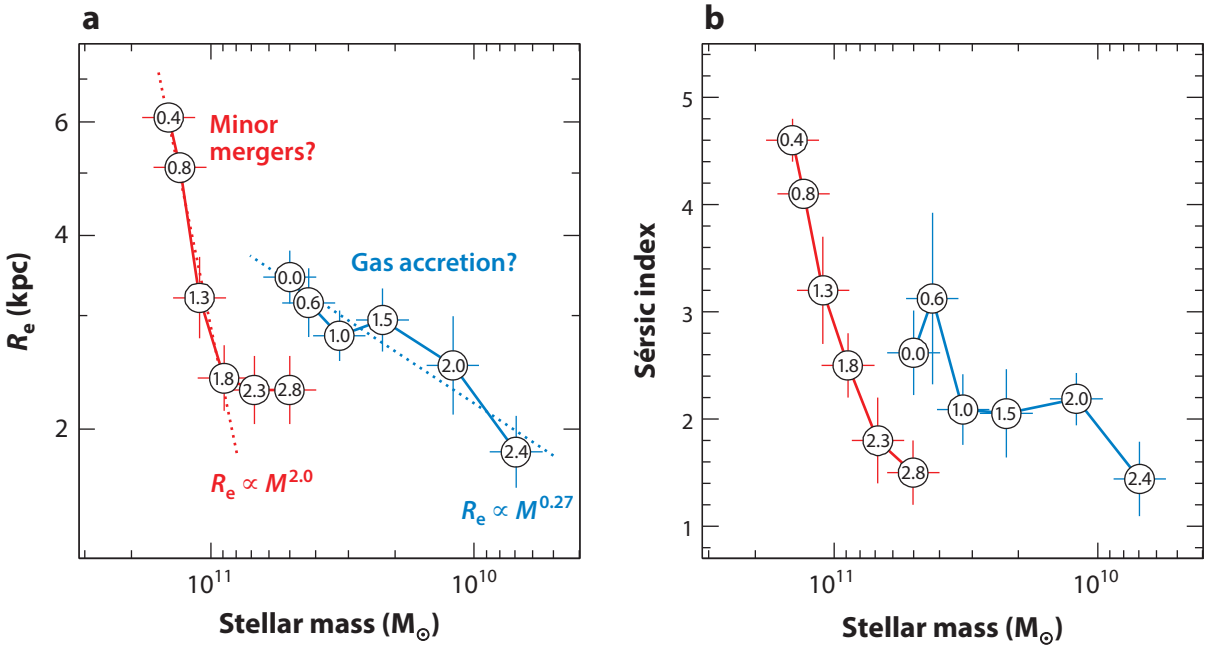


Figure 28

Size and profile evolution. Effective radius R_e and Sérsic index as a function of redshift and stellar mass M_*. The numbers inside the circles give the redshift. The blue line is for the progenitors of galaxies with present-day masses of $M_* \approx 5 \times 10^{10}$ $M_\odot$, and the red line is for present-day masses of $M_* \approx 3 \times 10^{11}$ $M_\odot$. The less-massive galaxies have undergone much less structural evolution than the present-day giant elliptical galaxies that populate the high-mass end of the mass function. Adapted with permission from van Dokkum et al. (2013).

7.1. Galaxy Evolution on the (M_*, R_e) Plane

From both IFS and high-redshift results, one can consistently describe the buildup of ETGs on the (M_*, R_e) diagram as in **Figure 29**. The average growth of the overall ETG population can be summarized as happening through two main channels. The first one is dominated by gas accretion. Spiral galaxies accrete gas or experience gas-rich minor mergers. The gas sinks toward the center, increases the galaxy mass, and at the same time builds a bulge. The presence of a bulge increases the likelihood of the galaxy having its star formation quenched. This makes the progenitor spiral end up as a passive fast-rotator ETG with a stellar disk and, on average, a larger bulge fraction. As a result of this first channel, all galaxy properties related to the star formation history vary, on average, along lines of nearly constant σ_e. The fact that at fixed mass, dense bulges have α-enhanced stellar population (**Figure 22**), indicates that bulge formation must be a rapid process, as expected during intense starbursts. The gas accretion and subsequent star formation leave the metallicity enhancements of the disks in fast rotators as a fossil record (**Figure 16**).

The second channel on the (M_*, R_e) diagram is dominated by mostly dry mergers, which move galaxies by increasing their sizes roughly proportionally to their masses, while leaving σ_e nearly unchanged. During dry mergers, the stellar population evolves passively and changes little for old systems. For this reason no significant trend is observed in galaxy properties related to the stellar population along this second channel. However, dry mergers leave an imprint in the nuclear profiles when supermassive BHs of the merging galaxies sink toward the center via dynamical

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

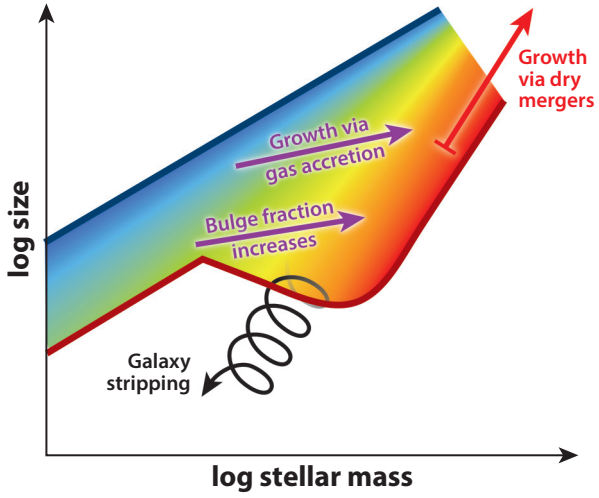


Figure 29

Galaxy evolution on the mass-size plane. Although the evolution of an individual galaxy is a complex combination of events, the observations indicate that the average evolution of ensembles of galaxies can be described by the following simple picture. The progenitors of fast-rotator early-type galaxies are star-forming disks. They grow their mass, while slowly increasing their sizes, predominantly by gas accretion (*purple lines*). During this accretion they grow a bulge, which increases the likelihood of the galaxy having its star formation quenched. They end up as bulge-dominated and passive fast rotators near the zone of exclusion (ZOE). Slow rotators build up most of their mass rapidly at high redshift and subsequently grow mostly by gas-poor (dry) merging, with their size growing at a rate proportional to their mass (*red line*). They end up in the top right of this diagram. The region below the ZOE is not empty: It includes ultra-compact dwarfs, which are the likely cores of normal galaxies that had their envelopes stripped by the gravitational field of larger galaxies. Combines elements from figure 15 of Cappellari et al. (2013a) and figure 28 of van Dokkum et al. (2015).

friction and eject stars in radial orbits, scouring a nuclear core or deficit (e.g., Faber et al. 1997, Milosavljević & Merritt 2001, Kormendy & Ho 2013).

Observations indicate that fast and slow rotators predominantly follow either one or the other channel. A key piece of information is provided by the environmental distribution (Section 5.1). If slow rotators formed by dry merging of fast rotators, one would expect to observed the following: (*a*) They would roughly follow their distribution, like fast rotators follow the spiral distribution. Instead, their distribution in clusters is very different, with the slow rotators near the cluster/group centers and the fast rotators following spiral galaxies (**Figure 26**). (*b*) Moreover, one would find core slow rotators along the whole sequence of passive ETGs, starting from $M_* \gtrsim 3 \times 10^{10}$ $M_\odot$, as they are still building up their mass, especially in the field. Instead, they suddenly appear only above $M_* \gtrsim 10^{11}$ $M_\odot$ (**Figure 21**).

I mention in passing that the region of the (M_*, R_e) diagram below the ZOE for $M_* \lesssim 3 \times 10^{10}$ $M_\odot$ is not actually devoid of stellar systems. It is populated by objects generally called dwarf ellipticals (dEs) and ultra-compact dwarfs (UCDs) (e.g., Misgeld & Hilker 2011, Norris et al. 2014). Evidence suggests they may be normal ETGs that fell well into the halos of larger galaxies and had their outer stellar envelopes stripped (e.g., Drinkwater et al. 2003). Integral field observations of the kinematics and population of these galaxies have just started and appear to confirm this interpretation (Ryś et al. 2014, Guérou et al. 2015). According to this picture, dEs and UCDs do not follow the main route of galaxy formation. However, I do not further discuss this interesting topic here.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

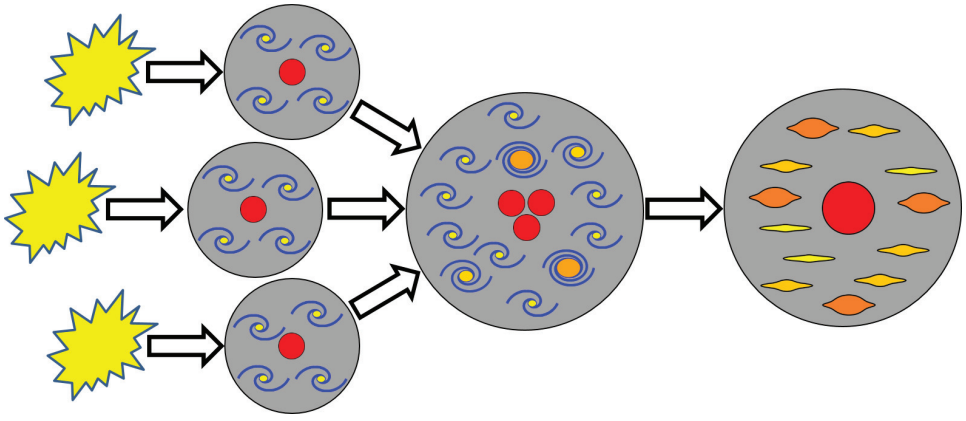


Figure 30

Hierarchical origin of fast and slow rotators. The progenitors of core slow rotators form in the high-redshift universe at the center of the largest dark matter overdensities. They quickly grow above the critical mass required for the gas to be heated by shocks or by feedback from active galactic nuclei. They become surrounded by a hot gaseous envelope that prevents further accretion. When groups merge to form massive clusters, slow rotators sink toward the center, where they merge to form more massive slow rotators. The same cannot happen to fast rotators, which have masses too small to efficiently sink to the center and velocities too fast to merge. They are quenched by the cluster and from then on their mass does not increase, but they are affected by tidal effects. Not included here is internal quenching, which appears associated with bulge growth and can act independently of environment. Symbols are as in **Figures 23** and **24**.

7.2. Hierarchical Origins of Fast- and Slow-Rotator Early-Type Galaxies

To understand how fast and slow rotators can follow separate evolutionary channels, one needs to consider the hierarchical growth of galaxies and clusters (e.g., De Lucia et al. 2012, Wilman & Erwin 2012). According to our current understanding of galaxy formation (e.g., Mo et al. 2010), after the Big Bang, the primordial dark matter distribution is imprinted with small fluctuations, and these inhomogeneities grow by gravitational instability to form dark matter halos. The primordial gas falls into a nearby halo and loses energy by radiating some of its energy. For this reason it sinks toward the halo center and forms a rotating disk. When the gas becomes sufficiently denser that the halo, it collapses into small clumps owing to radiative cooling and starts producing stars in stellar clusters.

The largest dark matter overdensities are able to acquire large amount of gas early on, when they are still at the center of their own halos and dominate the overall gravitational potential. They quickly reach the critical mass above which the infalling gas is shock-heated by the halo gravitational field (Kereš et al. 2005, Dekel & Birnboim 2006) and AGN feedback becomes effective (e.g., Kormendy & Ho 2013), and their star formation is suppressed. These massive galaxies are the progenitors of the slow rotators. The hot gas is actually observed in X-ray in local massive boxy ellipticals (Bender et al. 1989), in core galaxies (Kormendy et al. 2009), and in slow rotators in particular (Sarzi et al. 2013). When two groups merge during the hierarchical buildup of galaxies and clusters, the central galaxies in the two groups, which sit at rest near the center of mass of the halo, will efficiently sink to the center of the resulting larger cluster or group of galaxies via dynamical friction. These massive slow rotators will efficiently merge, owing to their large mass and small relative velocities, forming a more massive slow rotator with an inner core scoured by the resulting BH binary. In this way, a typical slow rotator is able to remain as such for the rest of its evolution (**Figure 30**).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

This picture is similar to the one that motivates the separation of central (i.e., the most massive) and satellite (i.e., the rest) galaxies in dark matter halos, when building theoretical descriptions of galaxy and halo properties (e.g., Zehavi et al. 2005, Zheng et al. 2005, Conroy et al. 2006). However, in the theoretical approaches every halo has a single central galaxy by definition, whereas in reality, a given cluster or a given massive dark matter halo may contain a handful of slow rotators, which were central galaxies of their respective halos when they formed but have yet to merge into a single object. For this reason the correspondence between central galaxies and slow rotators is only approximately correct.

The regions of lower dark matter density are unable to efficiently acquire gas and therefore grow more slowly and gradually. They form disk galaxies, which at high redshift are observed to have high gas fractions (Daddi et al. 2010, Tacconi et al. 2010), have large velocity dispersions, be turbulent compared with their local counterparts (Förster Schreiber et al. 2006, 2009; Genzel et al. 2006, 2008; Law et al. 2012; Kassin et al. 2012; Stott et al. 2016), and appear unstable to clump formation (Elmegreen et al. 2007, Genzel et al. 2011). These clumps may sink toward the center and start forming the bulge (Bournaud et al. 2007, Dekel et al. 2009), or the gas may simply spiral toward the nucleus to grow the bulge in starbursts. During the initial stages, when the velocity dispersion of their groups is quite small, gas-rich mergers can also happen. Isolated spirals are able to sustain nearly constant star formation for a long time, possibly with the contribution of the so called cosmological fountain effect (Fraternali & Binney 2008, Fraternali 2014). During this phase they lie on the so-called star-formation main sequence (Brinchmann et al. 2004, Salim et al. 2007).

However, the bulge growth (and σ_e or Σ_1 rise) increases the likelihood that the galaxy will have its star formation quenched. Some form of feedback is observed to stop the star formation and make the galaxy passive. The link between bulge fraction (or central mass density) and quenching is currently unclear. Outflows from supernovae or from a central AGN (for reviews, see Fabian 2012, Kormendy & Ho 2013, Heckman & Best 2014) must both play a role (Silk & Rees 1998, Granato et al. 2004, Bower et al. 2006, Croton et al. 2006, Hopkins et al. 2006), as may the stabilizing effect of the bulge itself (Martig et al. 2009, Ceverino et al. 2010), which may contribute to the observed decrease in the star formation efficiency of ETGs (Saintonge et al. 2012, Martig et al. 2013, Davis et al. 2014). Evidence of feedback in action within a field fast rotator was presented by Alatalo et al. (2011). These forms of internal feedback are needed to produce fast rotators in the field.

But environment has a more dramatic effect. In fact, as soon as one of these disk galaxies is acquired by a large halo, the galaxy's gravitational pull will be unable to acquire more gas, because the small halo will be orbiting at high speed within the ambient gas, which is at rest within the main halo. Moreover, at sufficiently high gas densities, their own gas, including their surrounding corona of hot gas and the resulting fountain effect (Fraternali & Binney 2008), will be stripped entirely (Boselli & Gavazzi 2006). As a result, when disk galaxies enter sufficiently massive clusters, they stop growing, because of both the lack of gas accretion and the inability to merge because of the high relative velocities of satellites. The environment will now only act via tidal perturbations due to the high-speed encounters, puffing up their disks (e.g., Moore et al. 1996). From then on, the galaxies will be fast-rotator ETGs. Their masses will stop growing during subsequent mergers of the parent group/cluster, while slow rotators will continue to grow in mass following the first dry-merging channel. This additional growth of slow rotators during the merging of clusters/groups tends to produce a gap between the mass distributions of the fast and slow rotators. It explains the observation that there is generally a gap in luminosity between the brightest galaxy in a massive cluster and the subsequently ranked galaxies (Sandage & Hardy 1973, Lauer et al. 2014), also clearly visible in **Figure 21** (*right panel*).

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

This picture is broadly consistent will all observational evidence, but nature is certainly not that simple. For example, central galaxies in clusters are not always passive (Liu et al. 2012), but the small fraction of star forming ones is consistent with the general picture. Moreover, the channel from star-forming spirals to bulge-dominated passive fast rotators does not always progress monotonically in one direction. Rejuvenation events can happen in ETGs (Kaviraj et al. 2007) and specifically in fast rotators (Young et al. 2014). In addition, the misalignment of the gas with respect to the stars indicates external accretion (Sarzi et al. 2006, Davis et al. 2011). Small fractions (relative to the stellar masses) of neutral gas are detected in 40% of ETGs in the field, especially at large radii (Morganti et al. 2006, Oosterloo et al. 2010, Serra et al. 2012). And tidal tails are also not uncommon in fast rotators (Duc et al. 2011, 2015). Finally, all trends illustrated in **Figure 22** have significant scatter, but the incidence of these events is fully consistent with the general picture I describe. In particular, the IFS observations allow one to exclude a scenario in which passive fast rotators can acquire major amounts of gas and become star-forming spirals again. These very bulge-dominated spirals are extremely rare (e.g., the Sombrero galaxy M104). Moreover, if these events were common, they would erase the clear empirical inverse dependency between bulge fraction (or σ_e) and star-formation indicators. Similarly one can exclude a scenario in which the slow rotators acquire a disk and become fast rotators. This is because the dynamics of bulges in fast rotators is well described by the simple axisymmetric JAM models over the whole mass range. This contrasts with the strikingly different dynamics and shape of slow rotators, as revealed by the IFS data.

8. CONCLUSIONS

This review was written at an ideal time, when studies of ETGs using the first generation of IFS surveys (SAURON, de Zeeuw et al. 2002; ATLAS^{3D}, Cappellari et al. 2011a; CALIFA, Sánchez et al. 2012), those targeting one galaxy at a time, were nearly complete. The second generation of IFS surveys (SAMI, Bryant et al. 2015; MaNGA, Bundy et al. 2015) just started but are actively acquiring data. A limited preview of the SAMI results is included in this review. At the same time, the new MUSE IFS (Bacon et al. 2010), on the 8.2-m Very Large Telescope of ESO (the European Southern Observatory), has started pushing the envelope of the data quality one can achieve on nearby galaxies (Emsellem et al. 2014, Krajnović et al. 2015).

For this reason, the goal of this review is to define the status of our knowledge of the structure, kinematics, and scaling relations of ETGs before the arrival of these new large IFS surveys. I aim to set a benchmark for assessing how much we have progressed.

Galaxy evolution can be studied via detailed observations in the nearby Universe or by using evolutionary studies as a function of redshift. Too often the groups working in one field are unaware of the results of the other and miss the opportunity to combine the two sets of information to advance our knowledge. This review tries to emphasize and illustrate the importance of synergy between the two approaches.

In the near future, we will be able to combine our knowledge about ETGs from the upcoming multiplexed IFS survey with the advances that the James Webb Space Telescope (Gardner et al. 2006) will bring, thanks to its ability to obtain deep near-infrared spectra, including IFS (not multiplexed), which targets the rest-frame optical spectra of ETGs free of atmospheric absorption. This will provide clean kinematics and stellar population observations for ETGs out to the peak of their assembly epoch around $z \sim 2$.

In the more distant future, with the upcoming class of 40-m telescopes, such as the European Extremely Large Telescope (E-ELT) and the Thirty Meter Telescope (TMT), coupled with multiplexed capabilities in the near-infrared it will become possible to perform IFS surveys similar

to the ones described here, out to $z \sim 2$. We can only look forward to witnessing these new developments.

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

SUMMARY POINTS

1. ETGs can be separated, using IFS kinematics, into two structurally homogeneous classes with and without disks: the fast and slow rotators, respectively. I show that there is a dichotomy, not a smooth transition, between the two classes.
2. This dichotomy broadly agrees with the previously identified photometric separation into, e.g., core versus power-law ETGs. But the kinematic classification is nearly independent of inclination and does not require high spatial resolution. This makes it ideal for large IFS surveys.
3. IFS revealed a close link between ETGs and spirals. Below $M_{\rm crit} \approx 2 \times 10^{11}$ $M_\odot$, fast-rotator ETGs form a parallel sequence in galaxy properties with spiral galaxies, whereas core slow rotators dominate above $M_{\rm crit}$.
4. In the spirals-to-fast-rotators sequence, suppression of star formation and the decrease of the molecular gas fraction are driven by the central mass density slope, or bulge mass fraction.
5. ETGs are dominated by stellar mass within R_e, and their $(M/L)_{\rm dyn}$ is mainly due to systematic variations in the stellar population, including the IMF.
6. The total mass density is well described by $\rho_{\rm tot} \propto r^{-2.2}$ from the center out to at least $4R_e$, with small scatter (in the currently explored mass range).
7. Slow rotators with cores are found near the centers of clusters/groups or subgroups within clusters. Fast rotators are distributed like spiral galaxies.
8. IFS observations and redshift evolution studies consistently and independently indicate a scenario in which the evolution of fast and slow rotators follows two distinct evolutionary channels: (*a*) gas-accretion driven and (*b*) dry-merger driven.

FUTURE ISSUES

1. What are the trends in population and kinematics at radii well beyond $R \gtrsim 2R_e$? Much information on galaxy assembly is contained in the stellar halos.
2. What is the physical mechanism making galaxies passive? Large IFS surveys of both spirals and ETGs can answer this long-standing question by combining spatially resolved gas and stellar observables with information about environment.
3. Do we understand stellar population and IMF in galaxies? Can we reliably predict stellar masses from spectra, or are there fundamental unsolved issues?
4. What are the trends in total density profiles at large radii where dark matter dominates? How do they relate to spirals and to detailed model predictions?
5. How do spatially resolved populations and kinematics evolve with redshift? Can we directly trace the assembly of mass and metals over time?

6. Can we follow the dark halo growth as a function of time using resolved IFS of the stars and gas dynamics?

DISCLOSURE STATEMENT

The author is not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

Many of the ideas presented in this review originated during more than a decade of enjoyable and productive collaboration, initially with the SAURON team and subsequently with the ATLAS^{3D} collaboration. I acknowledge with gratitude the numerous meetings and enlightening discussions with the members of the two teams. I am most grateful to my Scientific Editor John Kormendy and co-editor Sandy Faber for fruitful discussions and comments. I have been fortunate to be able to benefit from their insights and unique expertise on this subject. I thank Production Editors Roselyn Lowe-Webb and Anne Abramson, Illustration Editor Alan Stonebraker, and the production team at Annual Reviews for their excellent handling of my paper and figures. It is a particular pleasure to thank Eric Emsellem, Davor Krajnović, and Tim de Zeeuw for thoughtful suggestions. I thank Jesús Falcón-Barroso for providing the CALIFA ($\lambda_{R_e}, \varepsilon$) values. I acknowledge support from a Royal Society University Research Fellowship. I warmly thank my wife Christi Warner for making this review possible with her constant support and understanding.

LITERATURE CITED

Aihara H, Allende Prieto C, An D, et al. 2011. *Astrophys. J. Suppl.* 193:29
Alatalo K, Blitz L, Young LM, et al. 2011. *Astrophys. J.* 735:88
Andredakis YC, Peletier RF, Balcells M. 1995. *Mon. Not. R. Astron. Soc.* 275:874
Arnold JA, Romanowsky AJ, Brodie JP, et al. 2014. *Astrophys. J.* 791:80
Auger MW, Treu T, Bolton AS, et al. 2010a. *Astrophys. J.* 724:511–25
Auger MW, Treu T, Gavazzi R, et al. 2010b. *Astrophys. J. Lett.* 721:L163–67
Bacon R, Accardo M, Adjali L, et al. 2010. In *Ground-based and Airborne Instrumentation for Astronomy III*, ed. IS McLean, SK Ramsay, H Takami. *Proc. SPIE Conf. Ser.* 7735
Bacon R, Adam G, Baranne A, et al. 1995. *Astron. Astrophys. Suppl.* 113:347–57
Bacon R, Copin Y, Monnet G, et al. 2001. *Mon. Not. R. Astron. Soc.* 326:23–35
Bamford SP, Nichol RC, Baldry IK, et al. 2009. *Mon. Not. R. Astron. Soc.* 393:1324–52
Barnabè M, Czoske O, Koopmans LVE, Treu T, Bolton AS. 2011. *Mon. Not. R. Astron. Soc.* 415:2215–32
Barnes JE. 1992. *Astrophys. J.* 393:484–507
Barro G, Faber SM, Koo DC, et al. 2015. arXiv:1509.00469 [astro-ph.GA]
Bastian N, Covey KR, Meyer MR. 2010. *Annu. Rev. Astron. Astrophys.* 48:339–89
Begeman KG. 1989. *Astron. Astrophys.* 223:47–60
Behroozi PS, Conroy C, Wechsler RH. 2010. *Astrophys. J.* 717:379–403
Bell EF, van der Wel A, Papovich C, et al. 2012. *Astrophys. J.* 753:167
Belli S, Newman AB, Ellis RS. 2014. *Astrophys. J.* 783:117
Bender R. 1988. *Astron. Astrophys.* 193:L7–10
Bender R, Burstein D, Faber SM. 1992. *Astrophys. J.* 399:462–77
Bender R, Doebereiner S, Moellenhoff C. 1988. *Astron. Astrophys. Suppl.* 74:385–426
Bender R, Saglia RP, Gerhard OE. 1994. *Mon. Not. R. Astron. Soc.* 269:785–813
Bender R, Surma P, Doebereiner S, Moellenhoff C, Madejsky R. 1989. *Astron. Astrophys.* 217:35–43

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Bendinelli O. 1991. *Astrophys. J.* 366:599–604
Bernardi M, Roche N, Shankar F, Sheth RK. 2011. *Mon. Not. R. Astron. Soc.* 412:L6–10
Bernardi M, Shankar F, Hyde JB, et al. 2010. *Mon. Not. R. Astron. Soc.* 404:2087–122
Bernardi M, Sheth RK, Annis J, et al. 2003. *Astron. J.* 125:1866–81
Bershady MA, Verheijen MAW, Swaters RA, et al. 2010. *Astrophys. J.* 716:198–233
Bertin G, Ciotti L, Del Principe M. 2002. *Astron. Astrophys.* 386:149–68
Bertola F, Capaccioli M. 1975. *Astrophys. J.* 200:439–45
Bezanson R, van Dokkum PG, Tal T, et al. 2009. *Astrophys. J.* 697:1290–98
Bezanson R, van Dokkum PG, van de Sande J, et al. 2013. *Astrophys. J. Lett.* 779:L21
Binggeli B, Sandage A, Tarenghi M. 1984. *Astron. J.* 89:64–82
Binney J. 1978. *Mon. Not. R. Astron. Soc.* 183:501–14
Binney J. 2005. *Mon. Not. R. Astron. Soc.* 363:937–42
Binney J, de Vaucouleurs G. 1981. *Mon. Not. R. Astron. Soc.* 194:679–91
Binney J, Tremaine S. 2008. *Galactic Dynamics: Second Edition*. Princeton, NJ: Princeton Univ. Press
Blanton MR, Moustakas J. 2009. *Annu. Rev. Astron. Astrophys.* 47:159–210
Blanton MR, Roweis S. 2007. *Astron. J.* 133:734–54
Blumenthal GR, Faber SM, Primack JR, Rees MJ. 1984. *Nature* 311:517–25
Bois M, Emsellem E, Bournaud F, et al. 2011. *Mon. Not. R. Astron. Soc.* 416:1654–79
Bolton AS, Burles S, Koopmans LVE, Treu T, Moustakas LA. 2006. *Astrophys. J.* 638:703–24
Bolton AS, Burles S, Treu T, Koopmans LVE, Moustakas LA. 2007. *Astrophys. J. Lett.* 665:L105–8
Borriello A, Salucci P, Danese L. 2003. *Mon. Not. R. Astron. Soc.* 341:1109–20
Boselli A, Gavazzi G. 2006. *Publ. Astron. Soc. Pac.* 118:517–59
Boselli A, Gavazzi G. 2014. *Astron. Astrophys. Rev.* 22:74
Bosma A. 1978. *The distribution and kinematics of neutral hydrogen in spiral galaxies of various morphological types*. PhD thesis, Groningen Univ.
Bournaud F, Elmegreen BG, Elmegreen DM. 2007. *Astrophys. J.* 670:237–48
Bower RG, Benson AJ, Malbon R, et al. 2006. *Mon. Not. R. Astron. Soc.* 370:645–55
Boylan-Kolchin M, Ma CP, Quataert E. 2006. *Mon. Not. R. Astron. Soc.* 369:1081–89
Brinchmann J, Charlot S, White SDM, et al. 2004. *Mon. Not. R. Astron. Soc.* 351:1151–79
Brodie JP, Romanowsky AJ, Strader J, et al. 2014. *Astrophys. J.* 796:52
Bruzual G, Charlot S. 2003. *Mon. Not. R. Astron. Soc.* 344:1000–28
Bryant JJ, Owers MS, Robotham ASG, et al. 2015. *Mon. Not. R. Astron. Soc.* 447:2857–79
Bundy K, Bershady MA, Law DR, et al. 2015. *Astrophys. J.* 798:7
Burbidge EM, Burbidge GR, Prendergast KH. 1959. *Astrophys. J.* 130:739–48
Burstein D, Bender R, Faber S, Nolthenius R. 1997. *Astron. J.* 114:1365–92
Byun YI, Grillmair CJ, Faber SM, et al. 1996. *Astron. J.* 111:1889–1900
Caon N, Capaccioli M, D'Onofrio M. 1993. *Mon. Not. R. Astron. Soc.* 265:1013–21
Capaccioli M. 1989. In *World of Galaxies*, ed. HG Corwin Jr., L Bottinelli, pp. 208–27. New York: Springer-Verlag
Capak P, Abraham RG, Ellis RS, et al. 2007. *Astrophys. J. Suppl.* 172:284–94
Cappellari M. 2002. *Mon. Not. R. Astron. Soc.* 333:400–10
Cappellari M. 2008. *Mon. Not. R. Astron. Soc.* 390:71–86
Cappellari M. 2013. *Astrophys. J. Lett.* 778:L2
Cappellari M, Bacon R, Bureau M, et al. 2006. *Mon. Not. R. Astron. Soc.* 366:1126–50
Cappellari M, Copin Y. 2003. *Mon. Not. R. Astron. Soc.* 342:345–54
Cappellari M, di Serego Alighieri S, Cimatti A, et al. 2009. *Astrophys. J. Lett.* 704:L34–39
Cappellari M, Emsellem E. 2004. *Publ. Astron. Soc. Pac.* 116:138–47
Cappellari M, Emsellem E, Bacon R, et al. 2007. *Mon. Not. R. Astron. Soc.* 379:418–44
Cappellari M, Emsellem E, Krajnović D, et al. 2011a. *Mon. Not. R. Astron. Soc.* 413:813–36
Cappellari M, Emsellem E, Krajnović D, et al. 2011b. *Mon. Not. R. Astron. Soc.* 416:1680–96
Cappellari M, McDermid RM. 2005. *Class. Quantum Gravity* 22:347–53
Cappellari M, McDermid RM, Alatalo K, et al. 2012. *Nature* 484:485–88
Cappellari M, McDermid RM, Alatalo K, et al. 2013a. *Mon. Not. R. Astron. Soc.* 432:1862–93

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Cappellari M, Romanowsky AJ, Brodie JP, et al. 2015. *Astrophys. J. Lett.* 804:L21
Cappellari M, Scott N, Alatalo K, et al. 2013b. *Mon. Not. R. Astron. Soc.* 432:1709–41
Cappellari M, Verolme EK, van der Marel RP, et al. 2002. *Astrophys. J.* 578:787–805
Carollo CM, Bschorr TJ, Renzini A, et al. 2013. *Astrophys. J.* 773:112
Carollo CM, Danziger IJ. 1994. *Mon. Not. R. Astron. Soc.* 270:523–69
Carollo CM, Franx M, Illingworth GD, Forbes DA. 1997. *Astrophys. J.* 481:710–34
Carter D. 1978. *Mon. Not. R. Astron. Soc.* 182:797–99
Cebrián M, Trujillo I. 2014. *Mon. Not. R. Astron. Soc.* 444:682–99
Cenarro AJ, Trujillo I. 2009. *Astrophys. J. Lett.* 696:L43–47
Ceverino D, Dekel A, Bournaud F. 2010. *Mon. Not. R. Astron. Soc.* 404:2151–69
Chabrier G. 2003. *Publ. Astron. Soc. Pac.* 115:763–95
Cheung E, Faber SM, Koo DC, et al. 2012. *Astrophys. J.* 760:131
Cid Fernandes R, Mateus A, Sodré L, Stasińska G, Gomes JM. 2005. *Mon. Not. R. Astron. Soc.* 358:363–78
Cimatti A, Cassata P, Pozzetti L, et al. 2008. *Astron. Astrophys.* 482:21–42
Cimatti A, Daddi E, Renzini A. 2006. *Astron. Astrophys.* 453:L29–33
Cimatti A, Daddi E, Renzini A, et al. 2004. *Nature* 430:184–87
Ciotti L. 1991. *Astron. Astrophys.* 249:99–106
Ciotti L, Lanzoni B, Renzini A. 1996. *Mon. Not. R. Astron. Soc.* 282:1–12
Colless M, Saglia RP, Burstein D, et al. 2001. *Mon. Not. R. Astron. Soc.* 321:277–305
Conroy C. 2013. *Annu. Rev. Astron. Astrophys.* 51:393–455
Conroy C, Dutton AA, Graves GJ, Mendel JT, van Dokkum PG. 2013. *Astrophys. J. Lett.* 776:L26
Conroy C, van Dokkum PG. 2012a. *Astrophys. J.* 747:69
Conroy C, van Dokkum PG. 2012b. *Astrophys. J.* 760:71
Conroy C, Wechsler RH, Kravtsov AV. 2006. *Astrophys. J.* 647:201–14
Conselice CJ. 2006. *Mon. Not. R. Astron. Soc.* 373:1389–408
Conselice CJ. 2014. *Annu. Rev. Astron. Astrophys.* 52:291–337
Contopoulos G. 1956. *Z. Astrophys.* 39:126–32
Cooper MC, Newman JA, Croton DJ, et al. 2006. *Mon. Not. R. Astron. Soc.* 370:198–212
Côté P, Ferrarese L, Jordán A, et al. 2007. *Astrophys. J.* 671:1456–65
Courteau S, Cappellari M, de Jong RS, et al. 2014. *Rev. Mod. Phys.* 86:47–119
Crane P, Stiavelli M, King IR, et al. 1993. *Astron. J.* 106:1371–93
Croton DJ, Springel V, White SDM, et al. 2006. *Mon. Not. R. Astron. Soc.* 365:11–28
Daddi E, Bournaud F, Walter F, et al. 2010. *Astrophys. J.* 713:686–707
Daddi E, Renzini A, Pirzkal N, et al. 2005. *Astrophys. J.* 626:680–97
Davies RL, Efstathiou G, Fall SM, Illingworth G, Schechter PL. 1983. *Astrophys. J.* 266:41–57
Davies RL, Kuntschner H, Emsellem E, et al. 2001. *Astrophys. J. Lett.* 548:L33–36
Davies RL, Sadler EM, Peletier RF. 1993. *Mon. Not. R. Astron. Soc.* 262:650–80
Davis M, Geller MJ. 1976. *Astrophys. J.* 208:13–19
Davis TA, Alatalo K, Sarzi M, et al. 2011. *Mon. Not. R. Astron. Soc.* 417:882–99
Davis TA, Young LM, Crocker AF, et al. 2014. *Mon. Not. R. Astron. Soc.* 444:3427–45
de Jong RS. 1996. *Astron. Astrophys.* 313:45–64
de Lorenzi F, Gerhard O, Coccato L, et al. 2009. *Mon. Not. R. Astron. Soc.* 395:76–96
de Lorenzi F, Gerhard O, Saglia RP, et al. 2008. *Mon. Not. R. Astron. Soc.* 385:1729–48
De Lucia G, Weinmann S, Poggianti BM, Aragón-Salamanca A, Zaritsky D. 2012. *Mon. Not. R. Astron. Soc.* 423:1277–92
de Rijcke S, Michielsen D, Dejonghe H, Zeilinger WW, Hau GKT. 2005. *Astron. Astrophys.* 438:491–505
de Vaucouleurs G. 1948. *Ann. Astrophys.* 11:247–87
de Vaucouleurs G. 1959. *Handb. Phys.* 53:275
de Vaucouleurs G. 1963. *Astrophys. J. Suppl.* 8:31–97
de Vaucouleurs G, de Vaucouleurs A, Corwin HG Jr., et al. 1991. *Third Reference Catalog of Bright Galaxies*. Berlin: Springer-Verlag
de Zeeuw PT, Bureau M, Emsellem E, et al. 2002. *Mon. Not. R. Astron. Soc.* 329:513–30
Deason AJ, Belokurov V, Evans NW, McCarthy IG. 2012. *Astrophys. J.* 748:2

Dekel A, Birnboim Y. 2006. *Mon. Not. R. Astron. Soc.* 368:2–20
Dekel A, Cox TJ. 2006. *Mon. Not. R. Astron. Soc.* 370:1445–53
Dekel A, Sari R, Ceverino D. 2009. *Astrophys. J.* 703:785–801
Delaye L, Huertas-Company M, Mei S, et al. 2014. *Mon. Not. R. Astron. Soc.* 441:203–23
D'Eugenio F, Houghton RCW, Davies RL, Dalla Bontà E. 2013. *Mon. Not. R. Astron. Soc.* 429:1258–66
Djorgovski S, Davis M. 1987. *Astrophys. J.* 313:59–68
D'Onofrio M, Capaccioli M, Caon N. 1994. *Mon. Not. R. Astron. Soc.* 271:523–29
Dressler A. 1980. *Astrophys. J.* 236:351–65
Dressler A, Lynden-Bell D, Burstein D, et al. 1987. *Astrophys. J.* 313:42–58
Dressler A, Oemler A Jr., Couch WJ, et al. 1997. *Astrophys. J.* 490:577–91
Drinkwater MJ, Gregg MD, Hilker M, et al. 2003. *Nature* 423:519–21
Duc PA, Cuillandre JC, Karabal E, et al. 2015. *Mon. Not. R. Astron. Soc.* 446:120–43
Duc PA, Cuillandre JC, Serra P, et al. 2011. *Mon. Not. R. Astron. Soc.* 417:863–81
Dutton AA, Macciò AV, Mendel JT, Simard L. 2013. *Mon. Not. R. Astron. Soc.* 432:2496–511
Dutton AA, Treu T. 2014. *Mon. Not. R. Astron. Soc.* 438:3594–602
Elmegreen DM, Elmegreen BG, Ravindranath S, Coe DA. 2007. *Astrophys. J.* 658:763–77
Emsellem E, Bacon R, Monnet G, Poulain P. 1996. *Astron. Astrophys.* 312:777–96
Emsellem E, Cappellari M, Krajnović D, et al. 2011. *Mon. Not. R. Astron. Soc.* 414:888–912
Emsellem E, Cappellari M, Krajnović D, et al. 2007. *Mon. Not. R. Astron. Soc.* 379:401–17
Emsellem E, Cappellari M, Peletier RF, et al. 2004. *Mon. Not. R. Astron. Soc.* 352:721–43
Emsellem E, Dejonghe H, Bacon R. 1999. *Mon. Not. R. Astron. Soc.* 303:495–514
Emsellem E, Krajnović D, Sarzi M. 2014. *Mon. Not. R. Astron. Soc.* 445:L79–83
Emsellem E, Monnet G, Bacon R. 1994. *Astron. Astrophys.* 285:723–38
Evans NW, Collett JL. 1994. *Astrophys. J. Lett.* 420:L67–70
Faber SM, Dressler A, Davies RL, Burstein D, Lynden-Bell D. 1987. In *Nearly Normal Galaxies: From the Planck Time to the Present*, ed. SM Faber, pp. 175–83. New York: Springer-Verlag
Faber SM, Jackson RE. 1976. *Astrophys. J.* 204:668–83
Faber SM, Tremaine S, Ajhar EA, et al. 1997. *Astron. J.* 114:1771–96
Fabian AC. 2012. *Annu. Rev. Astron. Astrophys.* 50:455–89
Falcón-Barroso J, Lyubenova M, van de Ven G. 2015. In *Galaxy Masses as Constraints of Formation Models*, ed. M Cappellari, S Courteau. *Proc. IAU Symp.* 311:77–88. Cambridge, UK: Cambridge Univ. Press
Falcón-Barroso J, van de Ven G, Peletier RF, et al. 2011. *Mon. Not. R. Astron. Soc.* 417:1787–816
Fang JJ, Faber SM, Koo DC, Dekel A. 2013. *Astrophys. J.* 776:63
Fasano G, Poggianti BM, Couch WJ, et al. 2000. *Astrophys. J.* 542:673–83
Fasano G, Vio R. 1991. *Mon. Not. R. Astron. Soc.* 249:629–33
Fernández Lorenzo M, Sulentic J, Verdes-Montenegro L, Argudo-Fernández M. 2013. *Mon. Not. R. Astron. Soc.* 434:325–35
Ferrarese L, Côté P, Jordán A, et al. 2006. *Astrophys. J. Suppl.* 164:334–434
Ferrarese L, Merritt D. 2000. *Astrophys. J. Lett.* 539:L9–12
Ferrarese L, van den Bosch FC, Ford HC, Jaffe W, O'Connell RW. 1994. *Astron. J.* 108:1598–609
Ferreras I, La Barbera F, de la Rosa IG, et al. 2013. *Mon. Not. R. Astron. Soc.* 429:L15–19
Fogarty LMR, Scott N, Owers MS, et al. 2014. *Mon. Not. R. Astron. Soc.* 443:485–503
Fogarty LMR, Scott N, Owers MS, et al. 2015. *Mon. Not. R. Astron. Soc.* 454:2050–66
Forbes DA, Lasky P, Graham AW, Spitler L. 2008. *Mon. Not. R. Astron. Soc.* 389:1924–36
Forbes DA, Ponman TJ, Brown RJN. 1998. *Astrophys. J. Lett.* 508:L43–46
Forestell AD, Gebhardt K. 2010. *Astrophys. J.* 716:370–83
Förster Schreiber NM, Genzel R, Bouché N, et al. 2009. *Astrophys. J.* 706:1364–428
Förster Schreiber NM, Genzel R, Lehnert MD, et al. 2006. *Astrophys. J.* 645:1062–75
Foster C, Arnold JA, Forbes DA, et al. 2013. *Mon. Not. R. Astron. Soc.* 435:3587–91
Franx M, Illingworth G. 1990. *Astrophys. J. Lett.* 359:L41–45
Franx M, Illingworth G, de Zeeuw T. 1991. *Astrophys. J.* 383:112–34
Franx M, Illingworth G, Heckman T. 1989. *Astron. J.* 98:538–76
Franx M, Illingworth GD. 1988. *Astrophys. J. Lett.* 327:L55–59

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Franx M, van Dokkum PG, Förster Schreiber NM, et al. 2008. *Astrophys. J.* 688:770–88
Fraternali F. 2014. In *Setting the Scene for Gaia and LAMOST - The Current and Next Generations of Surveys and Models*, ed. S Feltzing, G Zhao, NA Walton, P Whitelock. *Proc. IAU Symp.* 298:228–39. Cambridge, UK: Cambridge Univ. Press
Fraternali F, Binney JJ. 2008. *Mon. Not. R. Astron. Soc.* 386:935–44
Gardner JP, Mather JC, Clampin M, et al. 2006. *Space Sci. Rev.* 123:485–606
Gavazzi R, Treu T, Rhodes JD, et al. 2007. *Astrophys. J.* 667:176–90
Gebhardt K, Bender R, Bower G, et al. 2000a. *Astrophys. J. Lett.* 539:L13–16
Gebhardt K, Richstone D, Kormendy J, et al. 2000b. *Astron. J.* 119:1157–71
Gebhardt K, Richstone D, Tremaine S, et al. 2003. *Astrophys. J.* 583:92–115
Geha M, Blanton MR, Yan R, Tinker JL. 2012. *Astrophys. J.* 757:85
Genzel R, Burkert A, Bouché N, et al. 2008. *Astrophys. J.* 687:59–77
Genzel R, Newman S, Jones T, et al. 2011. *Astrophys. J.* 733:101
Genzel R, Tacconi LJ, Eisenhauer F, et al. 2006. *Nature* 442:786–89
Gerhard O. 1996. In *Spiral Galaxies in the Near-IR*, ed. D Minniti, HW Rix, pp. 138–46. Berlin: Springer-Verlag
Gerhard O. 2013. In *The Intriguing Life of Massive Galaxies*, ed. D Thomas, A Pasquali, I Ferreras. *Proc. IAU Symp.* 295:211–20 Cambridge, UK: Cambridge Univ. Press
Gerhard O, Jeske G, Saglia RP, Bender R. 1998. *Mon. Not. R. Astron. Soc.* 295:197–215
Gerhard O, Kronawitter A, Saglia RP, Bender R. 2001. *Astron. J.* 121:1936–51
Gerhard OE, Binney JJ. 1996. *Mon. Not. R. Astron. Soc.* 279:993–1004
Giavalisco M, Ferguson HC, Koekemoer AM, et al. 2004. *Astrophys. J. Lett.* 600:L93–98
Giovanelli R, Haynes MP, Chincarini GL. 1986. *Astrophys. J.* 300:77–92
Glass L, Ferrarese L, Côté P, et al. 2011. *Astrophys. J.* 726:31
Glazebrook K, Abraham RG, McCarthy PJ, et al. 2004. *Nature* 430:181–84
Gnedin OY, Kravtsov AV, Klypin AA, Nagai D. 2004. *Astrophys. J.* 616:16–26
González Delgado RM, García-Benito R, Pérez E, et al. 2015. *Astron. Astrophys.* 581:A103
Graham A, Colless M. 1997. *Mon. Not. R. Astron. Soc.* 287:221–39
Graham AW. 2001. *Astron. J.* 121:820–40
Graham AW. 2012. *Astrophys. J.* 746:113
Graham AW. 2013. In *Planets, Stars and Stellar Systems. Volume 6: Extragalactic Astronomy and Cosmology*, ed. TD Oswalt, WC Keel, pp. 91–139. Berlin: Springer
Graham AW, Erwin P, Trujillo I, Asensio Ramos A. 2003. *Astron. J.* 125:2951–63
Graham AW, Guzmán R. 2003. *Astron. J.* 125:2936–50
Graham AW, Worley CC. 2008. *Mon. Not. R. Astron. Soc.* 388:1708–28
Granato GL, De Zotti G, Silva L, Bressan A, Danese L. 2004. *Astrophys. J.* 600:580–94
Graves GJ, Faber SM, Schiavon RP. 2009. *Astrophys. J.* 698:1590–608
Greene JE, Janish R, Ma CP, et al. 2015. *Astrophys. J.* 807:11
Grogin NA, Kocevski DD, Faber SM, et al. 2011. *Astrophys. J. Suppl.* 197:35
Guérou A, Emsellem E, McDermid RM, et al. 2015. *Astrophys. J.* 804:70
Guo Q, White S, Li C, Boylan-Kolchin M. 2010. *Mon. Not. R. Astron. Soc.* 404:1111–20
Häring N, Rix HW. 2004. *Astrophys. J. Lett.* 604:L89–92
Hausman MA, Ostriker JP. 1978. *Astrophys. J.* 224:320–36
Heckman TM, Best PN. 2014. *Annu. Rev. Astron. Astrophys.* 52:589–660
Hernquist L, Spergel DN, Heyl JS. 1993. *Astrophys. J.* 416:415–24
Hogg DW. 1999. arXiv:astro-ph/9905116
Hopkins PF, Bundy K, Hernquist L, Wuyts S, Cox TJ. 2010. *Mon. Not. R. Astron. Soc.* 401:1099–117
Hopkins PF, Hernquist L, Cox TJ, et al. 2006. *Astrophys. J. Suppl.* 163:1–49
Houghton RCW, Davies RL, D'Eugenio F, et al. 2013. *Mon. Not. R. Astron. Soc.* 436:19–33
Hubble EP. 1926. *Astrophys. J.* 64:321–69
Hubble EP. 1936. *Realm of the Nebulae*. New Haven, CT: Yale Univ. Press
Hudson MJ, Lucey JR, Smith RJ, Steel J. 1997. *Mon. Not. R. Astron. Soc.* 291:488–504
Huertas-Company M, Mei S, Shankar F, et al. 2013. *Mon. Not. R. Astron. Soc.* 428:1715–42

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Huertas-Company M, Pérez-González PG, Mei S, et al. 2015. *Astrophys. J.* 809:95
Humphrey PJ, Buote DA. 2010. *Mon. Not. R. Astron. Soc.* 403:2143–51
Hyde JB, Bernardi M. 2009. *Mon. Not. R. Astron. Soc.* 394:1978–90
Illingworth G. 1977. *Astrophys. J. Lett.* 218:L43–47
Jeans JH. 1915. *Mon. Not. R. Astron. Soc.* 76:70–84
Jeans JH. 1922. *Mon. Not. R. Astron. Soc.* 82:122–32
Jedrzejewski RI. 1987. *Mon. Not. R. Astron. Soc.* 226:747–68
Jedrzejewski RI, Schechter PL. 1988. *Astrophys. J. Lett.* 330:L87–91
Jimmy Tran KV, Brough S, et al. 2013. *Astrophys. J.* 778:171
Jorgensen I, Franx M, Kjaergaard P. 1996. *Mon. Not. R. Astron. Soc.* 280:167–85
Kassin SA, Weiner BJ, Faber SM, et al. 2012. *Astrophys. J.* 758:106
Kauffmann G, Heckman TM, White SDM, et al. 2003. *Mon. Not. R. Astron. Soc.* 341:54–69
Kauffmann G, White SDM, Heckman TM, et al. 2004. *Mon. Not. R. Astron. Soc.* 353:713–31
Kaviraj S, Schawinski K, Devriendt JEG, et al. 2007. *Astrophys. J. Suppl.* 173:619–42
Keres D, Katz N, Weinberg DH, Davé R. 2005. *Mon. Not. R. Astron. Soc.* 363:2–28
Khochfar S, Burkert A. 2006. *Astron. Astrophys.* 445:403–12
Khochfar S, Emsellem E, Serra P, et al. 2011. *Mon. Not. R. Astron. Soc.* 417:845–62
Khochfar S, Silk J. 2006. *Astrophys. J. Lett.* 648:L21–24
Koekemoer AM, Faber SM, Ferguson HC, et al. 2011. *Astrophys. J. Suppl.* 197:36
Koopmans LVE, Bolton A, Treu T, et al. 2009. *Astrophys. J. Lett.* 703:L51–54
Kormendy J. 1977. *Astrophys. J.* 218:333–46
Kormendy J. 1982a. In *Morphology and Dynamics of Galaxies*, *12th Advanced Course of the Swiss Society of Astronomy and Astrophysics*, ed. L Martinet, M Mayor, pp. 113–288. Sauverny: Geneva Obs
Kormendy J. 1982b. *Astrophys. J.* 257:75–88
Kormendy J. 1985. *Astrophys. J.* 295:73–79
Kormendy J. 2016. In *Galactic Bulges*, ed. E Laurikainen, R Peletier, D Gadotti, pp. 431–77. Berlin: Springer
Kormendy J, Bender R. 1996. *Astrophys. J. Lett.* 464:L119–22
Kormendy J, Bender R. 2012. *Astrophys. J. Suppl.* 198:2
Kormendy J, Djorgovski S. 1989. *Annu. Rev. Astron. Astrophys.* 27:235–77
Kormendy J, Fisher DB, Cornell ME, Bender R. 2009. *Astrophys. J. Suppl.* 182:216–309
Kormendy J, Ho LC. 2013. *Annu. Rev. Astron. Astrophys.* 51:511–653
Kormendy J, Illingworth G. 1982. *Astrophys. J.* 256:460–80
Krajnović D, Alatalo K, Blitz L, et al. 2013a. *Mon. Not. R. Astron. Soc.* 432:1768–95
Krajnović D, Bacon R, Cappellari M, et al. 2008. *Mon. Not. R. Astron. Soc.* 390:93–117
Krajnović D, Cappellari M, de Zeeuw PT, Copin Y. 2006. *Mon. Not. R. Astron. Soc.* 366:787–802
Krajnović D, Cappellari M, Emsellem E, McDermid RM, de Zeeuw PT. 2005. *Mon. Not. R. Astron. Soc.* 357:1113–33
Krajnović D, Emsellem E, Cappellari M, et al. 2011. *Mon. Not. R. Astron. Soc.* 414:2923–49
Krajnović D, Karick AM, Davies RL, et al. 2013b. *Mon. Not. R. Astron. Soc.* 433:2812–39
Krajnović D, Weilbacher PM, Urrutia T, et al. 2015. *Mon. Not. R. Astron. Soc.* 452:2–18
Kroupa P. 2001. *Mon. Not. R. Astron. Soc.* 322:231–46
Kuntschner H, Emsellem E, Bacon R, et al. 2010. *Mon. Not. R. Astron. Soc.* 408:97–132
La Barbera F, Ferreras I, Vazdekis A. 2015. *Mon. Not. R. Astron. Soc.* 449:L137–41
Lablanche PY, Cappellari M, Emsellem E, et al. 2012. *Mon. Not. R. Astron. Soc.* 424:1495–1521
Lambas DG, Maddox SJ, Loveday J. 1992. *Mon. Not. R. Astron. Soc.* 258:404–14
Läsker R, van den Bosch RCE, van de Ven G, et al. 2013. *Mon. Not. R. Astron. Soc.* 434:L31–35
Lauer TR. 1985. *Mon. Not. R. Astron. Soc.* 216:429–38
Lauer TR. 2012. *Astrophys. J.* 759:64
Lauer TR, Ajhar EA, Byun YI, et al. 1995. *Astron. J.* 110:2622–54
Lauer TR, Faber SM, Gebhardt K, et al. 2005. *Astron. J.* 129:2138–85
Lauer TR, Faber SM, Richstone D, et al. 2007a. *Astrophys. J.* 662:808–34
Lauer TR, Gebhardt K, Faber SM, et al. 2007b. *Astrophys. J.* 664:226–56
Lauer TR, Postman M, Strauss MA, Graves GJ, Chisari NE. 2014. *Astrophys. J.* 797:82

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Laurikainen E, Salo H, Buta R, Knapen JH, Comerón S. 2010. *Mon. Not. R. Astron. Soc.* 405:1089–118
Law DR, Shapley AE, Steidel CC, et al. 2012. *Nature* 487:338–40
Leauthaud A, Tinker J, Bundy K, et al. 2012. *Astrophys. J.* 744:159
Li H, Li R, Mao S, et al. 2016. *Mon. Not. R. Astron. Soc.* 455:3680–92
Liu FS, Mao S, Meng XM. 2012. *Mon. Not. R. Astron. Soc.* 423:422–36
Łokas EL, Mamon GA. 2003. *Mon. Not. R. Astron. Soc.* 343:401–12
Longhetti M, Saracco P, Severgnini P, et al. 2007. *Mon. Not. R. Astron. Soc.* 374:614–26
Ma CP, Greene JE, McConnell N, et al. 2014. *Astrophys. J.* 795:158
Magorrian J, Tremaine S, Richstone D, et al. 1998. *Astron. J.* 115:2285–305
Magoulas C, Springob CM, Colless M, et al. 2012. *Mon. Not. R. Astron. Soc.* 427:245–73
Maltby DT, Aragón-Salamanca A, Gray ME, et al. 2010. *Mon. Not. R. Astron. Soc.* 402:282–94
Maraston C, Strömbäck G. 2011. *Mon. Not. R. Astron. Soc.* 418:2785–811
Marconi A, Hunt LK. 2003. *Astrophys. J. Lett.* 589:L21–24
Martig M, Bournaud F, Teyssier R, Dekel A. 2009. *Astrophys. J.* 707:250–67
Martig M, Crocker AF, Bournaud F, et al. 2013. *Mon. Not. R. Astron. Soc.* 432:1914–27
Martín-Navarro I, Pérez-González PG, Trujillo I, et al. 2015a. *Astrophys. J. Lett.* 798:L4
Martín-Navarro I, Vazdekis A, La Barbera F, et al. 2015b. *Astrophys. J. Lett.* 806:L31
Masters KL, Mosleh M, Romer KA, et al. 2010. *Mon. Not. R. Astron. Soc.* 405:783–99
Matković A, Guzmán R. 2005. *Mon. Not. R. Astron. Soc.* 362:289–300
McConnell NJ, Ma CP, Gebhardt K, et al. 2011. *Nature* 480:215–18
McDermid RM, Alatalo K, Blitz L, et al. 2015. *Mon. Not. R. Astron. Soc.* 448:3484–513
McDermid RM, Cappellari M, Alatalo K, et al. 2014. *Astrophys. J. Lett.* 792:L37
McDermid RM, Emsellem E, Shapiro KL, et al. 2006. *Mon. Not. R. Astron. Soc.* 373:906–58
McLaughlin DE. 1999. *Astrophys. J. Lett.* 512:L9–12
Melnick J, Sargent WLW. 1977. *Astrophys. J.* 215:401–7
Mihos JC, Hernquist L. 1994. *Astrophys. J. Lett.* 431:L9–12
Milosavljević M, Merritt D. 2001. *Astrophys. J.* 563:34–62
Misgeld I, Hilker M. 2011. *Mon. Not. R. Astron. Soc.* 414:3699–710
Mo H, van den Bosch FC, White S. 2010. *Galaxy Formation and Evolution*. Cambridge, UK: Cambridge Univ. Press
Monnet G, Bacon R, Emsellem E. 1992. *Astron. Astrophys.* 253:366–73
Moore B, Katz N, Lake G, Dressler A, Oemler A. 1996. *Nature* 379:613–16
Moran SM, Ellis RS, Treu T, et al. 2007. *Astrophys. J.* 671:1503–22
Morganti L, Gerhard O, Coccato L, Martinez-Valpuesta I, Arnaboldi M. 2013. *Mon. Not. R. Astron. Soc.* 431:3570–88
Morganti R, de Zeeuw PT, Oosterloo TA, et al. 2006. *Mon. Not. R. Astron. Soc.* 371:157–69
Moster BP, Naab T, White SDM. 2013. *Mon. Not. R. Astron. Soc.* 428:3121–38
Moster BP, Somerville RS, Maulbetsch C, et al. 2010. *Astrophys. J.* 710:903–23
Murphy JD, Gebhardt K, Adams JJ. 2011. *Astrophys. J.* 729:129
Muzzin A, Marchesini D, Stefanon M, et al. 2013. *Astrophys. J.* 777:18
Naab T, Burkert A. 2003. *Astrophys. J.* 597:893–906
Naab T, Johansson PH, Ostriker JP. 2009. *Astrophys. J. Lett.* 699:L178–82
Naab T, Oser L, Emsellem E, et al. 2014. *Mon. Not. R. Astron. Soc.* 444:3357–87
Napolitano NR, Pota V, Romanowsky AJ, et al. 2014. *Mon. Not. R. Astron. Soc.* 439:659–72
Napolitano NR, Romanowsky AJ, Capaccioli M, et al. 2011. *Mon. Not. R. Astron. Soc.* 411:2035–53
Navarro JF, Frenk CS, White SDM. 1996. *Astrophys. J.* 462:563–75
Newman AB, Ellis RS, Bundy K, Treu T. 2012. *Astrophys. J.* 746:162
Newman AB, Ellis RS, Treu T. 2015. *Astrophys. J.* 814:26
Newman AB, Treu T, Ellis RS, et al. 2013a. *Astrophys. J.* 765:24
Newman JA, Cooper MC, Davis M, et al. 2013b. *Astrophys. J. Suppl.* 208:5
Nipoti C, Londrillo P, Ciotti L. 2003. *Mon. Not. R. Astron. Soc.* 342:501–12
Norris MA, Kannappan SJ, Forbes DA, et al. 2014. *Mon. Not. R. Astron. Soc.* 443:1151–72
Ocvirk P, Pichon C, Lancon A, Thiébaut E. 2006. *Mon. Not. R. Astron. Soc.* 365:74–84

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Oemler A Jr. 1974. *Astrophys. J.* 194:1–20
Onodera M, Daddi E, Gobat R, et al. 2010. *Astrophys. J. Lett.* 715:L6–11
Oosterloo T, Morganti R, Crocker A, et al. 2010. *Mon. Not. R. Astron. Soc.* 409:500–14
Oser L, Naab T, Ostriker JP, Johansson PH. 2012. *Astrophys. J.* 744:63
Padilla ND, Strauss MA. 2008. *Mon. Not. R. Astron. Soc.* 388:1321–34
Pahre MA, Djorgovski SG, de Carvalho RR. 1998. *Astron. J.* 116:1591–605
Paturel G, Petit C, Prugniel P, et al. 2003. *Astron. Astrophys.* 412:45–55
Peletier RF, Davies RL, Illingworth GD, Davis LE, Cawson M. 1990. *Astron. J.* 100:1091–142
Peng Yj, Lilly SJ, Kovač K, et al. 2010. *Astrophys. J.* 721:193–221
Pinkney J, Gebhardt K, Bender R, et al. 2003. *Astrophys. J.* 596:903–29
Poggianti BM, Calvi R, Bindoni D, et al. 2013. *Astrophys. J.* 762:77
Poggianti BM, Desai V, Finn R, et al. 2008. *Astrophys. J.* 684:888–904
Porter LA, Somerville RS, Primack JR, Johansson PH. 2014. *Mon. Not. R. Astron. Soc.* 444:942–60
Posacki S, Cappellari M, Treu T, Pellegrini S, Ciotti L. 2015. *Mon. Not. R. Astron. Soc.* 446:493–509
Postman M, Franx M, Cross NJG, et al. 2005. *Astrophys. J.* 623:721–41
Postman M, Geller MJ. 1984. *Astrophys. J.* 281:95–99
Prugniel P, Simien F. 1996. *Astron. Astrophys.* 309:749–59
Prugniel P, Simien F. 1997. *Astron. Astrophys.* 321:111–22
Querejeta M, Eliche-Moral MC, Tapia T, et al. 2015. *Astron. Astrophys.* 579:L2
Raskutti S, Greene JE, Murphy JD. 2014. *Astrophys. J.* 786:23
Remus RS, Burkert A, Dolag K, et al. 2013. *Astrophys. J.* 766:71
Renzini A. 2006. *Annu. Rev. Astron. Astrophys.* 44:141–92
Renzini A, Ciotti L. 1993. *Astrophys. J. Lett.* 416:L49–52
Richstone DO, Tremaine S. 1988. *Astrophys. J.* 327:82–88
Rix H, Franx M, Fisher D, Illingworth G. 1992. *Astrophys. J. Lett.* 400:L5–8
Rix HW, de Zeeuw PT, Cretton N, van der Marel RP, Carollo CM. 1997. *Astrophys. J.* 488:702–19
Robertson B, Bullock JS, Cox TJ, et al. 2006. *Astrophys. J.* 645:986–1000
Romanowsky AJ, Douglas NG, Arnaboldi M, et al. 2003. *Science* 301:1696–98
Romanowsky AJ, Kochanek CS. 1997. *Mon. Not. R. Astron. Soc.* 287:35–50
Rousseeuw P, Van Driessen K. 2006. *Data Min. Knowl. Discov.* 12:29–45
Rubin VC, Ford WK Jr. 1970. *Astrophys. J.* 159:379–403
Rubin VC, Ford WK Jr., Thonnard N. 1980. *Astrophys. J.* 238:471–87
Rubin VC, Graham JA, Kenney JDP. 1992. *Astrophys. J. Lett.* 394:L9–12
Rusli SP, Thomas J, Saglia RP, et al. 2013. *Astron. J.* 146:45
Rybicki GB. 1987. In *Structure and Dynamics of Elliptical Galaxies*, ed. PT de Zeeuw. *Proc. IAU Symp.* 127:397–98. Dordrecht: D. Reidel
Ryden B. 1992. *Astrophys. J.* 396:445–52
Ryś A, van de Ven G, Falcón-Barroso J. 2014. *Mon. Not. R. Astron. Soc.* 439:284–99
Saintonge A, Tacconi LJ, Fabello S, et al. 2012. *Astrophys. J.* 758:73
Salim S, Rich RM, Charlot S, et al. 2007. *Astrophys. J. Suppl.* 173:267–92
Salpeter EE. 1955. *Astrophys. J.* 121:161–67
Sánchez SF, Kennicutt RC, Gil de Paz A, et al. 2012. *Astron. Astrophys.* 538:A8
Sandage A. 1961. *The Hubble Atlas*. Washington, D.C.: Carnegie Inst.
Sandage A. 1975. In *Galaxies and the Universe*, ed. A Sandage, M Sandage, J Kristian, pp. 1–36. Chicago: Chicago Univ. Press
Sandage A. 2005. *Annu. Rev. Astron. Astrophys.* 43:581–624
Sandage A, Freeman KC, Stokes NR. 1970. *Astrophys. J.* 160:831–44
Sandage A, Hardy E. 1973. *Astrophys. J.* 183:743–58
Sarzi M, Alatalo K, Blitz L, et al. 2013. *Mon. Not. R. Astron. Soc.* 432:1845–61
Sarzi M, Falcón-Barroso J, Davies RL, et al. 2006. *Mon. Not. R. Astron. Soc.* 366:1151–200
Sarzi M, Ledo HR, Dotti M. 2015. *Mon. Not. R. Astron. Soc.* 453:1070–78
Schechter P. 1976. *Astrophys. J.* 203:297–306
Schechter PL, Gunn JE. 1979. *Astrophys. J.* 229:472–84

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Schoenmakers RHM, Franx M, de Zeeuw PT. 1997. *Mon. Not. R. Astron. Soc.* 292:349–64
Schwarzschild M. 1979. *Astrophys. J.* 232:236–47
Scodeggio M, Gavazzi G, Belsole E, Pierini D, Boselli A. 1998. *Mon. Not. R. Astron. Soc.* 301:1001–18
Scott N, Cappellari M, Davies RL, et al. 2009. *Mon. Not. R. Astron. Soc.* 398:1835–57
Scott N, Cappellari M, Davies RL, et al. 2013a. *Mon. Not. R. Astron. Soc.* 432:1894–913
Scott N, Davies RL, Houghton RCW, et al. 2014. *Mon. Not. R. Astron. Soc.* 441:274–88
Scott N, Fogarty LMR, Owers MS, et al. 2015. *Mon. Not. R. Astron. Soc.* 451:2723–34
Scott N, Graham AW, Schombert J. 2013b. *Astrophys. J.* 768:76
Scoville N, Abraham RG, Aussel H, et al. 2007. *Astrophys. J. Suppl.* 172:38–45
Serra P, Oosterloo T, Morganti R, et al. 2012. *Mon. Not. R. Astron. Soc.* 422:1835–62
Sérsic JL. 1968. *Atlas de galaxias australes*. Córdoba: Obs. Astron. Univ. Nac. Córdoba
Shen S, Mo HJ, White SDM, et al. 2003. *Mon. Not. R. Astron. Soc.* 343:978–94
Shetty S, Cappellari M. 2014. *Astrophys. J. Lett.* 786:L10
Sil'chenko OK. 1999. *Astron. J.* 117:2725–35
Sil'chenko OK, Vlasyuk VV, Burenkov AN. 1997. *Astron. Astrophys.* 326:941–49
Silk J, Rees MJ. 1998. *Astron. Astrophys.* 331:L1–4
Simien F, de Vaucouleurs G. 1986. *Astrophys. J.* 302:564–78
Skibba RA, Bamford SP, Nichol RC, et al. 2009. *Mon. Not. R. Astron. Soc.* 399:966–82
Skrutskie MF, Cutri RM, Stiening R, et al. 2006. *Astron. J.* 131:1163–83
Smith GP, Treu T, Ellis RS, Moran SM, Dressler A. 2005. *Astrophys. J.* 620:78–87
Smith RJ. 2014. *Mon. Not. R. Astron. Soc.* 443:L69–73
Smith RJ, Lucey JR, Carter D. 2012a. *Mon. Not. R. Astron. Soc.* 426:2994–3007
Smith RJ, Lucey JR, Price J, Hudson MJ, Phillipps S. 2012b. *Mon. Not. R. Astron. Soc.* 419:3167–80
Somerville RS, Davé R. 2015. *Annu. Rev. Astron. Astrophys.* 53:51–113
Sonnenfeld A, Treu T, Marshall PJ, et al. 2015. *Astrophys. J.* 800:94
Spiniello C, Trager SC, Koopmans LVE. 2015. *Astrophys. J.* 803:87
Spiniello C, Trager SC, Koopmans LVE, Chen YP. 2012. *Astrophys. J. Lett.* 753:L32
Spolaor M, Proctor RN, Forbes DA, Couch WJ. 2009. *Astrophys. J. Lett.* 691:L138–41
Springel V, White SDM, Jenkins A, et al. 2005. *Nature* 435:629–36
Springob CM, Magoulas C, Proctor R, et al. 2012. *Mon. Not. R. Astron. Soc.* 420:2773–84
Stanford SA, Eisenhardt PR, Dickinson M. 1998. *Astrophys. J.* 492:461–79
Stark AA. 1977. *Astrophys. J.* 213:368–73
Statler TS. 1987. *Astrophys. J.* 321:113–52
Stott JP, Swinbank AM, Johnson HL, et al. 2016. *Mon. Not. R. Astron. Soc.* 457:1888–904
Strateva I, Ivezić Ž, Knapp GR, et al. 2001. *Astron. J.* 122:1861–74
Syer D, Tremaine S. 1996. *Mon. Not. R. Astron. Soc.* 282:223–33
Tacconi LJ, Genzel R, Neri R, et al. 2010. *Nature* 463:781–84
Tasca LAM, Kneib J, Iovino A, et al. 2009. *Astron. Astrophys.* 503:379–98
Thomas D, Maraston C, Bender R, Mendes de Oliveira C. 2005. *Astrophys. J.* 621:673–94
Thomas J, Jesseit R, Saglia RP, et al. 2009. *Mon. Not. R. Astron. Soc.* 393:641–52
Thomas J, Saglia RP, Bender R, et al. 2011. *Mon. Not. R. Astron. Soc.* 415:545–62
Toft S, Gallazzi A, Zirm A, et al. 2012. *Astrophys. J.* 754:3
Toft S, van Dokkum P, Franx M, et al. 2007. *Astrophys. J.* 671:285–302
Tojeiro R, Heavens AF, Jimenez R, Panter B. 2007. *Mon. Not. R. Astron. Soc.* 381:1252–66
Tortora C, La Barbera F, Napolitano NR, de Carvalho RR, Romanowsky AJ. 2012. *Mon. Not. R. Astron. Soc.* 425:577–94
Tortora C, Napolitano NR, Romanowsky AJ, Capaccioli M, Covone G. 2009. *Mon. Not. R. Astron. Soc.* 396:1132–50
Tortora C, Romanowsky AJ, Napolitano NR. 2013. *Astrophys. J.* 765:8
Tremblay B, Merritt D. 1996. *Astron. J.* 111:2243–47
Treu T, Auger MW, Koopmans LVE, et al. 2010. *Astrophys. J.* 709:1195–202
Treu T, Ellis RS, Kneib J, et al. 2003. *Astrophys. J.* 591:53–78
Trujillo I, Burkert A, Bell EF. 2004a. *Astrophys. J. Lett.* 600:L39–42

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12

Trujillo I, Conselice CJ, Bundy K, et al. 2007. *Mon. Not. R. Astron. Soc.* 382:109–20
Trujillo I, Erwin P, Asensio Ramos A, Graham AW. 2004b. *Astron. J.* 127:1917–42
Trujillo I, Feulner G, Goranova Y, et al. 2006a. *Mon. Not. R. Astron. Soc.* 373:L36–40
Trujillo I, Förster Schreiber NM, Rudnick G, et al. 2006b. *Astrophys. J.* 650:18–41
Valluri M, Ferrarese L, Merritt D, Joseph CL. 2005. *Astrophys. J.* 628:137–52
Valluri M, Merritt D, Emsellem E. 2004. *Astrophys. J.* 602:66–92
van de Sande J, Kriek M, Franx M, et al. 2011. *Astrophys. J. Lett.* 736:L9
van de Sande J, Kriek M, Franx M, et al. 2013. *Astrophys. J.* 771:85
van den Bergh S. 1976. *Astrophys. J.* 206:883–87
van den Bergh S. 2007. *Astron. J.* 134:1508–14
van den Bosch FC. 1997. *Mon. Not. R. Astron. Soc.* 287:543–55
van den Bosch RCE, Gebhardt K, Gültekin K, et al. 2012. *Nature* 491:729–31
van den Bosch RCE, van de Ven G. 2009. *Mon. Not. R. Astron. Soc.* 398:1117–28
van den Bosch RCE, van de Ven G, Verolme EK, Cappellari M, de Zeeuw PT. 2008. *Mon. Not. R. Astron. Soc.* 385:647–66
van der Marel RP. 1991. *Mon. Not. R. Astron. Soc.* 253:710–26
van der Marel RP, Cretton N, de Zeeuw PT, Rix HW. 1998. *Astrophys. J.* 493:613–31
van der Marel RP, van Dokkum PG. 2007. *Astrophys. J.* 668:756–71
van der Wel A, Franx M, van Dokkum PG, et al. 2014. *Astrophys. J.* 788:28
van Dokkum PG, Conroy C. 2010. *Nature* 468:940–42
van Dokkum PG, Conroy C. 2012. *Astrophys. J.* 760:70
van Dokkum PG, Franx M, Fabricant D, Illingworth GD, Kelson DD. 2000. *Astrophys. J.* 541:95–111
van Dokkum PG, Franx M, Kriek M, et al. 2008. *Astrophys. J. Lett.* 677:L5–8
van Dokkum PG, Leja J, Nelson EJ, et al. 2013. *Astrophys. J. Lett.* 771:L35
van Dokkum PG, Nelson EJ, Franx M, et al. 2015. *Astrophys. J.* 813:23
van Dokkum PG, Whitaker KE, Brammer G, et al. 2010. *Astrophys. J.* 709:1018–41
Vazdekis A, Ricciardelli E, Cenarro AJ, et al. 2012. *Mon. Not. R. Astron. Soc.* 424:157–71
Walcher CJ, Wisotzki L, Bekeraité S, et al. 2014. *Astron. Astrophys.* 569:A1
Weijmans AM, Cappellari M, Bacon R, et al. 2009. *Mon. Not. R. Astron. Soc.* 398:561–74
Weijmans AM, de Zeeuw PT, Emsellem E, et al. 2014. *Mon. Not. R. Astron. Soc.* 444:3340–56
Weijmans AM, Krajnović D, van de Ven G, et al. 2008. *Mon. Not. R. Astron. Soc.* 383:1343–58
White SDM. 1983. In *Internal Kinematics and Dynamics of Galaxies*, ed. E Athanassoula. *Proc. IAU Symp.* 100:337–44
White SDM, Rees MJ. 1978. *Mon. Not. R. Astron. Soc.* 183:341–58
Wilkinson A, Sharples RM, Fosbury RAE, Wallace PT. 1986. *Mon. Not. R. Astron. Soc.* 218:297–329
Wilkinson DM, Maraston C, Thomas D, et al. 2015. *Mon. Not. R. Astron. Soc.* 449:328–60
Wilman DJ, Erwin P. 2012. *Astrophys. J.* 746:160
Wilman DJ, Oemler A, Mulchaey JS, et al. 2009. *Astrophys. J.* 692:298–308
Wisnioski E, Förster Schreiber NM, Wuyts S, et al. 2015. *Astrophys. J.* 799:209
Wolf C, Aragón-Salamanca A, Balogh M, et al. 2009. *Mon. Not. R. Astron. Soc.* 393:1302–23
Yldrm A, van den Bosch RCE, van de Ven G, et al. 2015. *Mon. Not. R. Astron. Soc.* 452:1792–816
Young CK, Currie MJ. 1994. *Mon. Not. R. Astron. Soc.* 268:L11–15
Young LM, Bureau M, Cappellari M. 2008. *Astrophys. J.* 676:317–34
Young LM, Bureau M, Davis TA, et al. 2011. *Mon. Not. R. Astron. Soc.* 414:940–67
Young LM, Scott N, Serra P, et al. 2014. *Mon. Not. R. Astron. Soc.* 444:3408–26
Zehavi I, Zheng Z, Weinberg DH, et al. 2005. *Astrophys. J.* 630:1–27
Zheng Z, Berlind AA, Weinberg DH, et al. 2005. *Astrophys. J.* 633:791–809
Zirm AW, van der Wel A, Franx M, et al. 2007. *Astrophys. J.* 656:66–72

Downloaded from www.annualreviews.org. WIB6417 - Max-Planck-Gesellschaft (ar-271826) IP: 130.183.139.54 On: Thu, 01 Oct 2026 08:17:12